Wealth Dynamics and Path Dependence *

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Abstract

I show that in an open economy, financial constraints reduce the speed of transition towards comparative advantage, and can make the steady-state specialization depend on the initial distribution of wealth. In a dynamic multisector general equilibrium model with heterogeneous entrepreneurs, each firm's ability to expand depends on market prices, persistent productivity shocks, and its current wealth. As a result, sectoral production and growth depend on how the distribution of wealth aligns with productivity. In a closed economy, relative prices respond to local scarcity: when the most productive entrepreneurs in a sector have lower levels of wealth, the resulting lower supply increases the sector's relative price and accelerates wealth accumulation through higher profits, which allows the sector to catch up. In an open economy, prices are instead tied to world prices, and sectors favored by the initial wealth distribution grow faster and larger. In a quantitative version of the model calibrated to data from Peru, relaxing financial constraints to mimic US levels considerably accelerates reallocation: after five years, the change in sectoral real-output composition is nearly as large as after twenty years under Peruvian borrowing limits.

Keywords: Wealth Distribution, Sectoral Heterogeneity, Persistence, Misallocation
JEL Codes: D31, D53, F11, F43, O11

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1 Introduction

> "Peru needs products from advanced countries, not experiments with Peruvian products."
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> *President Augusto B. Leguía, to the Peruvian entrepreneur Juan Alberto Grieve.*[1]

Structural transformation – defined as the reallocation of economic activity across broad sectors of the economy – has proceeded at varying paces across developing countries. For instance, Ministerio de la Producción (2014) reports that, in the last five decades, Peru's export basket changed very little, whereas South Korea's shifted towards products that are more diverse, complex, and specialized.[2] One interpretation of these differences is benign and can be attributed to comparative advantage stemming from differences in productivity or factor endowments. Comparative advantage theories often ignore, however, the existence of substantial distortions in the allocation of productive resources, especially in developing countries (Restuccia & Rogerson, 2008; Hsieh & Klenow, 2009). These distortions raise the possibility that observed specialization reflects not only underlying comparative advantage, but also frictions that prevent the expansion of productive firms. One of such frictions are financial constraints: In the presence of financial constraints, a firm's ability to expand depends on its ability to self-finance. As a result, production levels and relative sector sizes depend directly on how wealth is distributed across firms, and how that distribution evolves. In this paper, I study how wealth distribution dynamics under financial frictions and trade integration translate into specialization paths, and how initial differences can adjust or persist over time.

The paper proceeds in three steps. First, I develop a theory of specialization dynamics under financial constraints and endogenous wealth dynamics by extending Moll (2014) to a multisector environment. Entrepreneurs own firms in a given sector, face idiosyncratic productivity risk, and have access to capital subject to collateral constraints. As a result, endogenous wealth accumulation shapes firm expansion and the allocation of resources both within and across sectors over time. Unlike the original one-sector version, this extension allows me to study endogenous sectoral relative prices and reallocation under alternative trade regimes.

Using this model, I study the equilibrium dynamics under two sets of fundamentals. In the first one, two sectors are symmetric in every aspect except for the initial distribution of wealth. In the second, the sectors share the same initial conditions, but the productivity process in one

[1] Juan Alberto Grieve (1878–1950) was a Peruvian inventor who designed and built the first internal-combustion engine in South America and one of the first automobiles in the region. His vehicles, adapted to rough local conditions at the time, were priced at around half the cost of importing comparable European models. Despite this, his project failed to scale. Although anecdotal, this episode illustrates how limited access to finance may constrain the expansion or development of productive activities, despite efforts for innovation or productivity gains potential. See https://www.britannica.com/biography/Juan-Alberto-Grieve.

[2] In South Korea, the share of manufactured goods in merchandise exports increased from 18% in 1962 to nearly 90% in 2020. In Peru, by contrast, it increased from less than 1% to only about 9% over the same period. Source: World Bank, World Development Indicators.

sector reverts to a higher mean. In both cases, I compare outcomes under autarky and free trade. These exercises illustrate two related results in the simplest possible environment. In an open economy, specialization patterns can depend on initial conditions even when standard sources of comparative advantage do not predict differences between sectors. Moreover, when comparative advantage does predict differences, adjustment can be very gradual.

In the model, the sectoral wealth growth rates provide a useful summary to understand how distributional dynamics affect the transition and long-run specialization of the economy. Concretely, a sectoral wealth growth rate is the wealth-weighted average of individual firms' net saving rates, which are in turn a function of common prices faced by the economy and the sector, and each firm's idiosyncratic productivity. This way, in addition to standard sources of comparative advantage, each sector's realized accumulation depends on the distribution of wealth, its alignment with productivity, and the degree in which prices respond in general equilibrium.

Consider the first case, in which two sectors differ only in that a higher share of wealth is owned by the most productive firms in one sector. Because of financial constraints, this affects the capital effectively employed in each sector and, consequently, their productive capacity. In a closed economy, relative prices favor scarcity. Productive firms in the sector with lower wealth-productivity alignment benefit from higher prices, which increase their profits and saving rates. These eventually translate into higher sectoral wealth growth rates that allow the disadvantaged sector to catch up. In a small open economy, however, prices are taken from international markets and do not reflect local relative scarcity. If those relative prices are fixed in a way that they never favor the disadvantaged sector, initial differences will persist in the long run. Although self-financing allows productive firms in both sectors to accumulate wealth, the absence of compensating relative-price adjustments can prevent the initially disadvantaged sector from experiencing growth rates that would allow it to catch up. As a result, the sector with a lower alignment could account for a smaller share of output in the long run, even though both sectors have otherwise the same fundamentals.[3]

In the second case, both sectors share the same initial conditions instead, but one has higher long-run productivity potential. Both in autarky and in a small open economy, production shifts toward the more productive sector. In a closed economy, the endogenous relative-price adjustments resulting from demand for goods from both sectors limit the extent of specialization. In the small open economy case, trade enables specialization to move further toward the more productive sector. However, financial constraints make this adjustment gradual instead of immediate, even when the direction of comparative advantage is clear.

Second, I document three facts for Peru that suggests that the mechanisms I describe in the

[3]In the illustrative case with constant returns to scale, these differences persist even in the steady state. If generalized to incorporate decreasing returns to scale, as I do in the quantitative section, the growth rate gaps between sectors reduce in response to reductions in profits of the larger sector, as expansion reduces profitability. This would allow the sector with lower alignment to catch up. Conversely, other mechanisms not incorporated in the paper, such as learning-by-doing or other sources of increasing returns, could further accentuate the persistence of initial differences.

paper are empirically relevant. Using firm-level data from Peru – *Encuesta Económica Anual* (EEA) (Instituto Nacional de Estadística e Informática (INEI), 2001-2024) – I show that there is a substantial and persistent within-sector dispersion in the marginal revenue product of capital (MRPK), and important contemporary and dynamic cross-sectoral differences, especially between tradable and non-tradable sectors. The first-year autocorrelation of measured productivity shocks, in the 0.65–0.87 range, suggests a large persistence in these. However, the correlation between capital and measured productivity is negative in all sectors, and the subsequent MRPK trends differ, as non-tradable sectors experience higher productivity growth. To evaluate the specialization path of Peru, relative to the rest of the world, I complement the analysis using BACI bilateral trade data (CEPII, 2026; Gaulier & Zignago, 2010). I describe that the large dispersion in MRPK and potential gains from capital reallocation coexist with observed specialization patterns that have been very persistent in the last thirty years.

Third, I quantify the empirical importance of the mechanisms suggested by the theory using the data from Peru, which, like many developing countries, experiences a combination of access to world markets and policy objectives towards diversification. I adapt the quantitative model to reflect characteristics of the Peruvian economy that are empirically relevant and might also influence specialization. First, I introduce sector-specific factors to generate a congestion mechanism that limits divergence in response to international price shocks. Second, I allow differentiation between domestic and foreign varieties. This results in demand for domestic varieties even when international price changes redirect production towards some favored sectors, and allows the model to match the high share of domestic consumption expenditure shares in many sectors observed in national-accounts. Third, I allow differences between tradable and non-tradable sectors, and incorporate sectoral imbalances in the modeled sectors. Finally, I incorporate sector-specific tariffs on imports and exports. This allows me to incorporate in the model a proxy for international trade shocks measured through changes in the path of prices, and to provide a counterpart to the closed-economy vs SOE comparison in the baseline model by varying the degree of trade exposure through trade policy.

This quantitative analysis complements the theoretical analysis with two broad findings. First, financial development accelerates sectoral reallocation and raises consumption. When external financing limits are relaxed to mimic levels measured in the US, the change in sectoral real-output composition after five years is nearly as large as after twenty years under the existing borrowing limits in Peru. Moreover, wage-financed consumption increases by approximately 20% throughout the twenty years simulated. Second, lower trade exposure induced by higher import tariffs weakens the relationship between initial accumulation advantages (partly determined by initial wealth allocation) and subsequent changes in sectoral output shares. However, wage-financed consumption is lower throughout all the modeled transition, by 9.4% at the start and 5.6% at the end, under high tariffs[4] relative to zero import tariffs. In addition, aggregate real consumption for

[4]I use a 145% rate motivated by the aggressive rates during the US-China trade conflicts around April 2025.

entrepreneurs by the end of those twenty years is 9.29% lower. These losses are especially large in sectors that benefit the most from trade. For instance, in Agroindustry it falls by 34.55%. Nevertheless, even sectors that expand and gain larger output shares as a result of protectionism experience reductions in consumption.

Related Literature. This paper builds mainly on the macroeconomics literature on entrepreneurship with financial frictions, including Restuccia and Rogerson (2008), Hsieh and Klenow (2009), Midrigan and Xu (2014), Buera, Kaboski, and Shin (2011), and Moll (2014). I extend their mechanisms to a dynamic multisector environment with an external sector. As a result, wealth distributions shape not only aggregate productivity but also the evolution of sectoral specialization. By doing this, I introduce a mechanism in which initial conditions, such as wealth inequality or misallocation, can have long-run effects on production.

I show that history can shape long-run specialization through the endogenous evolution of wealth, even in the absence of learning-by-doing or increasing returns (Krugman, 1987; Stokey, 1988). The static mechanism is closely related to Wynne (2005), where financial conditions and aggregate wealth can limit the capacity of an economy to specialize in its comparative advantage. The dynamic mechanism is connected to Itskhoki and Moll (2019), where gradual wealth accumulation under financial frictions can delay the reallocation of resources towards activities with stronger latent comparative advantage. My contribution is to show how distribution-related mismatches between productivity potential and wealth evolve dynamically across sectors and how the price regime implied by trade integration determines whether they are eventually reversed or if they instead become a source of long-run specialization. Under international goods prices, the resulting wealth differences operate similarly to endogenous differences in effective factor supplies, which connects with the logic presented in Rybczynski (1955) in a dynamic setting.

This paper also relates to the literature that studies the interaction of international trade and finance. For instance, Antràs and Caballero (2009) study the interaction between aggregate trade flows and financial development, while Beck (2002) examines how financial development affects comparative advantage in manufacturing, and Manova (2008) studies how credit constraints affect firms' selection into exporting. These strands of research have been incorporated to quantify the resulting effects of financial development on the gains from trade (Kohn, Leibovici, & Szkup, 2022, 2023). Closer to this paper are Finlay (2026), who incorporates the persistence of productivity shocks to analyze the effects of financial frictions on trade to further quantify the gains from trade resulting from reallocation towards export production; and Buera and Shin (2017) who study how financial frictions shape productivity growth and international capital flows during transitions from one steady state to another following economic reforms. Relative to these efforts, I contribute with a framework that shows that the distribution of wealth across firms, and its endogenous evolution given financial development, can influence which sectors end up being more productive and larger, and how persistent specialization patterns are.

2 A Multisector Model of Wealth Dynamics and Specialization

This section presents the common environment of a multisector extension of the heterogeneous-agent model with financial frictions in Moll (2014). The key addition is that production takes place in multiple sectors whose outputs are combined through a CES demand system, so that sectoral allocations and relative prices are jointly determined in general equilibrium. To characterize sectoral and aggregate dynamics, I introduce wealth-weighted distribution objects that jointly summarize both the allocation of wealth across sectors, and its alignment with productivity types within sectors. These objects provide a tractable representation of how wealth heterogeneity affects sectoral production and its evolution over time.

2.1 Setup, Agents, and Final Consumption

Consider an economy with many sectors indexed by $i \in I$. Time is continuous and denoted by t. When convenient, time arguments will be omitted from prices and policy functions. There are two types of infinitely lived agents: workers (w) and entrepreneurs (e). The total mass of workers is fixed at L. Each worker is endowed with one efficiency unit of labor that they supply inelastically but can allocate across sectors. Workers do not have access to a savings technology, so they live "hand-to-mouth", as in Kaplan and Violante (2014).

Entrepreneurs are attached to a given sector, in which they operate a production technology using capital and labor.[5] In addition, entrepreneurs make forward-looking decisions over consumption and savings based on their stock of financial wealth a. Financial wealth can be transformed into productive capital k or turned into consumption expenditure.

I assume that all sectoral output $Y_i(t)$ is transformed into a final composite good through a constant elasticity of substitution (CES) aggregator:

$$Y(t) = \left(\sum_{i \in I} \upsilon_i^{\frac{1}{\eta}} Y_i(t)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \tag{1}$$

where η denotes the elasticity of substitution across sectors and υ_i are sector-specific weights. This implies the standard CES demand system:

$$D_i(t) = \upsilon_i \left(\frac{p_i(t)}{P(t)} \right)^{-\eta} D(t), \qquad P(t)^{1-\eta} = \sum_{i \in I} \upsilon_i p_i(t)^{1-\eta}. \tag{2}$$

Here, $D_i(t)$ denotes sectoral demand and $D(t)$ denotes total absorption, which includes consumption by both types of agents, the replacement of depreciated capital, and the accumulation of fi-

[5] The difference in mobility between workers and entrepreneurs is similar to the assumption used in Kleinman, Liu, and Redding (2023), where capitalists in a spatial economy are attached to their regions. This restriction helps incorporate forward-looking behavior while keeping the state space tractable.

nancial wealth.

Workers supply their labor efficiency units to entrepreneurs in a competitive labor market at wage $w(t)$. For each hand-to-mouth worker, consumption expenditure is determined by their labor income, so that $P(t)c_w(t) = w(t)\bar{L}$. Entrepreneurs' behavior is explained below.

2.2 Production, Technology, and Budgets

Each entrepreneur has access to a technology of production that allows them to use k units of capital and ℓ units of labor to produce

$$y_i = (zk)^{\alpha_i}\ell^{1-\alpha_i} \tag{3}$$

units of output of the good in sector i, where $\alpha_i \in (0,1)$. Capital depreciates at the same rate δ in every sector. Productivity z follows a continuous-time, sector-specific ergodic Markov process:

$$dz = \mu_i(z)\,dt + \sigma_i(z)\,dW, \tag{4}$$

where $\mu_i(z)$ and $\sigma_i(z)$ denote the drift and diffusion functions, respectively, W is a standard Brownian motion, and the process is assumed to admit a stationary distribution. Following Moll (2014), in the numerical exercises, I specialize log productivity to follow a sector-specific Ornstein–Uhlenbeck process. See Appendix E for details.

Entrepreneurs hire workers at wages $w(t)$ in a competitive labor market, and rent capital from other entrepreneurs in a competitive capital rental market at a rental rate $R(t)$, which is equal to the user cost of capital $R(t) = r(t) + \delta$. For an entrepreneur in sector i, and current productivity z, wealth $a(t)$ evolves according to:

$$\dot{a} = p_i(zk)^{\alpha_i}\ell^{1-\alpha_i} - w\ell - (r+\delta)k + ra - Pc \tag{5}$$

Entrepreneurs face financial constraints that limit their ability to transform wealth into productive capital. In particular, capital choices are subject to a collateral constraint of the form

$$k \leq \lambda_i a, \tag{6}$$

where $\lambda_i \geq 1$ governs the degree of financial frictions in sector i. A higher λ_i corresponds to looser borrowing constraints and more developed financial intermediation.

This constraint implies that entrepreneurs cannot freely arbitrage between the marginal product of capital and its user cost. In particular, when the constraint binds, capital is proportional to internal wealth, so that more productive but less wealthy entrepreneurs may operate below their efficient scale.

2.3 Entrepreneur's Problem

Entrepreneurs choose consumption and savings dynamically, taking prices $\{p_i(t), w(t), r(t)\}$ as given. Their decisions determine the evolution of wealth and their future production capacity.

Let $V_i(a, z; t)$ denote the value function of an entrepreneur in sector i with wealth a and productivity z at time t. The entrepreneur solves

$$V_i(a, z; t) = \max_{\{c(s), k(s), \ell(s)\}_{s \geq t}} \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)} u(c(s))\, ds\right], \tag{7}$$

subject to the law of motion for wealth, Equation 5; the collateral constraint, Equation 6, and the stochastic process for productivity, Equation 4. Throughout, preferences are logarithmic, so that $u(c) = \ln c$.

The entrepreneur's problem can be decomposed into a static and a dynamic component. The static component determines optimal input choices (k, ℓ) at each point in time, given wealth a, productivity z, and prices, and is summarized by the profit function $\Pi_i(a, z)$ defined in Equation 8. The dynamic component consists of choosing consumption and savings over time, taking as given the profit function and the stochastic evolution of productivity.

This separation is possible because input choices (k, ℓ) affect the entrepreneur's problem only through current-period profits and do not enter the law of motion for wealth except through their impact on current income. Given prices and states (a, z), the static problem determines the maximal flow payoff $\Pi_i(a, z)$ independently of intertemporal considerations. The dynamic problem then consists of allocating this flow of resources over time through consumption and savings decisions. As a result, solving the static and dynamic problems sequentially yields the solution to the joint problem.

Static problem. It is convenient to define the entrepreneur's static profit function as the solution to the within-period problem:

$$\Pi_i(a, z) = \max_{k, \ell} \left\{ p_i (zk)^{\alpha_i} \ell^{1-\alpha_i} - w\ell - (r + \delta)k \;\text{ s.t. }\; k \leq \lambda_i a \right\}. \tag{8}$$

This formulation summarizes the effect of prices, productivity, and financial constraints on current income. Using this definition, the law of motion for wealth can be written as

$$\dot{a} = \Pi_i(a, z) + ra - Pc. \tag{9}$$

Lemma 1 (Static allocation)**.** *Given prices $\{p_i, w, r\}_{i \in I}$, the solution to the entrepreneurs' static problem*

implies that capital demand, labor demand, and profits are given by

$$k_i(a,z) = \begin{cases} \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise}, \end{cases}$$
$$\ell_i(a,z) = \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1}{\alpha_i}} z\, k_i(a,z),$$
$$\Pi_i(a,z) = \lambda_i a \cdot \max\left\{\pi_i z - (r+\delta),\, 0\right\},$$

where $\pi_i = p_i \alpha_i \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1-\alpha_i}{\alpha_i}}$, *and the productivity cutoff* $\underline{z}_i$ *satisfies* $\pi_i \underline{z}_i = r + \delta$.

Proof in Appendix A.1.

Dynamic problem. Using the static allocation results, the entrepreneur's problem reduces to a consumption-savings problem with a stochastic return on wealth. In particular, the law of motion for wealth can be written as

$$\dot{a}(t) = \Gamma_i(z,t)\, a(t) - P(t)c(t), \tag{10}$$

where

$$\Gamma_i(z,t) \equiv \lambda_i \max\left\{\pi_i(t) z - (r(t)+\delta),\, 0\right\} + r(t). \tag{11}$$

The entrepreneur's dynamic problem is therefore

$$\max_{\{c(s)\}_{s\geq t}} \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)} \ln(c(s))\, ds\right] \quad \text{s.t.} \quad \dot{a}(s) = \Gamma_i(z,s)\, a(s) - P(s)c(s), \tag{12}$$

and can be represented by a Hamilton-Jacobi-Bellman (HJB) equation provided in Appendix A.2. Under logarithmic preferences, the solution of the problem implies that consumption expenditure is proportional to wealth:

$$P(t)c(t) = \rho a(t), \tag{13}$$

Lemma 2 (Wealth accumulation)**.** *Under logarithmic preferences, individual wealth evolves according to*

$$\dot{a}(t) = s_i(z,t) a(t), \tag{14}$$

where

$$s_i(z,t) \equiv \Gamma_i(z,t) - \rho$$

denotes the net saving rate of entrepreneurs with productivity z *in sector* i*. Proof in Appendix A.2.*

2.4 Wealth Distributions and Aggregation

Let $g_i(a, z; t)$ denote the joint density of entrepreneurs with wealth a and productivity z in sector i at time t. Aggregate wealth in the economy and in sector i are given by

$$A(t) = \sum_{i \in I} A_i(t), \qquad A_i(t) = \int a \, g_i(a, z; t) \, da \, dz. \tag{15}$$

Definition 3 (Wealth distributions)**.** Relative to aggregate wealth quantities $A(t)$ and $A_i(t)$, define the following wealth-weighted distributions:

- The share of aggregate wealth held in sector i: $O_i(t) = \frac{A_i(t)}{A(t)}$.
- The within-sector wealth-weighted productivity density: $o_i(z, t) = \frac{1}{A_i(t)} \int a \, g_i(a, z; t) \, da$.
- The sector-specific wealth density over productivity: $\omega_i(z, t) = \frac{1}{A(t)} \int a \, g_i(a, z; t) \, da = O_i(t) \, o_i(z, t)$.
- The economy-wide wealth density $\omega(z, t) = \sum_{i \in I} \omega_i(z, t)$.

These objects satisfy

$$\int o_i(z, t) \, dz = 1, \qquad \int \omega_i(z, t) \, dz = O_i(t), \qquad \text{and } \sum_{i \in I} O_i(t) = 1. \tag{16}$$

In addition, they imply cumulative wealth share distributions within sectors $\Omega_i(z, t) \equiv \int_0^z o_i(x, t) \, dx$, and for the economy as a whole $\Omega(z, t) \equiv \sum_{i \in I} O_i(t) \cdot \Omega_i(z, t) = \sum_{i \in I} \int_0^z \omega_i(x, t) \, dx$.

The objects $O_i(t)$ and $o_i(z, t)$ summarize, respectively, the distribution of wealth across sectors and across productivity levels within sectors. Their product, $\omega_i(z, t)$, characterizes the distribution of aggregate wealth productivity levels in sector i. The collection $\{\omega_i(z, t)\}_{i \in I}$ jointly characterizes the allocation of wealth across all sectors and productivity levels. Together with aggregate wealth and prices, $\{\omega_i(z, t)\}_{i \in I}$ is sufficient to determine aggregate allocations and production.

Lemma 4 (Aggregation)**.** *Because optimal policies are linear in wealth, aggregate allocations depend only on the wealth-weighted distribution over productivity, and tracking $g_i(a, z)$ is not necessary to characterize them. In particular, for any individual policy or payoff that satisfies $x_i(a, z) = a x_i(z)$,*

$$\int x_i(a, z) \, g_i(a, z; t) \, da \, dz = A(t) \int x_i(z) \, \omega_i(z, t) \, dz. \tag{17}$$

3 Closed Economy

I first close the environment described in Section 2 as a closed economy, in which goods prices adjust endogenously to clear domestic goods markets. I define the corresponding equilibrium and then characterize how wealth distributions, sectoral allocations, and relative prices evolve over time through a stylized illustrative example with two sectors.

3.1 Closed Economy Equilibrium

Definition 5 (Closed economy equilibrium)**.** Given initial wealth level $A(0)$, its initial distribution characterized by $\{\omega_i(z,0)\}_{i\in I}$, and labor endowment $\bar{L}$, a competitive closed economy equilibrium is a path of prices $\{w(t), r(t), p_i(t)\}_{i\in I, t\geq 0}$, allocation rules $\{k_i(z,t), \ell_i(z,t), c_i(z,t)\}_{i\in I, t\geq 0}$, and wealth distributions $\{\omega_i(z,t)\}_{i\in I, t\geq 0}$, such that:

1. Entrepreneurs optimize given prices.
2. Factor markets clear:

$$\bar{L} = A(t)\sum_{i\in I}\int \ell_i(z,t)\,\omega_i(z,t)\,dz, \quad A(t) = A(t)\sum_{i\in I}\int k_i(z,t)\,\omega_i(z,t)\,dz.$$

3. Goods markets clear:

$$Y_i(t) = D_i(t), \qquad \forall i \in I,$$

   where sectoral output is $Y_i(t) = A(t)\int y_i(z,t)\,\omega_i(z,t)\,dz$.

4. Sectoral demands satisfy the CES system

$$D_i(t) = \upsilon_i \left(\frac{p_i(t)}{P(t)}\right)^{-\eta} D(t),$$

   with price index $P(t)^{1-\eta} = \sum_{i\in I} \upsilon_i p_i(t)^{1-\eta}$.

5. Aggregate absorption satisfies

$$P(t)D(t) = P(t)C(t) + \dot{A}(t) + \delta A(t),$$

   where aggregate consumption expenditure is $P(t)C(t) = w(t)\bar{L} + \rho A(t)$.

6. Wealth distributions evolve consistently with individual wealth accumulation and the productivity process according to the law of motion characterized below.

3.2 Aggregation and Dynamics

Proposition 6 (Wealth distributions, sectoral productivity, and accumulation)**.** *Following Lemma 4, sectoral output can be written as*

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i},$$

where $Z_i(t) = \left(E_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}$ *. Likewise, sectoral wealth evolves according to*

$$\dot{A}_i(t) = \alpha_i p_i(t) Y_i(t) + r(t)\big[A_i(t) - K_i(t)\big] - \delta K_i(t) - \rho A_i(t).$$

Proof and factor-price conditions in Appendix A.3.

Sectoral productivity and output depend on how wealth is allocated across productivity types within each sector, while sectoral wealth accumulation depends on production income, prices, and net asset positions. Relative to the one-sector environment in Moll (2014), sectoral wealth and sectoral capital do not need to coincide: a sector can use more capital than the wealth held by its entrepreneurs, or it can hold more wealth than the capital used in production. As a result, the common interest rate affects sectoral wealth accumulation through a net asset position channel. An increase in the interest rate raises wealth accumulation in sectors that are net lenders, with $A_i(t) > K_i(t)$, and lowers it in sectors that are net borrowers, with $A_i(t) < K_i(t)$.

3.3 Evolution of Wealth Shares

Given the stochastic evolution of productivity and the optimal saving policy, the individual state (z, a) evolves according to

$$\begin{aligned} dz &= \mu_i(z)\, dt + \sigma_i(z)\, dW, \\ da &= s_i(z,t)\, a\, dt, \end{aligned} \tag{18}$$

where $s_i(z,t) \equiv \Gamma_i(z,t) - \rho$ denotes the net growth rate of wealth implied by optimal behavior.

These dynamics induce a law of motion for the joint distribution $g_i(a, z; t)$ via the associated Kolmogorov Forward equation. As shown in Appendix B, applying this equation to the joint process and exploiting the linearity of policies in wealth implies that the wealth-weighted distribution $\omega_i(z,t)$ satisfies

$$\frac{\partial \omega_i(z,t)}{\partial t} = \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} \right] \omega_i(z,t) - \frac{\partial}{\partial z}\left[\mu_i(z)\omega_i(z,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\left[\sigma_i^2(z)\omega_i(z,t)\right]. \tag{19}$$

The first term captures differential wealth accumulation across productivity levels, relative to aggregate wealth growth. The remaining terms reflect the stochastic evolution of productivity.

This law of motion is the point at which the model departs most directly from the one-sector case in Moll (2014). In Moll's framework, wealth accumulation changes the economy-wide wealth-weighted productivity distribution. Here, the same force operates within each sector, but it also governs the evolution of sectoral wealth shares. In particular, the within-sector distribution evolves according to

$$\frac{\partial o_i(z,t)}{\partial t} = \left[s_i(z,t) - \bar{s}_i(t) \right] o_i(z,t) - \frac{\partial}{\partial z}\left[\mu_i(z) o_i(z,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\left[\sigma_i^2(z) o_i(z,t)\right], \tag{20}$$

where $\bar{s}_i(t) = \int s_i(z,t) o_i(z,t)\, dz$ is the average net saving rate in sector i, while sectoral wealth shares evolve according to

$$\dot{O}_i(t) = \left[\bar{s}_i(t) - \frac{\dot{A}(t)}{A(t)} \right] O_i(t). \tag{21}$$

This distinction is central because sectoral wealth shares are an important determinant of specialization in the model, and their evolution depends on sector-specific saving rates shaped by relative prices and the common interest rate and propensity to consume implied by ρ. At the same time, these allow to summarize differences in realized growth advantage between sectors.

Lemma 7 (Sectoral wealth growth and realized accumulation advantage)**.** *Sectoral wealth grows at the wealth-weighted average of individual net saving rates:*

$$\frac{\dot{A}_i(t)}{A_i(t)} = \bar{s}_i(t) \equiv \int \underbrace{s_i(z,t)}_{\text{accumulation opportunities}} \times \underbrace{o_i(z,t)}_{\text{wealth allocation}} \, dz. \tag{22}$$

That is, $\bar{s}_i(t)$ summarizes the sector's realized accumulation advantage. It combines the opportunities available to each productivity type with the allocation of wealth across those types. Moreover, for any two sectors i and j,

$$\frac{d}{dt} \log \left(\frac{O_i(t)}{O_j(t)} \right) = \bar{s}_i(t) - \bar{s}_j(t). \tag{23}$$

3.4 Steady State

A steady state is an equilibrium in which sectoral wealth levels and wealth distributions are time-invariant. Formally, for all sectors $i \in I$,

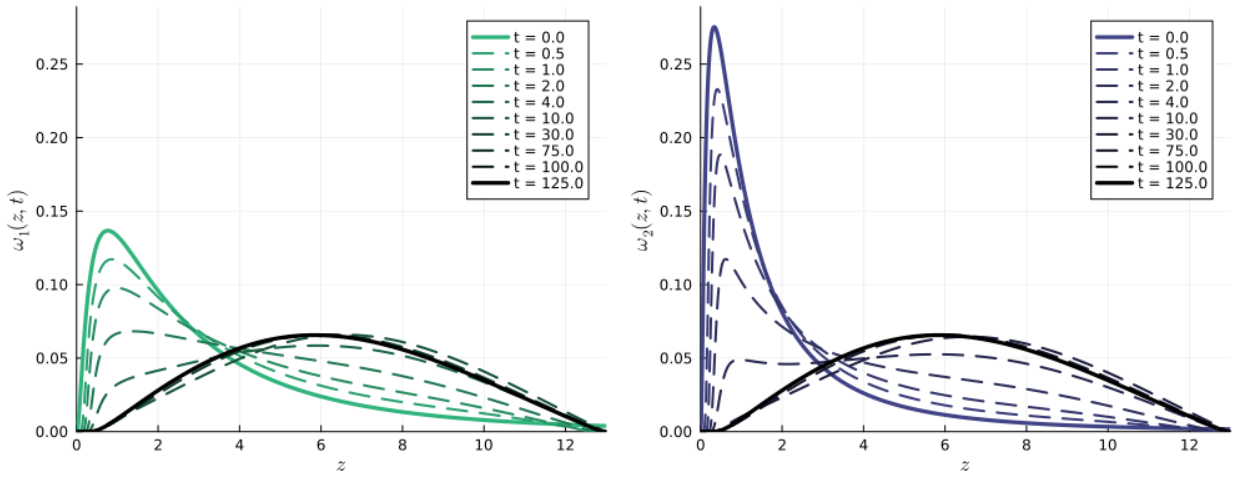
$$\dot{A}_i(t) = 0, \quad g_i(a,z;t) = g_i(a,z), \quad \omega_i(z,t) = \omega_i(z), \quad \text{and} \quad O_i(t) = O_i. \tag{24}$$

In a steady state, sectoral productivity and expenditure shares are constant, so specialization is characterized by the stationary allocation of wealth across sectors and productivity states. Appendix A.4 provides the formal steady-state characterization with conditions for sectoral output, factor prices, capital income, and TFP.

3.5 Two-Sector Illustration: Same Long-Run Potential

I now use the model to show an example that illustrates the model mechanisms in a closed economy. In particular, it will describe how endogenous relative prices mediate the transition path. In this exercise, and in the one presented in Section 4.3, I use the parameterization summarized in Appendix E. Unless otherwise noted, parameters are common across sectors and chosen to generate symmetry in every dimension not explicitly stated to vary. Appendix D describes the numerical procedure, including the fixed-point solution for equilibrium prices and the finite-difference approximation used to update the wealth-share distributions.

Consider an economy with two sectors, $i \in \{1,2\}$ demanded as imperfect substitutes ($\eta > 1$). Let's assume both sectors are identical in fundamentals and endowments and differ only in the initial alignment between productivity and wealth. In particular, the initial correlation between log



(a) Sector 1 (b) Sector 2

Figure 1: Wealth distribution dynamics in a two-sector closed economy.

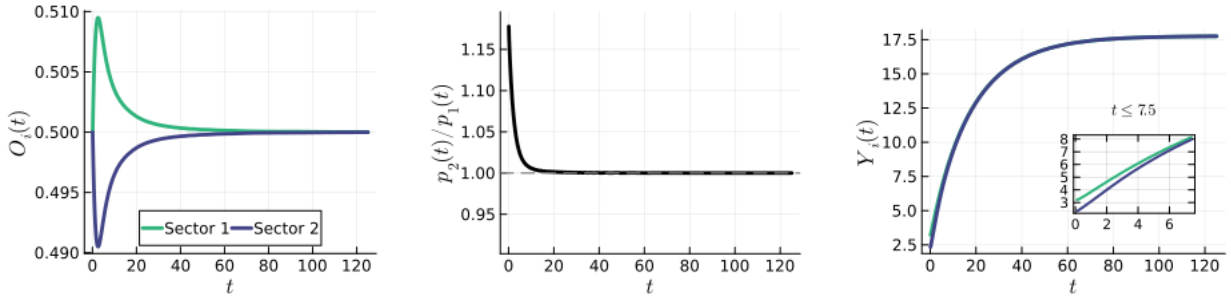
Notes: The figure shows the evolution of the wealth-weighted productivity distributions $\omega_i(z,t)$ in the two-sector closed economy presented in Section 3.5. Panel (a) corresponds to sector 1, which starts with a stronger positive alignment between wealth and productivity, while panel (b) corresponds to sector 2. Each curve reports the distribution at a different point along the transition. Parameters are common across sectors except for their initial wealth-productivity alignment.

wealth and log productivity is set to $\text{Corr}(\log a, \log z)_{1,0} = 0.8$ in sector 1 and $\text{Corr}(\log a, \log z)_{2,0} = 0.2$ in sector 2.

Wealth-distribution dynamics. The solid colored lines in both panels of Figure 1 represent the initial wealth distribution for each sector, $\omega_1(z,0)$ and $\omega_2(z,0)$. Both sectors start with the same total wealth, then the total share for each is equivalent to the integral of each corresponding curve and equal to $O_1(0) = O_2(0) = 0.5$. Even though sector 1 has a better wealth-productivity alignment, both sectors start with a relative large share of wealth in the hands of low productivity entrepreneurs. However, the entrepreneurs with higher productivity realizations z are able to make higher profits and have a larger effective saving rate. The ability to self-finance allows both sectors to eventually transition towards a stationary wealth distribution represented by the solid black line $\omega_1(z,125) = \omega_2(z,125)$, which despite transitional differences implies a return to an equal share of total wealth $O_1(125) = O_2(125) = 0.5$.

Sectoral reallocation and relative prices. Figure 2 shows the dynamics that accompany and explain the distributional transitions in Figure 1 and their aggregate consequences. At $t = 0$, even though both sectors start with the same aggregate wealth, the initial alignment advantage of sector 1 translates into a slightly larger supply, represented in panel (c); and an initially higher average saving rate, $\bar{s}_1(0) > \bar{s}_2(0)$, reflected in the initial increase in its wealth share in panel (a). This

advantage is transitory because relative prices adjust, as shown in panel (b). When sector 2 has a lower relative supply, its benefits from a higher relative price. This helps highly productive entrepreneurs in the initially disadvantaged sector to benefit from higher profits and saving rates, which after enough within-sector rellocation generates $\bar{s}_2(t) \geq s_1(t)$ in the transition period. This allows sector 2 to accumulate wealth and grow at a faster rate and catch up with sector 1. In the long-run, as differences between sectors disappear, the rate of growth of both sectors converge, $\bar{s}_2(\infty) = \bar{s}_1(\infty)$.



(a) Aggregate wealth (b) Relative Price (c) Sectoral Output

Figure 2: Aggregate dynamics in a two-sector closed economy.

Notes: The figure shows aggregate transition dynamics in the two-sector closed economy presented in Section 3.5. Panel (a) reports sectoral shares of aggregate entrepreneurial wealth, $O_i(t) = A_i(t)/A(t)$. Panel (b) reports the relative goods price, $p_2(t)/p_1(t)$. Panel (c) reports sectoral output, $Y_i(t)$. The inset in panel (c) shows the first ten periods of the transition. Parameters are common across sectors except for their initial wealth-productivity alignment.

Convergence in the closed economy. In the long run, both sectors achieve the same long-run wealth share and distribution that reflects their same long run potential. The initial asymmetry between sectors produced by differences in wealth allocation has a transitory effect in the specialization of the economy.[6] Although lowered and slowed down by financial frictions, long-run aggregate output doesn't depend on initial differences in this closed economy, consistent with the current literature consensus (Buera & Shin, 2013; Moll, 2014; Buera, Kaboski, & Shin, 2015). The current exercise has emphasized that endogenous relative prices intermediate this result in an economy with multiple sectors. In the next section, I show that initial conditions have long run consequences when relative prices are tied to external conditions in an open economy.

4 Small Open Economy

In this section, the structure from the previous two sections is preserved almost entirely. But equilibrium is determined as a small open economy in goods with financial autarky. That is, sectoral

[6] A similar result could have been obtained if in the example the within-sector distribution for both sectors where the same and the total wealth on each was different.

goods prices $\{p_i^*(t)\}_{i\in I}$ are exogenously given by world markets, instead of being endogenously determined by market clearing. Resulting excesses of demand or supply at the sector level are assumed to be met by exports and imports from rest of the world. Trade balances at the sector level are assumed to cancel out to keep aggregate trade balance, and therefore financial autarky.

4.1 Equilibrium

Definition 8 (Small open economy equilibrium with financial autarky)**.** Given initial wealth level $A(0)$, its initial distribution characterized by $\{\omega_i(z,0)\}_{i\in I}$, labor endowment $\bar{L}$, and a sequence of world prices $\{p_i^*(t)\}_{i\in I,t\geq 0}$, a competitive small open economy equilibrium with financial autarky is a path of factor prices $\{w(t), r(t)\}_{t\geq 0}$, allocation rules $\{k_i(z,t), \ell_i(z,t), c_i(z,t)\}_{i\in I,t\geq 0}$, net exports $\{NX_i(t)\}_{i\in I,t\geq 0}$, and wealth distributions $\{\omega_i(z,t)\}_{i\in I,t\geq 0}$, such that:

1. Entrepreneurs optimize given world prices and factor prices.

2. Factor markets clear:

$$\bar{L} = A(t)\sum_{i\in I}\int \ell_i(z,t)\,\omega_i(z,t)\,dz, \quad A(t) = A(t)\sum_{i\in I}\int k_i(z,t)\,\omega_i(z,t)\,dz.$$

3. Goods markets satisfy

$$Y_i(t) = D_i(t) + NX_i(t), \qquad \forall i \in I,$$

where domestic absorption satisfies the CES demand system $D_i(t) = \upsilon_i \left(\frac{p_i^*(t)}{P^*(t)}\right)^{-\eta} D(t)$, with price index $P^*(t)^{1-\eta} = \sum_{i\in I} \upsilon_i \left(p_i^*(t)\right)^{1-\eta}$.

4. External balance holds:

$$\sum_{i\in I} p_i^*(t) NX_i(t) = 0.$$

5. Aggregate absorption satisfies

$$P^*(t)D(t) = P^*(t)C(t) + \dot{A}(t) + \delta A(t),$$

where aggregate consumption expenditure is $P^*(t)C(t) = w(t)\bar{L} + \rho A(t)$.

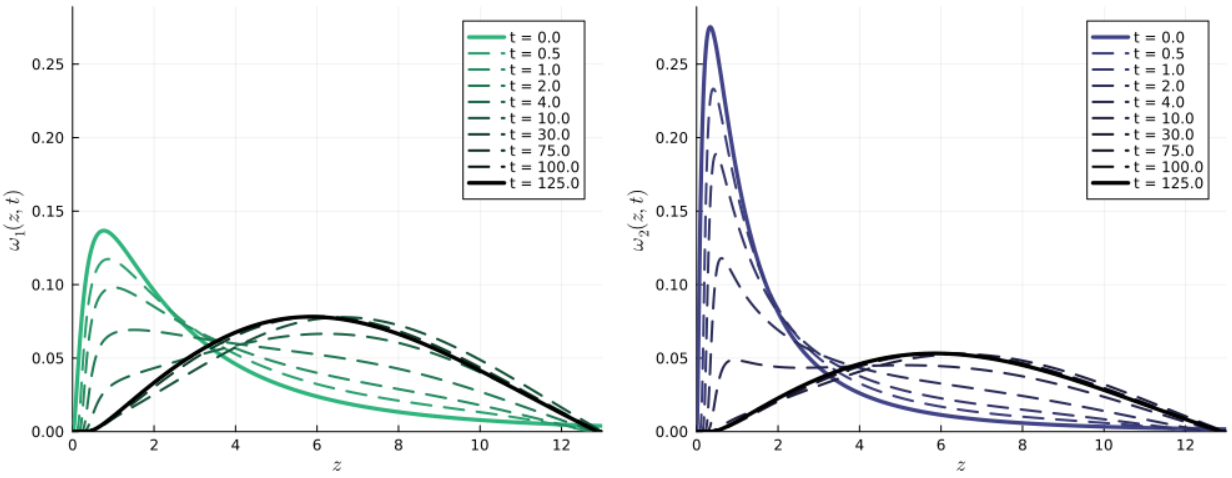
6. Wealth distributions evolve consistently with individual wealth accumulation and the productivity process.

Under this equilibrium definition, the wealth accumulation laws previously exposed are almost entirely preserved too. The only difference is that, since good prices are taken as given, saving rates and wealth allocation dynamics will have an exogenous component that won't respond to local differences but instead tied to external conditions.

4.2 Two-Sector Illustration: Same Long-Run Potential, Closed Economy vs. SOE

Let's revisit now the example introduced in Section 3.5, but in now under the SOE regime as just defined above. Starting off from the same initial conditions, let's suppose that the economy trades frictionlessly with the rest of the world at a fixed relative price $p_2^*/p_1^* = 1$.

Wealth-distribution dynamics. The solid colored lines in both panels of Figure 3 coincide with the corresponding panels in Figure 1, since the initial state is the same. Through realized savings, both sectors move from their initial low-alignment wealth distribution towards the end of transition wealth distributions represented by the solid black lines $\omega_1(z, 125)$ and $\omega_2(z, 125)$. Unlike in the closed economy case, these distributions are not equal. Although the within-sector wealth-weighted productivity distributions converge to the same distribution ($o_1(z, 125) = o_2(z, 125)$), sector 1's initial advantage translates into a larger long-run share of total wealth $O_1(125) \approx 0.6 > O_2(125) \approx 0.4$. As consequence, since $\omega_i(z, t) = O_i(t)o_i(z, t)$, sector 1 has a larger long-run relative share of total wealth across productivity levels $\omega_1(z, 125)/\omega_2(z, 125) \approx 3/2$.



(a) Sector 1 (b) Sector 2

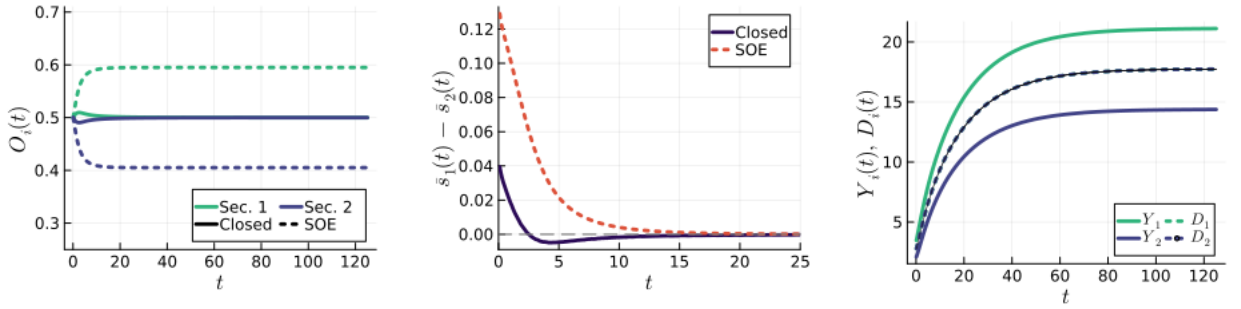
Figure 3: Total wealth sectoral shares distribution dynamics in SOE with two sectors.

Notes: The figure shows the transition of the sector-specific wealth-weighted productivity distributions in the two-sector small open economy. Differences in initial wealth-productivity alignment generate different selection dynamics across sectors, without endogenous price adjustment, these differences produce persistent long-run differences.

Sectoral accumulation and saving rates. Figure 4 shows the aggregate dynamics that accompany and explain the results in Figure 3, and their divergence with respect to the closed economy case. As anticipated in the previous paragraph, panel (a) shows that the initial wealth-share difference between sectors doesn't disappear but increases to a persistent level. Panel (b) summarizes the mechanisms behind this result following Lemma 7. Since in the SOE the relative price is fixed at

1, entrepreneurs with same productivity draw z experience the same accumulation opportunities regardless of the sector in which they operate. Sector 1's initial allocation advantage fully translates into a larger rate of growth while both sectors transition towards their stationary wealth distribution. Sector 2, eventually catches up in growth rates but not it in levels. In order to have done so, it would have needed to grow faster than sector 1 for some time. As a result, panel (c) shows that the economy becomes a net exporter for sector 1's good, and a net importer in sector 2's good.

Path dependence and exogenous prices. Panel (b) of Figure 4 also shows the role endogenous relative prices played for sector 2 to catch up to sector 1 in the closed economy. There, the scarcity-driven higher relative price of sector 2 made the accumulation opportunities for its entrepreneurs increase. This lowered the initial advantage of sector 1 relative to what would have happened if prices were fixed and soon it even reverses it. With $\bar{s}_1 - \bar{s}_2 < 0$ during part of the transition period, sector 2 is able to reach the same size as sector 1. In the small open economy, this never occurs. The initial growth advantage eventually disappears, but the level difference accumulated while it was positive persist generating specialization despite identical long-run fundamentals.



(a) Sector Wealth Shares: SOE vs Closed

(b) Saving Rates Difference: SOE vs Closed

(c) Output and Absorption (SOE)

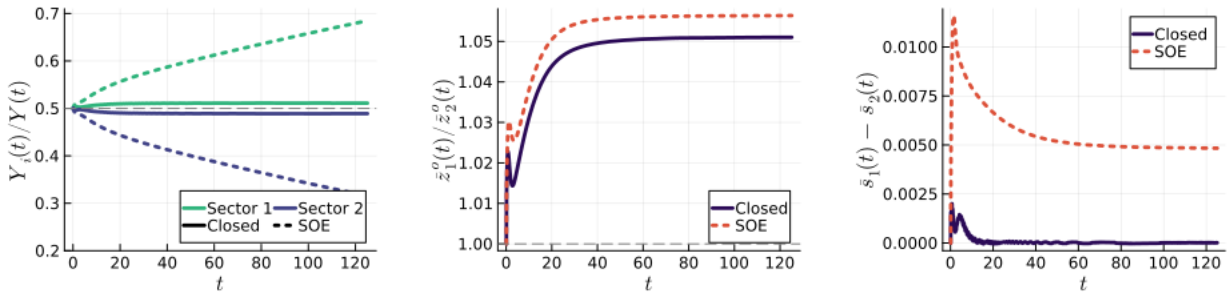
Figure 4: Two-sector aggregate dynamics SOE vs. closed economy

Notes: The figure compares transition dynamics under the closed-economy and small-open-economy regimes, starting from identical fundamentals and initial wealth-productivity distributions. Panel (a) reports sectoral wealth shares $O_i(t)$. Panel (b) reports the difference in average sectoral saving rates, $\bar{s}_1(t) - \bar{s}_2(t)$; the horizontal axis is restricted to the first 25 periods to highlight the early transition. Panel (c) reports sectoral output $Y_i(t)$ and domestic absorption $D_i(t)$ in the small open economy.

4.3 Two-Sector Illustration: Different Long-Run Potential, Closed Economy vs. SOE

Now I use the model to explore a follow-up question: what happens when sectors have different long-run productivity potential? In this example, both sectors share the same initial conditions, including wealth level and distribution, other fundamentals, and face equal world prices in the SOE case. However, the productivity process in sector 1 mean-reverts towards a higher long-run level of log-productivity, $\bar{\mu}_1 = 1 > \bar{\mu}_2 = 0$. That is, instead of varying history while keeping

long-run potential equal as in the previous exercise, now, given shared initial conditions, long-run potential varies.



(a) Output Shares (b) Relative Productivity (c) Saving Rates Difference

Figure 5: Two-sector dynamics with different long-run productivity potential

Notes: The figure compares transition dynamics in the closed economy and small open economy starting from identical initial conditions, with sector 1 having higher long-run productivity potential. Panel (a) reports sectoral output shares. Panel (b) reports the ratio of wealth-weighted mean productivity, $\bar{z}_1^o(t)/\bar{z}_2^o(t)$. Panel (c) reports the difference in average sectoral saving rates, $\bar{s}_1(t) - \bar{s}_2(t)$. All other parameters are common across sectors and regimes.

Adjustment towards productive advantage. The results of this exercise are summarized in Figure 5. As shown in panel (a), as a result of its higher long-run productivity potential, sector 1 gradually gains a larger output share in both regimes. In the closed economy, it stabilizes in a moderate long-run difference, as internal demand requires goods from both sectors and prices limit specialization in equilibrium such that $Y_1/Y_2 = (p_2/p_1)^\eta > 1$.[7]. Not only sector 1's good is relatively cheaper in autarky than at international prices, $(p_2/p_1)_{\text{autarky}} > (p_2^*/p_1^*) = 1$, which would imply a comparative advantage in sector 1. But panel (b) shows that opening up to trade allows further increase in the realized productivity advantage of sector 1 relative to sector 2. Specialization now moves in the direction implied by fundamentals, but the movement towards it is gradual.

Saving rates and slow specialization. Figure 5, panel (c) helps to explain why the specialization is moderate in the closed economy, and why it takes so long in the SOE. First, in the closed economy, the productivity gains of sector 1 are countered by the scarcity-driven price gains of sector 2. Despite an initial difference, relative price differences leads both sectors to grow at the same speed, which prevents wealth reallocation despite the primitive productivity difference. Second, in the SOE, although sector 1 has a consistent advantage, the gap in average saving rates soon becomes relatively small during the transition. Financial frictions make the economy slow to realize a productive advantage even when its direction is clear.

[7]Notice that the CES demand system implies $\frac{p_i(t)D_i(t)}{P(t)D(t)} = \upsilon_i \left(\frac{p_i(t)}{P(t)}\right)^{1-\eta}$, $\upsilon_1 = \upsilon_2$ in the exercise, and in the closed economy output satisfies absorption $Y_i(t) = D_i(t)$.

4.4 Summary and Interpretation

The model and illustrations show two main results related to the persistence of specialization. Different wealth allocation in sectors with the same long-run potential will generate persistent specialization under exogenous prices given by international markets. When sectors have different long-run potential, financial frictions will slow the movement of the economy towards specialization in its comparative advantage.

In the appendix, I include additional exercises that complement the intuition provided by the previous results. Appendix G develops a simplified model that shows analytically how fixed world prices can sustain persistent specialization in a small open economy environment. In Appendix F.1, I use the main text model to study cases in which the home economy has a comparative advantage with the rest of the world based on alternative values of p_2^*/p_1^*, that favor one of the sectors relative to the other.

5 Empirical Evidence

The examples and comparisons discussed in the previous sections are intentionally illustrative and provide stark results, as the environment abstracts from many realistic margins and nuances. Now I turn to evidence from Peru that is informed by firm-level, trade, household, national-accounts, and international price and macroeconomic series data. In this section, I briefly summarize the data sources and provide some empirical counterparts of the model mechanisms. Then, Section 6 will introduce richer Peru-specific features to the model to evaluate the quantitative importance of the mechanisms once those features are included.

5.1 Data

The main source of information is the Annual Economic Survey (*Encuesta Económica Anual*, EEA) conducted by Instituto Nacional de Estadística e Informática (INEI) (2001-2024), which INEI uses to collect economic and financial information for official national and sectoral statistics in Peru.[8] It is mandatory for all formal firms that are selected or that have net sales above a certain threshold for the corresponding year's sample. In practice, it serves as a census for all medium and large firms. The format of the survey resembles standard financial statements that include information on sales, revenues, costs, exports, value added, salaries, staff numbers, fixed assets, value of plant and equipment, and other assets and liabilities. I use this dataset to measure firm-side heterogeneity in terms of productivity, capital distribution, factor shares, and related indicators. To proxy capital stock, I use two different approaches depending on the firm size as categorized by the survey. For small firms, the capital stock is proxied by the total fixed assets as reported on their general

[8] It covers firms in all sectors except for those in mining and extractive industries, which instead report the results of their activities to the Ministry of Energy and Mines.

balance sheet. For a medium to big firm, which tends to include financial investments not involved in the productive process as fixed assets, it is proxied by the value of the stock of equipment.

To measure aggregate sectoral output, trade balance composition, and domestic expenditure shares, I use Peru's 2007 Input–Output Table (Instituto Nacional de Estadística e Informática (INEI), 2007) together with the annual Supply and Use Tables for 2007–2024 (Instituto Nacional de Estadística e Informática (INEI), 2007–2024). These sources provide the national-accounts information used to construct sectoral aggregates and expenditure shares and to discipline the external-balance conditions used in the quantitative analysis. I approximate the evolution of external prices using import-price series from the Penn World Table (PWT 11.0) (Feenstra, Inklaar, & Timmer, 2025, 2015), and measure changes in trade policy using Most-Favored-Nation (MFN) tariff rates from the UNCTAD TRAINS database accessed through the World Integrated Trade Solution (WITS) (World Bank and United Nations Conference on Trade and Development (UNCTAD), 2026). I also combine PWT data with World Bank World Development Indicators (World Bank, 2026) to construct country-level measures of financial development used to discipline the model's financial-friction parameter. The patterns of revealed comparative advantage in Figure 7 are constructed from BACI bilateral trade data (CEPII, 2026; Gaulier & Zignago, 2010). Finally, I use the Peruvian household survey, *Encuesta Nacional de Hogares* (ENAHO), (Instituto Nacional de Estadística e Informática (INEI), 2004-2023) to construct sectoral employment shares and their evolution over time. Further details on the use of these data sources are provided in Appendix E.3.

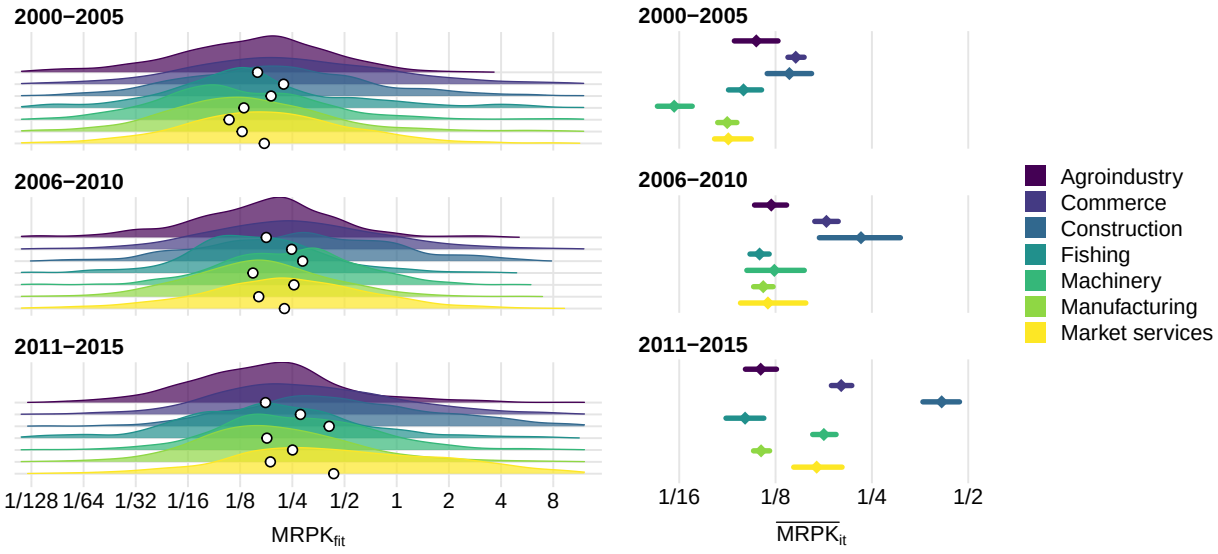
5.2 Capital Allocation and Sectoral Heterogeneity

Figure 6 describes the distribution and trends of the marginal revenue product of capital (MRPK) in the Peruvian sectors measured by the Annual Economic Survey (EEA). First, panel (a) shows that there is a substantial and persistent within-sector dispersion of MRPK in Peru. Second, there are important differences between sectors. Panel (b) plots sector level MRPK trends, measured by the capital-weighted mean of the firm-level MRPK. [9] These show that there is also important contemporary and dynamic between-sector differences. The Agroindustry, Fishing, and (non-Machinery) Manufacturing sectors started with similar levels of MRPK that stayed at almost the same level over the three five-year periods of reference. The only tradable sector has had experienced an important MRPK growth is Machinery, which more than doubled its initial mean level, even though it started as the lowest. All the non-tradable sectors, Commerce, Construction, and

[9] I summarize sector-level MRPK using a capital-weighted mean because it provides and empirical counterpart of the aggregate return to capital in the model. At the firm level, $\text{MRPK}_{fi} = \alpha_i p_i y_{fi}/k_{fi}$. Then, when firms within sector i face a common producer price and capital elasticity,

$$\frac{\sum_f k_{fi}\text{MRPK}_{fi}}{\sum_f k_{fi}} = \frac{\alpha_i p_i \sum_f y_{fi}}{\sum_f k_{fi}} = \frac{\alpha_i p_i Y_i}{K_i}.$$

This way, the capital-weighted mean aggregates firm-level MRPK into the sector-level object $\alpha_i p_i Y_i/K_i$ that appears in the model. Weighting by capital also measures the return associated with the allocation of the sector's capital stock, instead of giving the same influence to small and large firms.



(a) Distribution of MRPK

(b) Capital-weighted Mean of MRPK

Figure 6: Distribution and Capital-Weighted Mean of MRPK by Sector in Peru, 2000–2015

Notes: Panel (a) plots the distribution of firm-level MRPK within each sector and period. White circles denote the sector-period median. Panel (b) reports the capital-weighted mean MRPK for each sector-period, where firm-level MRPK is weighted by capital. Horizontal bars denote 95% percentile confidence intervals obtained from 1,000 bootstrap resamples of firm-year observations within each sector-period, with MRPK and capital resampled jointly. Construction details in Appendix E.3.

Services, have experienced an important growth as well, especially construction.

There are, of course, different potential frictions and shocks that can be behind these differences in addition to financial access: adjustment costs, risk, markups, fixed factors, sector-specific rents, etc. However, Figure 6 shows that capital is distributed across firms facing very different marginal returns, which also have different trends among sectors.

5.3 Wealth-Productivity Alignment Across Sectors

Table 1 complements Figure 6 by describing the dispersion and persistence of firm-level productive opportunities and their alignment with capital during the first years of the sample.[10] I focus on 2000-2003 data to mirror the initial conditions considered in the illustrative model, and to exclude later period changes that may reflect endogenous outcomes of the mechanisms that will be under

[10] For this exercise, I approximate firm-level productivity using a within-sector-year return-shifter proxy,

$$\log z_{fit} \approx \widehat{x}_{fit} = \log va_{fit}^{nom} - \log k_{fit} - \overline{\left(\log va_{fit}^{nom} - \log k_{fit}\right)}_{it}.$$

That is, I measure each firm's nominal value added per unit of capital relative to the corresponding sector-year average. This demeaning process removes common movements in sectoral prices and aggregate conditions, so that $\widehat{x}_{fit}$ captures cross-firm differences in productivity within a sector at a given point in time. Appendix E.3 provides further details on its construction.

Table 1: Firm Heterogeneity and Wealth–Productivity Alignment by Sector, 2000–2003

Panel A. Baseline Measures

Sector	(1) Std. $\log k_{fit}$	(2) Std. $\log z_{fit}$	(3) Corr. $(\log z_{fit}, \log k_{fit})$	(4) Autocorr. $\log z_{fit}$
Agroindustry	1.941	1.074	-0.293	0.671
Commerce	1.639	1.226	-0.488	0.826
Construction	1.881	1.480	-0.568	0.803
Fishing	2.436	1.873	-0.536	0.562
Machinery	1.965	1.081	-0.675	0.818
Manufacturing	1.691	1.140	-0.526	0.821
Market Services	1.968	1.066	-0.408	0.835

Panel B. Robustness Measures

Sector	(5) Corr. $(\log z_{fit}, \log k_{fi,t-1})$	(6) Corr. $(\log z^H_{fit}, \log k_{fit})$	(7) Corr. $(\log z^H_{fit}, \log k_{fi,t-1})$	(8) Autocorr. $\log z^H_{fit}$
Agroindustry	-0.289	0.064	0.061	0.463
Commerce	-0.448	-0.297	-0.268	0.777
Construction	-0.555	-0.415	-0.407	0.714
Fishing	-0.126	-0.570	-0.041	0.523
Machinery	-0.602	-0.219	-0.177	0.392
Manufacturing	-0.500	-0.214	-0.201	0.587
Market Services	-0.417	0.069	0.056	0.820

Notes: Capital k_{fi} and productivity z_{fi} are constructed from firm-level information in the EEA. Columns (1) and (2) report within-sector dispersion in log capital and log productivity, respectively. Columns (3) and (5) report the within-sector correlations between log productivity and contemporaneous and lagged log capital, respectively. Column (4) reports the first-order within-firm autocorrelation of log productivity. Columns (6)–(8) report the corresponding correlations and autocorrelation using the alternative productivity measure z^H_{fi}. All columns are calculated using the same sample of 3,745 observations to keep them comparable. Appendix E.3 provides details on variable construction, sample selection, and the mapping of these moments into the quantitative model.

study.

Panel A of Table 1 summarizes the main measures to describe firm productive heterogeneity and wealth-productivity alignment by sector in the reference years. Column (1) indicates that Fishing has the highest dispersion of capital, followed by Market Services and Machinery, while Commerce and Manufacturing show the lowest dispersion. Column (2) shows a large dispersion of the productivity proxy, consistent with Figure 6, and a different ranking between sectors than column (1). Column (4) shows a high autocorrelation of this productivity measure, which supports the modeling of persistence in productivity differences that this paper inherited from Moll (2014). Finally, column (3) shows a striking takeaway. In all sectors, the firms with the highest measured productivity opportunities are typically those with lower capital stocks. In addition, the magnitude of this relationship varies across sectors (e.g., from about -0.29 in Agroindustry to -0.68 in Machinery).

The empirical measure of z_{fit} used in the reported statistics is my preferred productivity measure for descriptive purposes for three reasons: its simplicity, its construction does not require esti-

mating factor elasticities, and, by assumption, Corr. $(\log z_{fit}, \log k_{fit}) \approx$ Corr. $(\log MRPK_{fit}, \log k_{fit})$.[11] However, this measure might be affected by division bias[12] and the implicit assumption that other inputs adjust optimally.[13] To address these concerns, Panel B of Table 1 reports robustness measures calculated using lagged capital $\log k_{fi,t-1}$ and an alternative measure z^H_{fit} that is calculated using the sectoral labor elasticity and allows labor to deviate from the optimum implied by the model.[14] In column (5), the correlation remains negative in all sectors when lagged capital is used, with limited changes in magnitude except in Fishing. In column (6), using z^H_{fit}, they remain negative in five of seven sectors. When both adjustments are combined in column (7), the correlations remain substantially negative in Commerce, Construction, Machinery, and Manufacturing, while they are close to zero in the remaining sectors.

5.4 Sectoral Specialization and Long-Run Reallocation

Figure 7 examines how Peru's sectoral specialization has evolved over the a similarly broad recent period. To do so, I focus on the tradable sectors at the ISIC two-digit level and compare their relative importance in Peru in relation to their importance in the rest of the world as measured by the Revealed Comparative Advantage (RCA) index (Balassa, 1965).

The thirty years of data summarized in Figure 7 suggest that Peruvian specialization is very persistent, although clearly not immutable and with important sector-specific differences. First, most sectors in which Peru had RCA in the 90s are still the ones in which the country exhibits RCA in the 2020s; the majority of the dots are on the same side of the $RCA = 1$ lines at the start and in the end of the period and many of them are very close to the 45° line, even though a few of them moved significantly without changing sides. Second, the specific movements relative to the 45° line is heterogeneous across sectors. Metal ores, basic metals, agriculture and food products, remain among the highest RCA activities. Fishing is the only sector in which Peru has lost its RCA, while Forestry and Other Mining the only ones in which it experiences new RCA.

Overall, even though the previous evidence suggests there are heterogeneous productive opportunities, persistent capital return differences and weak alignment between capital and productivity, Figure 7 displays that observed specialization patterns are persistent too and when they change, they do not point towards a unified direction of reallocation.

[11] $MRPK_{fit} = \alpha_i VA_{fit}/k_{fit}$, so $\log MRPK_{fit} = \log \alpha_i + \log VA_{fit} - \log k_{fit}$. Since α_i is common within sectors, the correlation with capital is identical when both measures are expressed relative to their sector-year means.

[12] Measurement error in capital enters negatively in $\log z_{fit}$, through its denominator, and positively in $\log k_{fit}$. This potentially generates a negative correlation mechanically. Using lagged capital mitigates this concern if measurement errors are not serially correlated.

[13] The relationship between value added per unit of capital and the productivity shock in the model relies on assuming constant returns to scale and optimal adjustment of labor or sector-specific factors (see Section 6).

[14] If labor is allowed to deviate from its optimum, $\log z^H_{fit} = (\log VA_{fit} - \beta_i \log \ell_{fit})/(1-\beta_i) - \log k_{fit}$, where β_i is the sectoral labor elasticity. To see this, substitute the optimal demand for the sector-specific factor, $h_{fit} = m^H_{it} z_{fit} k_{fit}$, into the production function $y_{fit} = (z_{fit}k_{fit})^{\alpha_i} \ell^{\beta_i}_{fit} h^{\gamma_i}_{fit}$. Using $\alpha_i + \beta_i + \gamma_i = 1$ gives $\log VA_{fit} = (1-\beta_i)(\log z_{fit} + \log k_{fit}) + \beta_i \log \ell_{fit} + \kappa^H_{it}$, where κ^H_{it} is common within sector-years. Rearranging and removing the sector-year constant results in the expression above.

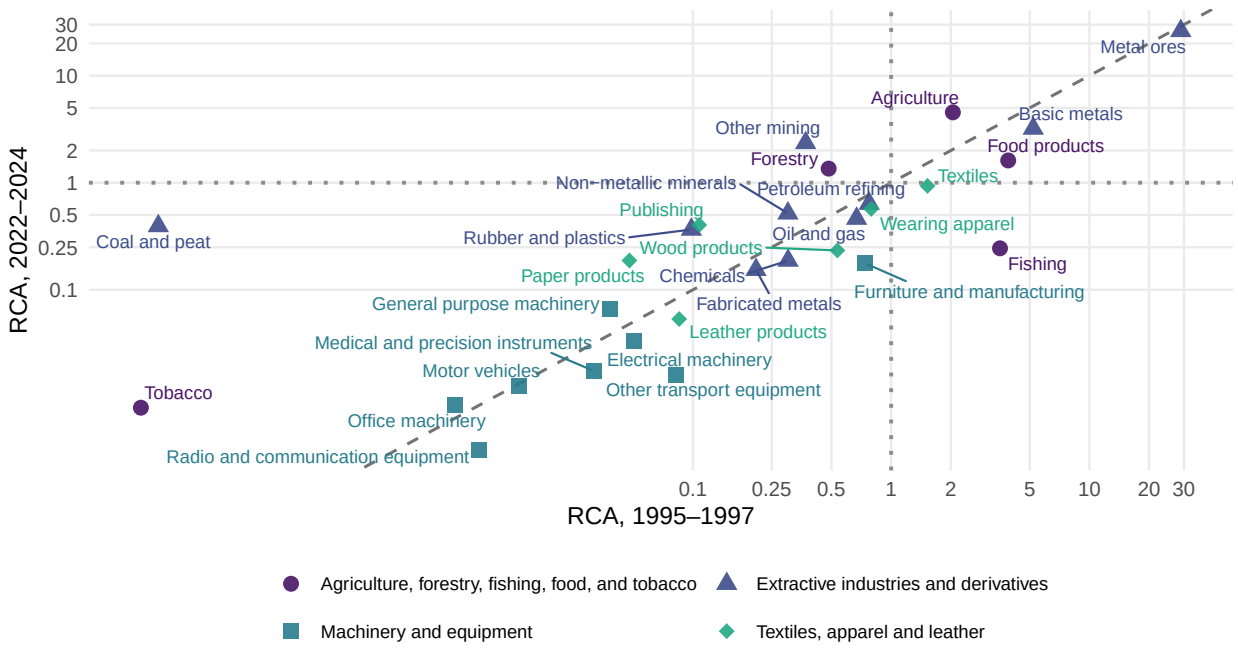


Figure 7: Sectoral Revealed Comparative Advantage in Peru, 1995–1997 vs. 2022–2024

Notes: The figure plots revealed comparative advantage (RCA) for Peruvian exports across ISIC Rev. 3 two-digit sectors in two periods: 1995–1997 (horizontal axis) and 2022–2024 (vertical axis). RCA is defined as $RCA_{i,t} = (X_{Peru,i,t}/X_{Peru,t}) \,/\, (X_{World,i,t}/X_{World,t})$ where $X_{\text{Peru},i,t}$ denotes Peruvian exports in sector i at time t, $X_{\text{Peru},t}$ total Peruvian exports, and $X_{\text{World},i,t}$ and $X_{\text{World},t}$ the corresponding world aggregates. Both axes are reported on a logarithmic scale. The dashed vertical and horizontal lines mark $RCA = 1$, the threshold for comparative advantage. The dashed 45-degree line indicates unchanged specialization between periods; sectors above (below) the line increased (decreased) their revealed comparative advantage over time. Colors and marker shapes identify broad industry groups. Trade data are from the BACI International Trade Database harmonized to ISIC Rev. 3 classifications.

6 Quantitative Application

Although the patterns discussed in Section 5 are informative and suggestive, they motivate quantitative analysis to answer questions that the descriptive evidence can't answer on its own: In particular, how much can initial wealth allocation contribute to the observed specialization patterns? How do financial development and trade exposure change that relationship once more realistic features are incorporated to the analysis?

6.1 Quantitative Environment

The quantitative analysis extends the stylized model in four dimensions overviewed below. Details and derivations are presented in Appendix C. The implementation combines parameters measured directly from Peruvian data, internally calibrated objects, observed transition paths, and a small set of externally disciplined parameters. Table 2 gives an overview of the main parameters and initial conditions. Further details and extended parameter tables are provided in Appendix E.3.

Table 2: Summary of Quantitative Calibration

Object	Value / Variation	Discipline
$\alpha_i, \beta_i, \gamma_i$	Sector-specific	EEA factor shares
σ_i, θ_i	Sector-specific	Dispersion and persistence of $\log z$
$O_i(0)$, $o_i(z,0)$	Sector-specific	Initial wealth allocation and alignment with $\log z$
η	1.1	Estimated using Peruvian Supply and Use Tables
$\eta^\star$	1.76	Boehm, Levchenko, and Pandalai-Nayar (2023)
λ	1.13	Implied by domestic credit and capital-output ratio
φ_i	Sector-specific	Domestic-origin expenditure shares
$\bar{H}_i^{\gamma_i} = \bar{H}$	1	Normalization

Notes: The table summarizes the main parameters and initial conditions used in the quantitative application. The EEA is Peru's Annual Economic Survey. Sector-specific productivity and initial-allocation moments are based on the empirical evidence described in Section 5. Observed paths for international prices, tariffs, and the net-export residual are introduced separately over the quantitative transition. Appendix E.3 provides full details on measurement, parameter construction, calibration targets, and data sources.

I. Sector-Specific Factors and Decreasing Returns to Scale. Estimates of labor and capital shares used in the quantitative exercise add up to less than one. I interpret the residual factor share as the remuneration of a sector-specific input. This allow the model to generate decreasing returns to scale at the sector level while preserving the tractability advantages of a CRS structure at the firm-level.

In the quantitative economy, an entrepreneur who owns firm f in sector i can produce according to the following function:

$$y_{fi} = (z_{fi}k_{fi})^{\alpha_i}\ell_{fi}^{\beta_i}h_{fi}^{\gamma_i}, \qquad \alpha_i + \beta_i + \gamma_i = 1, \tag{25}$$

where h_{fi} denotes a factor specific to sector i. This formulation follows the logic of specific-factors models (Jones, 1971; Samuelson, 1971; Eaton, 1987), but adapts their logic allowing each sector to combine mobile capital and labor with a fixed sector-specific factor. Unlike labor and capital, which can be used by any sector in the economy and are traded in integrated markets, the specific factor is only rented by firms in sector i and its aggregate supply is fixed at $\bar{H}_i$. Since this production function has constant returns to scale, optimal labor and specific-factor demands are proportional to $z_{fi}k_{fi}$. Entrepreneurs face the same collateral constraints as in the illustrative model, so active entrepreneurs choose $k_{fi} = \lambda_i a_{fi}$, and firm-level input demands and profits are linear in wealth. Hence, the quantitative model maintains the wealth-aggregation property. That is, wealth-weighted distributions remain sufficient to characterize sector-level aggregates. In particular, sector-level production is of the form

$$Y_i(t) = Z_i(t)K_i(t)^{\alpha_i}L_i(t)^{\beta_i}\bar{H}_i^{\gamma_i}, \tag{26}$$

where

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}\left[z \mid z \geq \bar{z}_i(t)\right]\right)^{\alpha_i}. \tag{27}$$

Since $\bar{H}_i$ is fixed, this sector-level production function shows decreasing returns in the mobile factors:

$$Y_i(t) = \widetilde{Z}_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i}, \qquad \alpha_i + \beta_i = 1 - \gamma_i < 1, \qquad \widetilde{Z}_i(t) = Z_i(t)\bar{H}_i^{\gamma_i}$$

As a sector expands, the rental price of its specific factor rises, which produces a congestion force and reduces profitability at the firm level even though the underlying firm-level technology is represented with CRS.

Assuming that $\bar{H}_i$ remains constant over the quantitative horizon is a strong assumption. This could be natural if it represents slowly adjustable inputs such as land. But if it represented sector-specific human capital, infrastructure, organizational capabilities or technologies that may benefit from learning-by-doing, $\bar{H}_i$ could instead accumulate over time and benefit from agglomeration. I abstract from these dynamics and hold $\bar{H}_i$ fixed to provide a conservative congestion force. As a sector expands, it must use additional mobile inputs over a fixed complementary factor, which weakens its advantage relative to the ones that are expanding less. I further normalize the effective scale of the specific factor to be common across sectors, so that differences in specialization are not mechanically generated by unobserved differences in sector-specific factor endowments.

II. Domestic Use and Production. Next, I relax the assumption that domestic and foreign goods are perfect substitutes. This allows the quantitative exercise to match the home bias observed in national-accounts expenditure shares and prevents international prices from mechanically determining the composition of domestic use. I do this by introducing differentiation à la Armington. Domestic use in a tradable sector is a CES composite of a domestic-origin variety, $D_i^H(t)$, and imports, $M_i(t)$:

$$D_i(t) = \left[\varphi_i^{1/\eta_i^\star}\left(D_i^H(t)\right)^{(\eta_i^\star-1)/\eta_i^\star} + (1-\varphi_i)^{1/\eta_i^\star} M_i(t)^{(\eta_i^\star-1)/\eta_i^\star}\right]^{\eta_i^\star/(\eta_i^\star-1)}.$$

Here, φ_i governs home bias and $\eta_i^\star$ is the elasticity of substitution between domestic-origin and imported varieties. Final users face the corresponding composite price $p_i^D(t)$, while domestic producers receive the domestic-origin price $p_i^H(t)$.

Domestic supply to its domestic demand and exports:

$$Y_i(t) = D_i^H(t) + X_i(t).$$

Since the domestic and foreign varieties are differentiated à la Armington, exports are required to be non negative and imports are a separate flow. That is, to guarantee equilibrium in the domestic variety in sector i, the quantity produced on it should be at least enough to satisfy its internal demand. Otherwise, the domestic price would adjust to enforce equilibrium. With access to foreign

markets, the following conditions summarize the underlying mechanism, and the value of exports.

$$p_i^H(t) \geq p_i^X(t), \qquad X_i(t) \geq 0, \qquad \left[p_i^H(t) - p_i^X(t)\right] X_i(t) = 0.$$

The domestic origin price $p_i^H(t)$ is bounded by the net export price $p_i^X(t)$. When the sector exports, these prices are equal. If the domestic demand for the domestic variety at $p_i^X(t)$ would be larger than what the sector is able to supply at that price, then domestic demand would instead exhaust domestic production, exports would be zero and the domestic-origin price may exceed the net export price. This way, international prices and tariffs discipline tradable producer prices, while still allowing domestic scarcity to affect profitability. For nontradable sectors, $Y_i(t) = D_i(t)$, and the producer and domestic-use prices always coincide with the price that clears the domestic market.

This hybrid price regime treatment, in which international conditions discipline tradable-goods prices while nontradable prices adjust to clear domestic markets, follows the dependent-economy tradition of Meade (1956), Salter (1959), Corden (1960), Swan (1963), and Dornbusch (1974).[15]

III. Tradable and Non-Tradable Sectors, and Trade Balance. I further adapt the quantitative analysis to Peru's sectoral and external-balance structure by allowing the model distinguish between tradable and nontradable sectors and to account for the net exports of omitted extractive industries through an exogenous net-export residual.

The modeled economy consists of seven broad sectors represented in the EEA. These are grouped in two. The set of tradable sectors $\mathcal{I}^T$ includes agroindustry, fishing, machinery and transport equipment, and other manufacturing. The set of non-tradable sectors $\mathcal{I}^N$ includes commerce, construction, and market services. In total, these sectors employed about 56.93% of the total employed population in Peru, and represented 71.39% of total gross production in 2007. The largest share of non-considered labor force are those engaged in primary agriculture, which engaged around 36.11% of the total employed population in the same year, although only represented 4.79% of total production.

The quantitative application excludes sectors whose underlying mechanisms that are different from the entrepreneurship logic the model emphasizes: public services (5.53% of employment, 6.24% of output), financial services (1.07% of employment, 2.74% of production), and oil and mining extractive sectors (0.38% of employment, 14.9% of production). Even though mining is the largest share of Peru's trade balance, the reason it has a persistent surplus, and an important share of production; it is not included in the analysis because it is weakly linked to the other productive sectors – employs less than 0.4% of workers, and its production is almost entirely destinated towards exports – and has a different organization structure. [16]

[15] See Schmitt-Grohé and Uribe (2021) for a modern treatment and discussion of the Salter-Swan model.

[16] In Peru, natural resources are owned by the state, and extraction operates through concession agreements with large and often transnational firms. Therefore, mining activity and its growth depend primarily on geological endowments and concession rights, and is backed up by large-scale external finance. See Daza (2025) who studies this institutional

I nevertheless retain the extractive sector's implications for national income and external balance by assuming it finances the trade deficit generated by the modeled tradable sectors. The net exports needed to guarantee trade balance according to the data, are incorporated through the exogenous residual $NX_{\text{res}}(t)$, which in the model is rebated lump-sum to hand-to-mouth income earners, and impose

$$\sum_{i \in \mathcal{I}^T} \left[p_i^X(t) X_i(t) - p_i^*(t) M_i(t) \right] + NX_{\text{res}}(t) = 0. \tag{28}$$

This split in the external balance between resource and non-resource components is similar to the approach in Bodenstein, Erceg, and Guerrieri (2011), specifically in the case of financial autarky which requires trade balance.

IV. Prices and Tariffs. Finally, I incorporate sector-specific tariffs on imports and exports and distinguish world prices from the prices relevant to domestic producers. In tradable sectors, domestic-origin prices are disciplined by international prices whenever domestic supply is sufficiently large to support exports, while domestic scarcity can generate additional price adjustment when the supply of the sector is low. Introducing tariffs also allows variation of policy-driven trade exposure.

The term $p_i^*(t)$ in Equation 28 represents the exogenous world-price for each tradable sector. Imports are purchased at tariff-inclusive prices as follows

$$p_i^M(t) = \left[1 + \tau_i^M(t) \right] p_i^*(t),$$

while domestic exporters receive the net-of-tariff price

$$p_i^X(t) = \frac{p_i^*(t)}{1 + \tau_i^X(t)}.$$

$\tau_i^M(t)$ represent domestically imposed tariffs, while $\tau_i^X(t)$ represent tariffs imposed abroad that affect the home economy exports.

Same as the trade balance residual introduced in Equation 28, domestic import-tariff revenues,

$$T^M(t) = \sum_{i \in \mathcal{I}^T} \tau_i^M(t) p_i^*(t) M_i(t),$$

are rebated lump-sum towards the hand-to-mouth agent following standard treatment in quantitative analysis of tariffs; see, for instance, Auer, Bonadio, and Levchenko (2020) and Alessandria, Ding, Khan, and Mix (2025).

structure and the domestic transmission of resource revenues related government spending, and Toma, Tomarchio, Velasquez-Cabrera, and Cuba (2026) who study its financial transmission.

6.2 Model Fit and Validation

Table 3 reports the model fit in the first period of the simulation. Given the measurements of initial conditions and other parameters described in Appendix E.3, I jointly calibrate initial relative world prices, treating the Agroindustry international good as the numeraire, and the outer CES sectoral weights to match domestic-expenditure shares from the 2007 Input-Output Table and the trade balances of the modeled tradable sectors as shares of aggregate output. In addition, I report sectoral labor-income shares measured in ENAHO as non-targeted validation moments.

The calibration reproduces the targeted domestic-expenditure shares essentially exactly and matches the sectoral trade-balance targets very closely. In domestic-expenditure shares the largest discrepancy is approximately 4×10^{-6} percentage points, while the largest trade-balance discrepancy is about 0.27 percentage points. The external-balance condition is closed with a net residual $NX_{\text{res}}(0)$ equal to approximately 2.46% of modeled output.

The final two columns of Table 3 provide an additional assessment of the model's calibration by comparing the non-targeted sectoral labor allocation with the data. Although there are differences for some sectors, particularly Commerce and Non-Machinery Manufacturing, the model broadly reproduces the cross-sectoral composition observed in the data. In particular, Market Services accounts for the largest labor-income share and Machinery and Transport Equipment for the smallest in both the data and the model. The largest discrepancy across sectors is approximately 6.7 percentage points, for Commerce.

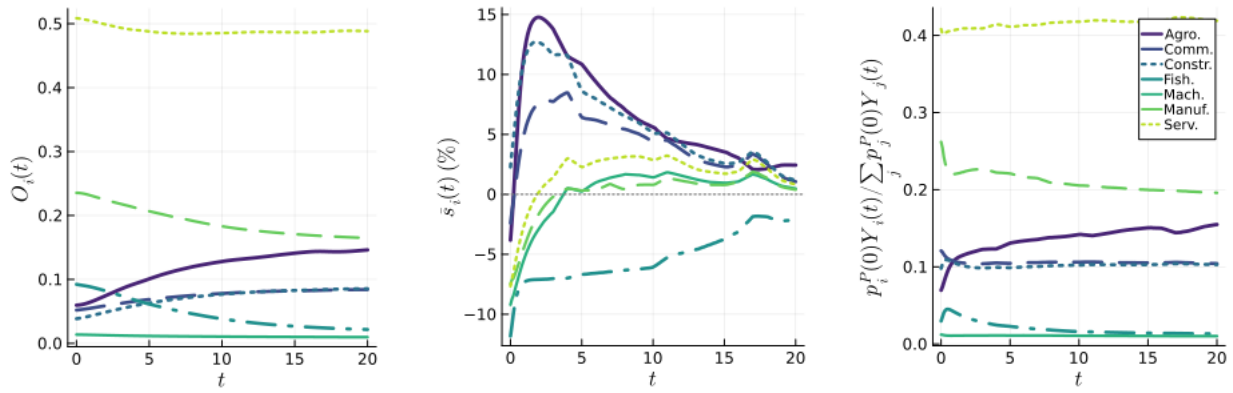
Table 3: Model Fit and Validation

Sector	Targeted moments				Validation (ENAHO)	
	D data (%)	D model (%)	tb_i data (%)	tb_i model (%)	$\frac{w_i L_i}{\sum_j w_j L_j}$ data (%)	$\frac{wL_i}{\sum_j wL_j}$ model (%)
Agroindustry	7.24	7.24	-0.27	-0.50	5.27	6.63
Commerce	11.63	11.63	–	–	16.28	9.54
Construction	9.43	9.43	–	–	10.71	9.55
Fishing	1.33	1.33	1.50	1.59	4.29	2.34
Machinery and Transport Equipment	6.67	6.67	-5.37	-5.14	0.79	1.23
Non-Machinery Manufacturing	24.47	24.47	1.68	1.42	22.70	26.16
Market Services	39.23	39.23	–	–	39.96	44.55

Notes: The first two columns report sectoral labor-income shares using ENAHO 2004 and the corresponding model values. The ENAHO measure is restricted to dependent workers and uses CIIU Rev. 4 sectors aligned with the modeled activities. D denotes domestic-use expenditure shares and tb sectoral trade balances as a share of aggregate output; these moments are targeted to match values on the first period. Additional details on sample construction and sector concordances are provided in Appendix E.3.

6.3 Baseline Transition Dynamics

As shown in Table 1, the misalignment between productivity and capital allocation is quite severe in the Peruvian economy. Besides financial frictions, there are arguably many reasons why such



(a) Sectoral Wealth Shares (b) Saving Rates (c) Real Output Composition

Figure 8: Transition dynamics in the quantitative implementation

Notes: The figure reports transition paths from the empirically informed initial sectoral wealth allocation and parametrization. Panel (a) reports each sector's share of aggregate entrepreneurial wealth. Panel (b) reports average sector level effective saving rates. Panel (c) reports sectoral shares of real GDP, using producer prices in the first period as reference.

misalignment existed, and they might still affect the actual economy. However, in this exercise, I take the distribution implied by those moments as an exogenous initial condition and let the quantitative economy operate through the mechanisms emphasized by the model. After calibrating the initial period, as previously summarized, I now impose data-driven paths for world prices, tariffs, and the trade-balance residual. I interpret the initial period of the quantitative economy as representing 2004 and let the transition evolve for 20 years, up to 2024.

Figure 8 reports the transition dynamics over this reference period. Panel (a) reports the evolution of sectoral wealth shares. The differences across series reflect the different effective saving rates experienced by the modeled sectors. In particular, Agroindustry, Commerce, and Construction increase their share of wealth over time; Fishing, Machinery, and Non-Machinery Manufacturing lose relative importance; and Market Services experiences a moderate decline.

Panel (b) shows that, consistent with 7, differences in sectoral effective saving rates are behind those wealth-share patterns and complement our understanding of the underlying mechanisms. Several sectors begin the transition with negative growth rates. In the model, this reflects the large initial misalignment between wealth and productive opportunities. An important mass of entrepreneurs choose to abandon production and instead lend their wealth in the unified capital market, where the most profitable sectors and productive entrepreneurs demand it most strongly. Over time, differences in profits and the reallocation of capital through wealth accumulation allow most sectors to move into positive growth rates. In Agroindustry, Commerce, and Construction, this happens relatively quickly. As the transition progresses, saving rates gradually converge across sectors.

Panel (c) reports the resulting transitions in sectoral real-output shares. In addition to the

changes in wealth shares shown in panel (a), the movements in this panel also reflect sectoral differences in technology, demand, and changes in within-sector distribution. The short-run movements in output composition can be larger than the corresponding changes in wealth shares because wealth is also being reallocated towards the most productive firms within each sector. This also helps explain why Market Services experiences a moderate decline in wealth share but a moderate increase in output share.

6.4 Financial Development and Specialization

In this section, I repeat the previous exercise varying only the value of the collateral constraint parameter, which as explained in Appendix E.3 is measured using data on aggregate capital and private credit. The main comparison moves this parameter from $\lambda = 1.13$, which corresponds to Peruvian levels, to a value of $\lambda = 2.29$, which corresponds to US levels. This change implies that the share of capital financed with external funds $1-1/\lambda$ rises from about 11.5% under the Peruvian calibration to about 56% under the US value. Everything else is held the same, including the initial calibration, the path of exogenous prices, tariffs, and trade balance residual. The effects of this parameter variation in sectoral output paths are summarized in Figure 9. The figures in Appendix F.3 summarize results under intermediate values of λ.

The left panel in Figure 9 shows the real output paths for the modeled sectors at both levels of financial development. The dashed lines plot the baseline paths under the Peruvian financial development level, and the solid ones the counterfactual. Increasing the financial development in the quantitative economy increases the aggregate production in the economy, but it also has heterogeneous effects among sectors. The transition towards the most profitable sectors, for instance Agroindustry, occurs faster.

To quantify the speed of this adjustment, I measure net sectoral shifts relative to the initial Peruvian composition using one-half the sum of absolute changes in real-output shares. In particular, I calculate the following index[17]:

$$R_Y^{\lambda}(t) = \frac{1}{2}\sum_{i\in\mathcal{I}}\left|\frac{p_i^P(0)Y_i^{\lambda}(t)}{\sum_{j\in\mathcal{I}}p_j^P(0)Y_j^{\lambda}(t)} - \frac{p_i^P(0)Y_i^{1.13}(0)}{\sum_{j\in\mathcal{I}}p_j^P(0)Y_j^{1.13}(0)}\right|. \tag{29}$$

The center panel in Figure 9 reports the effects summarized by this measure. In the initial period, increasing the collateral multiplier to the US value generates about 1.8 percentage points of net reallocation relative to the baseline composition with the Peruvian value. The main difference between both scenarios is in the speed of transition. With $\lambda = 1.13$, a large share of the adjustment is gradual over the whole period; whereas with $\lambda = 2.29$ the bulk of the adjustment happens early, and the sectoral composition is already close to its later shares by around year five. Concretely,

[17] Notice that it equals one-half the sum of absolute changes in sectoral output shares so that it doesn't double count shifts away from some sectors which are received by others.

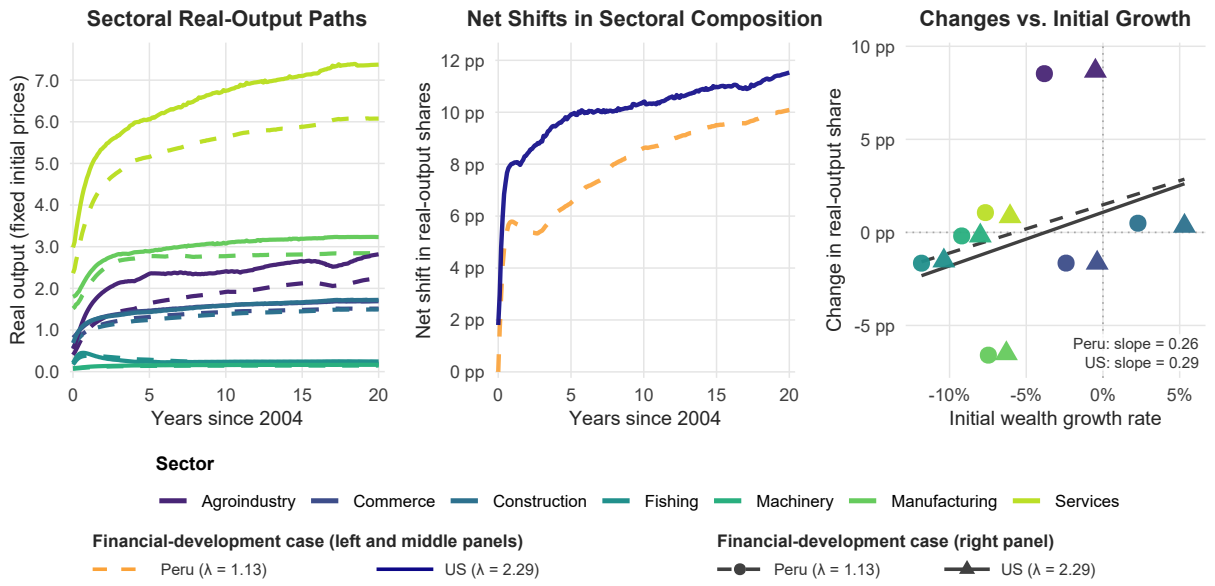
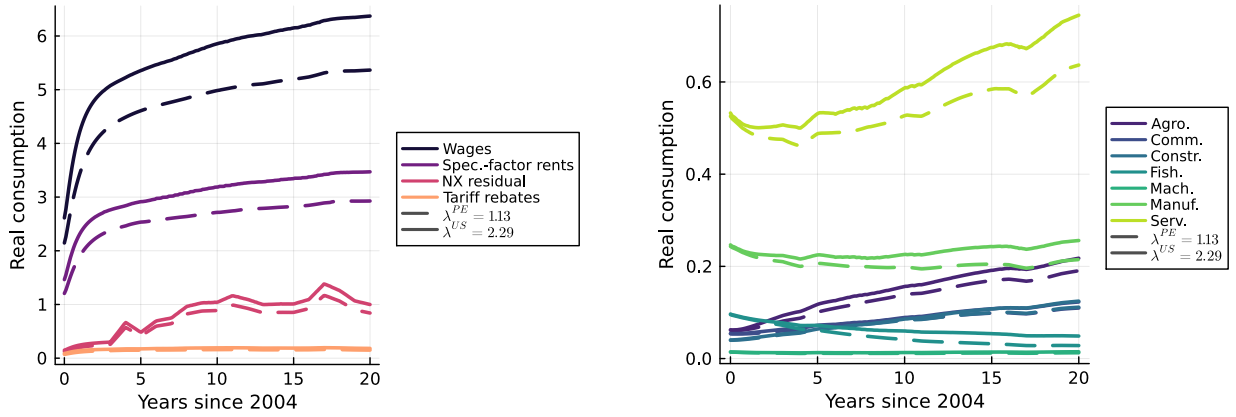


Figure 9: Sectoral specialization under alternative financial-development regimes

Notes: The figure compares the baseline Peruvian financial-development calibration, $\lambda = 1.13$, with one with the US value, $\lambda = 2.29$. Both counterfactuals start from the same initial conditions and face the same paths of world prices, tariffs, and NX_{res}; only the collateral multiplier differs. Real output is valued using the initial producer prices of the Peru baseline. The left panel plots sector level output series. The center panel plots one-half the sum of absolute differences between each scenario's sectoral real-output shares at time t and Peru's initial sectoral real-output shares. The right panel relates each sector's initial wealth-growth rate to its change in real-output share over the 20-year transition.

the net reallocation since the initial composition with $\lambda = 1.13$ is 6.5 pp, and with $\lambda = 2.29$ it is 9.9 pp. The underlying changes imply that, by the fifth year, 65% and 84% of the twenty-year increase in the reallocation index has already occurred under Peruvian and US borrowing limits, respectively. Moreover, the magnitude of the compositional adjustment with $\lambda = 2.29$ in five years is almost as large as the corresponding to the one with $\lambda = 1.13$ in twenty years from 2004 to 2024. More importantly, the more rapid growth and faster reallocation resulting from higher financial development imply a large increase in consumption by the end of the reference period, as discussed next.

Finally, Figure 9 right panel summarizes the relationship between initial growth levels at the sector level and the changes in real-output shares. First, it shows that the initial rates of wealth growth, driven by initial wealth-productivity alignment and all general equilibrium prices, are generally higher under US borrowing limits. Under both financial-development levels, the sectors with higher initial wealth-growth rates are the one that tend to experience larger increases in real-output shares. The relationship is of similar magnitude in both regimes, with estimated slopes of 0.26 under Peruvian borrowing limits and 0.29 under US borrowing limits.



(a) Hand-to-Mouth Households

(b) Entrepreneurs

Figure 10: Consumption Variation by Financial Development

Notes: The figure reports real consumption measured in units of the final CES consumption composite. That is, nominal consumption is deflated by the contemporaneous CES final-good price index, $P(t)$, implied by the equilibrium in each counterfactual. Panel (a) shows consumption for hand-to-mouth households by income source, and panel (b) shows entrepreneur's consumption by sector.

Real Consumption Effects. Figure 10 reports consumption paths in units of the final CES aggregator, for both the baseline simulation with $\lambda = 1.13$ and the one with $\lambda = 2.29$. Panel (a) in Figure 10 decomposes the consumption for hand-to-mouth households by income source. Total hand-to-mouth consumption is higher under greater financial development throughout all periods. Wage-financed consumption is its most important component and it is approximately 21.8% higher in the initial period and 18.7% higher in the final period under US borrowing limits than under the baseline Peruvian borrowing limits. Panel (b) shows corresponding improvements of entrepreneurs' consumption. Since entrepreneurs consume a fraction of their accumulated wealth, the increase in the initial period is modest (around 1.35%) and is a result of a reduction in the final-good price index. In the final period, the consumption gains are larger and heterogeneous across all sectors, with most of them ranging from 11.83% in construction to 19.28% in manufacturing, with the exception of a larger gain in fishing (73.45%).

6.5 Trade Protectionism and Specialization

Now, I study the quantitative economy under different levels of policy-induced trade exposure. To make differences in trade exposure between counterfactuals depend directly on tariffs, in these exercises I impose no residual trade balance ($NX_{\text{res}} = 0$).[18] In these simulations I also impose uniform import tariffs across tradable sectors. As in the previous section, the model is not recalibrated. Initial endowments, preferences, financial development, international price trends, and tariffs imposed abroad, are held the same as in Section 6.3 simulation. The main comparison moves from a

[18] If $NX_{\text{res}} > 0$, even if tariffs were massive, import values would need to be high to ensure trade balance.

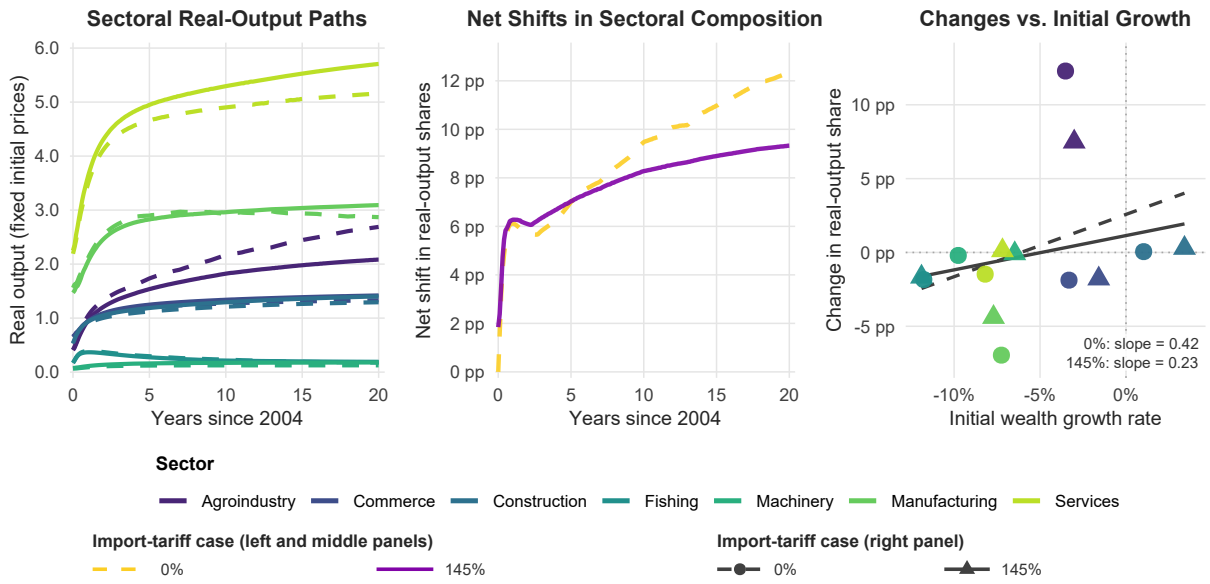


Figure 11: Sectoral specialization under alternative import tariff regimes

Notes: The figure compares sectoral specialization under a zero import tariff and a high-protection counterfactual with a uniform 145% import tariff on tradable sectors. Real output is valued using first-period producer prices in the 0% tariffs scenario. The left panel plots sector level output series. The center panel plots one-half the sum of absolute differences between each scenario's sectoral real-output shares at time t and Peru's initial sectoral real-output shares. The right panel relates each sector's initial wealth-growth rate to its change in real-output share over the 20-year transition.

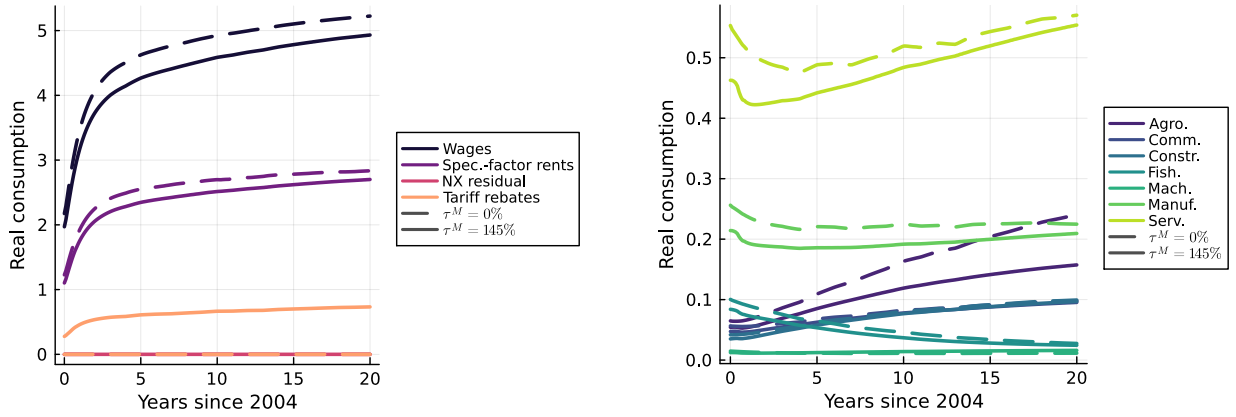
case with no import tariffs (0%) to one in which import tariffs are 145%.[19] This change implies a contraction of the size of the external sector, measured by the value of total imports (exports) as a share of output, from about 13.7% to around 4.6% in the final period. These higher tariffs are then not sufficient to generate autarky in the economy, but I use it as a high-protection reference.

Figure 11 summarizes the difference in results between the two reference tariff scenarios. As the figure shows, although the overall structure of the economy remains stable, the reduction of trade exposure produces changes in the composition of the economy that are sizable by the end of the evaluated period. Overall, higher protection weakens the association between initial advantages, approximated by initial growth rates, and subsequent changes in output shares.

The left panel of Figure 11 shows that even though the initial sector output levels of the economy differ only a little across both counterfactuals, there are considerable changes by the end period. Particularly, the increase in the share of Agroindustry is attenuated, mainly in favor of an increased importance of the Services sector.

As in the financial-development exercise, I measure net sectoral reallocation using the sum of absolute changes in real-output shares, and report them in the center panel of Figure 11. Here,

[19] This number is motivated by the aggressive rates during the US-China trade conflicts around April 2025. Additional results with alternative tariff levels are summarized in Appendix F.4.



(a) Hand-to-Mouth Households

(b) Entrepreneurs

Figure 12: Consumption Variation by Import Tariff Counterfactuals

Notes: The figure reports real consumption measured in units of the final CES consumption composite. That is, nominal consumption is deflated by the contemporaneous CES final-good price index, $P(t)$, implied by the equilibrium in each counterfactual. Panel (a) shows consumption for hand-to-mouth households by income source, and panel (b) shows entrepreneur's consumption by sector.

however, the reference composition is the initial allocation under zero import tariffs.[20] At impact, the higher tariffs produce approximately 1.8 percentage points of reallocation relative to the zero-tariff composition. Even though the magnitudes of compositional adjustment during the first five years is similar in both scenarios, by the end of the transition period the reallocation index reaches 12.3 percentage points under zero import tariffs, but only 9.3 under 145% import tariffs.

The right panel of Figure 11 reports the final changes in real-output shares over the period under analysis in relation to the initial growth rates. An increase in protectionism reduces the relationship of initial advantages with long-run composition. The estimated slope that relates initial growth rates and changes in real-output shares falls from 0.42 under zero import tariffs to 0.23 under 145% tariffs. In particular, Agroindustry – the sector which expands the most – doesn't grow as much in the less exposed scenario. This attenuation is consistent with the mechanism explored in the stylized illustration in Figure 5, in which greater responsiveness of domestic prices to local scarcity reduce between-sectors differences in profits. The reported differences in specialization and growth are accompanied by important changes in real consumption, which are discussed next.

Real Consumption Effects. Figure 12 reports consumption paths in units of the final CES aggregator, for both the 145% import-tariffs simulation and the no-import-tariffs simulation. Panel

[20] That is, the series are calculated using the following measure:

$$R_Y^\tau(t) = \frac{1}{2}\sum_{i\in\mathcal{I}}\left|\frac{p_i^{P,\tau=0}(0)Y_i^\tau(t)}{\sum_{j\in\mathcal{I}}p_j^{P,\tau=0}(0)Y_j^\tau(t)} - \frac{p_i^{P,\tau=0}(0)Y_i^0(0)}{\sum_{j\in\mathcal{I}}p_j^{P,\tau=0}(0)Y_j^0(0)}\right|.$$

(a) shows that consumption for hand-to-mouth households funded by wages and specific factor rents falls with higher protectionism. In particular, wage-funded consumption falls by 9.37% in the initial period and 5.60% in the final period. The corresponding decreases in consumption funded by specific-factor rents are 10.22% and 4.73%. However, tariff rebates could more than compensate for these losses in later periods. Panel (b) shows the consequences of protectionism in entrepreneurs' consumption. In the initial period, the tariffs-induced increase in the final-good price index reduces real consumption fall by 16.37% for entrepreneurs of all sectors. At the end of the reported transition, entrepreneurs' total real consumption falls by 9.29% relative to the free-trade case. The largest loss is in Agroindustry (34.55%), which was also the sector most affected in terms of output levels and shares. However, even entrepreneurs in sectors that seemed to be favored by protectionism ended up with lower consumption levels. For instance, end-of-period consumption by entrepreneurs in services and manufacturing falls by 2.87% and 6.84%, respectively.

7 Conclusion

The effects of trade integration, domestic financial conditions, and the initial distribution of productive wealth cannot be evaluated separately because the economy's sectoral composition responds endogenously to them. This paper makes this point by developing a theoretical framework that shows that financial frictions have effects on the relative size of sectors, both in the short- and long-run, and that the evolution of sectoral specialization depends on how relative prices reflect scarcity, and how this maps to wealth accumulation. Trade integration and financial constraints can make specialization history-dependent even when sectors have the same long-run productivity potential. When their long-run potential differs, the same constraints slow the adjustment of production towards comparative advantage.

A quantitative application to Peru using a richer environment shows that the mechanism survives the inclusion of empirically relevant features. Initial sectoral conditions matter for future specialization paths. Greater financial development and tariff-induced reductions in trade exposure reduce the relationship between initial accumulation advantages and subsequent sectoral changes, although only greater financial development improves the end-period consumption outcomes.

This paper leaves open several questions that future research could address. First, for tractability, I worked under financial autarky and an aggregate trade balance. A natural question to ask, given that Antràs and Caballero (2009) show that trade and capital mobility can become complements when financial development differs across countries, is how much international financial integration undoes or amplifies the wealth-driven specialization forces highlighted by this paper. Second, my open economy results are deliberately stark because international prices do not respond to the home economy's supply. Acemoglu and Ventura (2002) argued that faster-growing countries will face declining export prices. A couple of important questions could be answered through a full multicountry extension of this paper's framework: how large must a country or in-

dustry be before global price adjustment begins to undo wealth-induced path dependence? How does rapid growth in large countries affect path dependence in emerging economies? Finally, even though my analysis is mos4tly positive, I expect it to motivate policymakers and researchers to direct their attention towards normative questions related to how policy should respond to history-dependent specialization paths. Itskhoki and Moll (2019) and Hausmann and Rodrik (2003) have suggested, respectively, that policy can accelerate transitions towards latent or to-be-discovered comparative advantage. This paper suggests that the effectiveness of those policies may depend on how exposed sectors are to international prices.

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A Solution and Aggregation of the Baseline Model

A.1 Solution of the Static Problem of the Firm

The entrepreneur's static problem is:

$$\Pi_i(a,z) = \max_{k,\ell} \left\{ p_i (zk)^{\alpha_i} \ell^{1-\alpha_i} - w\ell - (r+\delta)k \;\text{ s.t. }\; k \leq \lambda_i a \right\}. \tag{A.1}$$

Step 1: Optimal labor demand. The first-order condition with respect to ℓ is:

$$\frac{\partial \Pi_i}{\partial \ell} = 0 \iff p_i(1-\alpha_i)(zk)^{\alpha_i}\ell^{-\alpha_i} = w \tag{A.2}$$

$$\iff \ell = \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1}{\alpha_i}} zk. \tag{A.3}$$

Step 2: Reduced-form profit function. Substituting optimal labor into profits yields:

$$\Pi_i(a,z) = \max_{k \leq \lambda_i a} \left\{ p_i (zk)^{\alpha_i} \ell^{1-\alpha_i} - w\ell - (r+\delta)k \right\} \tag{A.4}$$

$$= \max_{k \leq \lambda_i a} \left\{ \left[p_i \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1-\alpha_i}{\alpha_i}} - w \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1}{\alpha_i}} \right] zk - (r+\delta)k \right\}. \tag{A.5}$$

Define

$$\pi_i \equiv p_i \alpha_i \left(\frac{1-\alpha_i}{w/p_i}\right)^{\frac{1-\alpha_i}{\alpha_i}}, \tag{A.6}$$

so that profits can be written as

$$\Pi_i(a,z) = \max_{k \leq \lambda_i a} \left\{ \left[\pi_i z - (r+\delta) \right] k \right\}. \tag{A.7}$$

Step 3: Optimal capital choice. Since profits are linear in k, the optimal choice is at a corner:

$$k(a,z) = \begin{cases} \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{A.8}$$

The productivity cutoff $\underline{z}_i$ is defined by $\pi_i \underline{z}_i = r + \delta$.

Resulting profit function. Substituting optimal $k(a,z)$ yields:

$$\Pi_i(a,z) = \lambda_i \cdot \max\left\{ \pi_i z - (r+\delta),\, 0 \right\} a. \tag{A.9}$$

A.2 Solution of the Dynamic Problem of the Entrepreneur

The entrepreneur's dynamic problem

$$\max_{\{c(s)\}_{s \geq t}} \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)} \ln(c(s))\, ds \right] \quad \text{s.t.} \quad \dot{a}(s) = \Gamma_i(z,s)\, a(s) - P(s)c(s), \tag{A.10}$$

can be represented by the following Hamilton-Jacobi-Bellman (HJB) equation:

$$\rho V_i(a,z,t) = \max_c \left\{ \ln(c) + V_{i,a}(a,z,t) \left[\Gamma_i(z,t)a - P(t)c \right] + V_{i,t}(a,z,t) + \mathcal{L}_z V_i(a,z,t) \right\}, \tag{A.11}$$

where $\mathcal{L}_z$ denotes the infinitesimal generator associated with the productivity process in Equation 4:

$$\mathcal{L}_z V_i(a,z,t) \equiv \mu_i(z)\, V_{i,z}(a,z,t) + \frac{1}{2}\sigma_i(z)^2\, V_{i,zz}(a,z,t). \tag{A.12}$$

This section provides details on the solution of the entrepreneur's dynamic problem.

The first-order and envelope conditions associated with the Hamilton-Jacobi-Bellman equation are:

$$u'(c) = P(t)\, V_{i,a}(a,z,t), \tag{A.13}$$

$$\rho V_{i,a} = V_{i,a}\, \Gamma_i(z,t) + V_{i,at} + \mathcal{L}_z V_{i,a}. \tag{A.14}$$

The following subsections present two alternative derivations of the optimal consumption rule and the induced wealth dynamics. The first uses a Hamilton-Jacobi-Bellman approach with guess and verify, while the second uses a perfect foresight optimal control formulation.

A.2.1 Hamilton-Jacobi-Bellman with Guess and Verify Approach

Following Moll (2014), consider a guess-and-verify solution that exploits the homothetic structure of the problem. Suppressing sector indices for notational simplicity, the Hamilton–Jacobi–Bellman equation becomes

$$\rho V(a,z,t) = \max_c \left\{ \ln c + V_a(a,z,t) \left[\Gamma_i(z,t)a - P(t)c \right] + V_t(a,z,t) + \mathcal{L}_z V(a,z,t) \right\}.$$

The first-order condition with respect to consumption is

$$\frac{1}{c} = P(t) V_a(a,z,t),$$

or equivalently,

$$P(t)c(a,z,t) = \frac{1}{V_a(a,z,t)}.$$

Consider the guess

$$V(a,z,t) = D(z,t) + E(z,t)\log a.$$

Then

$$V_a(a,z,t) = \frac{E(z,t)}{a},$$

implying

$$P(t)c(a,z,t) = \frac{a}{E(z,t)}.$$

Substituting the guess and optimal consumption rule into the HJB equation and collecting terms proportional to $\log a$ yields

$$\rho E(z,t) = 1 + E_t(z,t).$$

A time-invariant solution therefore satisfies

$$E(z) = \frac{1}{\rho}.$$

Hence,

$$P(t)c(a,t) = \rho a(t).$$

Substituting into the wealth accumulation equation yields

$$\dot{a}(t) = \left[\Gamma_i(z,t) - \rho\right] a(t).$$

A.2.2 Perfect Foresight Approach

As an alternative, the problem can be solved using a standard optimal control approach. Consider the Hamiltonian:

$$\mathcal{H} = \log c(t) + \mu(t)\left[\Gamma_i(z,t)a(t) - P(t)c(t)\right].$$

The first-order condition with respect to consumption is:

$$\frac{\partial \mathcal{H}}{\partial c(t)} = 0 \quad \Longleftrightarrow \quad \frac{1}{c(t)} = \mu(t)P(t) \quad \Longleftrightarrow \quad \mu(t) = \frac{1}{P(t)c(t)}.$$

The costate equation is:

$$\dot{\mu}(t) = \rho\mu(t) - \frac{\partial \mathcal{H}}{\partial a(t)} = \rho\mu(t) - \mu(t)\Gamma_i(z,t).$$

Substituting for $\mu(t)$ yields:

$$\frac{(\dot{P(t)c(t)})}{P(t)c(t)} = \Gamma_i(z,t) - \rho.$$

Let $m(t) \equiv P(t)c(t)$. The costate equation can be further rewritten as

$$\dot{m}(t) = \left[\Gamma_i(z,t) - \rho\right] m(t).$$

Together with the law of motion for wealth,

$$\dot{a}(t) = \Gamma_i(z,t)a(t) - m(t),$$

this implies that the ratio $x(t) \equiv m(t)/a(t)$ evolves according to

$$\dot{x}(t) = x(t)\big(x(t) - \rho\big).$$

The optimal path consistent with the transversality condition is the stationary ratio $x(t) = \rho$, so that

$$P(t)c(t) = \rho a(t).$$

Hence,

$$\dot{a}(t) = \left[\Gamma_i(z,t) - \rho\right] a(t).$$

A.3 Aggregation and factor prices

This appendix derives the aggregation results used in Proposition 6. The derivation proceeds in four steps. First, I aggregate individual wealth accumulation to the sectoral level. Second, I derive sectoral capital and labor demand. Third, I obtain the sectoral production representation. Fourth, I derive the factor-price conditions.

Useful definitions. Let

$$X_i(t) \equiv \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz, \qquad M_i(t) \equiv 1 - \Omega_i(\underline{z}_i(t)).$$

where $M_i(t)$ is the share of sectoral wealth held by active entrepreneurs. Then sectoral capital demand is

$$K_i(t) = \lambda_i A_i(t) M_i(t),$$

and sectoral productivity is

$$Z_i(t) = \left(\frac{X_i(t)}{M_i(t)}\right)^{\alpha_i} = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

A.3.1 Sectoral wealth accumulation

Using the individual law of motion $\dot{a} = s_i(z,t)a$, sectoral wealth evolves as

$$\dot{A}_i(t) = \int \dot{a}\, g_i(a,z;t)\, da\, dz = \int s_i(z,t) a\, g_i(a,z;t)\, da\, dz.$$

Using the definition of the within-sector wealth-weighted density,

$$o_i(z,t) = \frac{1}{A_i(t)} \int a g_i(a,z;t)\, da,$$

this becomes

$$\dot{A}_i(t) = A_i(t) \int s_i(z,t) o_i(z,t)\, dz.$$

Aggregating across sectors,

$$\dot{A}(t) = \sum_{i \in I} \dot{A}_i(t) = \sum_{i \in I} A_i(t) \int s_i(z,t) o_i(z,t)\, dz = A(t) \sum_{i \in I} \int s_i(z,t) \omega_i(z,t)\, dz.$$

A.3.2 Sectoral input demands

From the static problem, active entrepreneurs satisfy

$$k_i(a,z;t) = \lambda_i a, \qquad \ell_i(a,z;t) = \left(\frac{\pi_i(t)}{p_i(t)\alpha_i}\right)^{\frac{1}{1-\alpha_i}} z \lambda_i a \quad \text{for } z \geq \underline{z}_i(t).$$

Therefore,

$$K_i(t) = \int_{\underline{z}_i(t)}^{\infty} \int_0^{\infty} \lambda_i a g_i(a, z; t)\, da\, dz = \lambda_i A_i(t) M_i(t),$$

and

$$L_i(t) = \left(\frac{\pi_i(t)}{p_i(t)\alpha_i} \right)^{\frac{1}{1-\alpha_i}} \lambda_i A_i(t) X_i(t).$$

Solving the last expression for $\pi_i(t)$ gives

$$\pi_i(t) = p_i(t)\alpha_i \left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i - 1} L_i(t)^{1-\alpha_i}.$$

A.3.3 Sectoral output and measured productivity

For active entrepreneurs, substituting optimal labor demand into production gives

$$y_i(a, z; t) = z \frac{\pi_i(t)}{p_i(t)\alpha_i} \lambda_i a.$$

Aggregating,

$$Y_i(t) = \lambda_i \frac{\pi_i(t)}{p_i(t)\alpha_i} A_i(t) X_i(t).$$

Using the expression for $\pi_i(t)$ above,

$$Y_i(t) = \left[\lambda_i A_i(t) X_i(t)\right]^{\alpha_i} L_i(t)^{1-\alpha_i}.$$

Since

$$K_i(t) = \lambda_i A_i(t) M_i(t),$$

this can be rewritten as

$$Y_i(t) = \left(\frac{X_i(t)}{M_i(t)} \right)^{\alpha_i} K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i}.$$

Thus,

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i},$$

where

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

A.3.4 Sectoral wealth law of motion

Using $s_i(z, t) = \Gamma_i(z, t) - \rho$, sectoral wealth growth is

$$\dot{A}_i(t) = A_i(t) \int \left[\lambda_i \max\{\pi_i(t) z - (r(t) + \delta), 0\} + r(t) - \rho\right] o_i(z, t)\, dz.$$

Splitting the integral at the cutoff $\underline{z}_i(t)$,

$$\dot{A}_i(t) = \lambda_i \pi_i(t) A_i(t) X_i(t) - (r(t) + \delta) K_i(t) + (r(t) - \rho) A_i(t).$$

Since

$$\lambda_i \pi_i(t) A_i(t) X_i(t) = \alpha_i p_i(t) Y_i(t),$$

we obtain

$$\dot{A}_i(t) = \alpha_i p_i(t) Y_i(t) + r(t) \left[A_i(t) - K_i(t) \right] - \delta K_i(t) - \rho A_i(t).$$

A.3.5 Factor-price conditions

Wage condition. The aggregate labor demand expression derived above is

$$L_i(t) = \left(\frac{\pi_i(t)}{p_i(t)\alpha_i} \right)^{\frac{1}{1-\alpha_i}} \lambda_i A_i(t) X_i(t),$$

where

$$X_i(t) \equiv \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz$$

and

$$\pi_i(t) = p_i(t)\alpha_i \left(\frac{1-\alpha_i}{w(t)/p_i(t)} \right)^{\frac{1-\alpha_i}{\alpha_i}}.$$

Substituting this expression for $\pi_i(t)$ into sectoral labor demand gives

$$\begin{aligned} L_i(t) &= \left[\left(\frac{1-\alpha_i}{w(t)/p_i(t)} \right)^{\frac{1-\alpha_i}{\alpha_i}} \right]^{\frac{1}{1-\alpha_i}} \lambda_i A_i(t) X_i(t) \\ &= \left(\frac{1-\alpha_i}{w(t)/p_i(t)} \right)^{\frac{1}{\alpha_i}} \lambda_i A_i(t) X_i(t). \end{aligned}$$

Raising both sides to the power α_i and rearranging yields

$$L_i(t)^{\alpha_i} = \frac{1-\alpha_i}{w(t)/p_i(t)} \left[\lambda_i A_i(t) X_i(t) \right]^{\alpha_i},$$

or equivalently,

$$\frac{w(t)}{p_i(t)} = (1-\alpha_i) \left[\lambda_i A_i(t) X_i(t) \right]^{\alpha_i} L_i(t)^{-\alpha_i}.$$

Using the sectoral production representation derived above,

$$Y_i(t) = \left[\lambda_i A_i(t) X_i(t) \right]^{\alpha_i} L_i(t)^{1-\alpha_i} = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i},$$

we have

$$\left[\lambda_i A_i(t) X_i(t) \right]^{\alpha_i} = Z_i(t) K_i(t)^{\alpha_i}.$$

Substituting this expression into the real wage equation gives

$$\frac{w(t)}{p_i(t)} = (1-\alpha_i) Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{-\alpha_i}.$$

Equivalently,

$$w(t)L_i(t) = (1-\alpha_i)p_i(t)Y_i(t).$$

Interest-rate condition. The productivity cutoff satisfies

$$\pi_i(t)\underline{z}_i(t) = r(t) + \delta.$$

Using

$$\pi_i(t) = p_i(t)\alpha_i \left(\frac{1-\alpha_i}{w(t)/p_i(t)}\right)^{\frac{1-\alpha_i}{\alpha_i}},$$

and the real wage condition derived above,

$$\frac{w(t)}{p_i(t)} = (1-\alpha_i)Z_i(t)K_i(t)^{\alpha_i}L_i(t)^{-\alpha_i},$$

we can rewrite the term inside the profit coefficient as

$$\frac{1-\alpha_i}{w(t)/p_i(t)} = Z_i(t)^{-1}K_i(t)^{-\alpha_i}L_i(t)^{\alpha_i}.$$

Therefore,

$$\begin{aligned} r(t)+\delta &= \underline{z}_i(t)p_i(t)\alpha_i \left[Z_i(t)^{-1}K_i(t)^{-\alpha_i}L_i(t)^{\alpha_i}\right]^{\frac{1-\alpha_i}{\alpha_i}} \\ &= p_i(t)\alpha_i\underline{z}_i(t)Z_i(t)^{\frac{\alpha_i-1}{\alpha_i}}K_i(t)^{\alpha_i-1}L_i(t)^{1-\alpha_i}. \end{aligned}$$

Since

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}\left[z \mid z \geq \underline{z}_i(t)\right]\right)^{\alpha_i},$$

define the selection wedge

$$\zeta_i(t) \equiv \frac{\underline{z}_i(t)}{Z_i(t)^{1/\alpha_i}} = \frac{\underline{z}_i(t)}{\mathbb{E}_{o_i,t}\left[z \mid z \geq \underline{z}_i(t)\right]} \in [0,1].$$

Then the interest-rate condition becomes

$$r(t) = p_i(t)\alpha_i\zeta_i(t)Z_i(t)K_i(t)^{\alpha_i-1}L_i(t)^{1-\alpha_i} - \delta.$$

Aggregate factor-price conditions. Aggregating across sectors and using labor and capital market clearing gives

$$w(t) = \frac{1}{\bar{L}}\sum_{i\in I}(1-\alpha_i)p_i(t)Y_i(t),$$

and

$$r(t) = \frac{1}{K(t)}\sum_{i\in I}p_i(t)\alpha_i\zeta_i(t)Y_i(t) - \delta,$$

where $K(t) = \sum_{i\in I}K_i(t)$.

A.3.6 Summary

Collecting the results, sectoral aggregates satisfy

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{1-\alpha_i},$$

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i},$$

and

$$\dot{A}_i(t) = \alpha_i p_i(t) Y_i(t) + r(t)\left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t).$$

These are the aggregation results stated in Proposition 6.

A.4 Steady State

A steady state is an equilibrium in which aggregate wealth and its distribution are time-invariant. Formally, for all sectors $i \in I$,

$$\dot{A}_i(t) = 0, \quad g_i(a, z; t) = g_i(a, z), \quad \omega_i(z, t) = \omega_i(z), \quad \text{and} \quad O_i(t) = O_i. \tag{A.15}$$

In this environment, sectoral output is given by

$$Y_i = Z_i K_i^{\alpha_i} L_i^{1-\alpha_i}, \tag{A.16}$$

where Z_i denotes sectoral productivity, which depends on the stationary distribution of active entrepreneurs.

Using the optimality conditions of firms and the definition of profits, capital income in each sector satisfies

$$\alpha_i p_i Y_i = (r + \delta) K_i - (r - \rho) A_i. \tag{A.17}$$

Aggregating across sectors, factor prices are pinned down by market clearing. The wage satisfies

$$w = \frac{1}{\bar{L}} \left(\sum_{i \in I} p_i Y_i - (\rho + \delta) K \right), \tag{A.18}$$

while the interest rate satisfies

$$r = \frac{1}{K} \sum_{i \in I} \zeta_i \left[(r + \delta) K_i - (r - \rho) A_i\right] - \delta. \tag{A.19}$$

In steady state, the objects underlying aggregate productivity admit a simple characterization. Since sectoral productivities and expenditure shares are constant over time, $\dot{\log} Z_i(t) = 0$ and $\dot{D}_i(t) = 0$ for all i, implying zero aggregate productivity growth, $\dot{\log} Z^A(t) = 0$. The Domar weights D_i become time-invariant and reflect the stationary allocation of expenditure across sectors, while sectoral productivity levels Z_i are pinned down by the stationary distribution of entrepreneurs and the corresponding selection thresholds.

As a result, the within–between decomposition reduces to a static accounting identity: aggregate productivity is fully characterized by the cross-sectional distribution of sectoral productivities and their Domar weights, both of which are endogenously determined in equilibrium through wealth dynamics, selection, and relative prices.

B Wealth Dynamics

This appendix derives the evolution of wealth shares used in the main text. The procedure follows three steps. First, it states the general Fokker–Planck equation that governs the evolution of densities associated with Itô diffusion processes. The derivation and intuition behind this equation are standard and can be found in Risken (1996); Gardiner (2009); Pavliotis (2014); Øksendal (2013).

Second, the Fokker–Planck equation is applied to the joint process for productivity and wealth to derive the law of motion for the wealth-share density. Third, these dynamics are decomposed into within-sector and across-sector components, which are used in the analysis in the main text.

B.1 The General Fokker-Planck Equation

Consider a d-dimensional Itô diffusion process $X_t \in \mathbb{R}^d$ governed by:

$$dX_t = \mu(X_t, t)\, dt + \Sigma(X_t, t)\, dW_t,$$

where:

- $\mu : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^d$ is the drift vector,
- $\Sigma : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^{d\times m}$ is the diffusion matrix,
- W_t is an m-dimensional Wiener process.

Then, the probability density $\phi(x, t)$ of X_t evolves according to the Fokker–Planck equation:

$$\frac{\partial \phi(x,t)}{\partial t} = -\nabla \cdot [\mu(x,t)\phi(x,t)] + \frac{1}{2}\sum_{i,j=1}^{d} \frac{\partial^2}{\partial x_i \partial x_j}\left[D_{ij}(x,t)\phi(x,t)\right],$$

where $D(x, t) = \Sigma(x, t)\Sigma(x, t)^\top$ is the diffusion (covariance) matrix.

B.2 Derivation of the Wealth Share Dynamics

Apply the general Fokker–Planck equation to the joint process $X_t = (z(t), a(t))$, where $\phi(x, t) = g_i(z, a, t)$:

1. Start from the bivariate system implied by: **sectoral productivity dynamics** and the **optimal savings policy** for sector i:

$$\begin{aligned} dz &= \mu_i(z)\, dt + \sigma_i(z)\, dW_t \\ da &= s_i(z,t)\, a\, dt \end{aligned}$$

   Note that wealth follows a deterministic drift (conditional on z), so there is no diffusion term in the a-dimension.

2. Then, $g_i(z, a, t)$ evolves according to the bivariate Fokker–Planck equation:

$$\frac{\partial g_i}{\partial t} = -\frac{\partial}{\partial z}[\mu_i(z) g_i] - \frac{\partial}{\partial a}[s_i(z,t) a g_i] + \frac{1}{2}\frac{\partial^2}{\partial z^2}[\sigma_i^2(z) g_i]$$

   Since shocks affect only productivity, there are no cross-partial diffusion terms involving a.

3. Define $\omega_i(z,t)$ as the wealth-share density over productivity z in sector i:

$$\omega_i(z,t) := \frac{1}{A(t)} \int_0^\infty a\, g_i(z,a,t)\, da$$

Then, its time derivative is:

$$\frac{\partial \omega_i(z,t)}{\partial t} = \frac{1}{A(t)} \int_0^\infty a \frac{\partial g_i}{\partial t}\, da - \frac{\dot{A}(t)}{A(t)} \omega_i(z,t)$$

4. Substituting from the Fokker–Planck equation into the expression for $\partial_t \omega_i(z,t)$ yields:

$$\frac{\partial \omega_i(z,t)}{\partial t} = \frac{1}{A(t)} \int_0^\infty a \left\{ -\frac{\partial}{\partial z}\big[\mu_i(z) g_i\big] - \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\big[\sigma_i^2(z) g_i\big] \right\} da - \frac{\dot{A}(t)}{A(t)} \omega_i(z,t).$$

Consider the term involving derivatives with respect to a:

$$\int_0^\infty a \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big]\, da.$$

Applying integration by parts,

$$\int_0^\infty u\, dv = [uv]_0^\infty - \int_0^\infty v\, du,$$

with $u = a$ and $dv = \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big] da$, yields

$$\int_0^\infty a \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big]\, da = \big[a\, s_i(z,t)\, a\, g_i(z,a,t)\big]_0^\infty - \int_0^\infty s_i(z,t)\, a\, g_i(z,a,t)\, da.$$

Assuming $a^2 g_i(a,z,t) \to 0$ as $a \to 0$ and $a \to \infty$; the boundary term vanishes, so

$$\int_0^\infty a \frac{\partial}{\partial a}\big[s_i(z,t)\, a\, g_i\big]\, da = -\int_0^\infty s_i(z,t)\, a\, g_i(z,a,t)\, da.$$

Similarly, derivatives with respect to z can be taken outside the integral:

$$\int_0^\infty a\, \frac{\partial}{\partial z}\big[\mu_i(z) g_i\big]\, da = \frac{\partial}{\partial z}\left[\mu_i(z) \int_0^\infty a\, g_i\, da\right],$$

and analogously for the second derivative term.

Using the definition of $\omega_i(z,t)$, this implies

$$\frac{\partial \omega_i(z,t)}{\partial t} = \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)}\right] \omega_i(z,t) - \frac{\partial}{\partial z}\left[\mu_i(z)\omega_i(z,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial z^2}\left[\sigma_i^2(z)\omega_i(z,t)\right]. \tag{B.1}$$

B.3 Decomposition of Wealth Dynamics

This appendix derives the decomposition of the evolution of wealth shares into within- and across-sector components.

Step 1: Evolution of wealth-weighted distribution. Recall that the wealth-weighted density $\omega_i(z,t)$ evolves according to

$$\frac{\partial \omega_i(z,t)}{\partial t} = \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} \right] \omega_i(z,t) - \frac{\partial}{\partial z} \left[\mu_i(z) \omega_i(z,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial z^2} \left[\sigma_i^2(z) \omega_i(z,t) \right]. \tag{B.2}$$

Step 2: Decomposition. Using the identity

$$\omega_i(z,t) = O_i(t)\, o_i(z,t), \tag{B.3}$$

its time derivative satisfies

$$\frac{\partial \omega_i(z,t)}{\partial t} = \dot{O}_i(t)\, o_i(z,t) + O_i(t)\, \frac{\partial o_i(z,t)}{\partial t}. \tag{B.4}$$

Substituting (B.3) and (B.4) into (B.2) and dividing by $O_i(t)$ yields

$$\begin{aligned} \frac{\dot{O}_i(t)}{O_i(t)}\, o_i(z,t) + \frac{\partial o_i(z,t)}{\partial t} = & \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} \right] o_i(z,t) \\ & - \frac{\partial}{\partial z} \left[\mu_i(z) o_i(z,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial z^2} \left[\sigma_i^2(z) o_i(z,t) \right]. \end{aligned} \tag{B.5}$$

Step 3: Law of motion for sectoral wealth. Integrating (B.2) over z and using that $\int \omega_i(z,t)\, dz = O_i(t)$, we obtain

$$\dot{O}_i(t) = \int s_i(z,t)\, \omega_i(z,t)\, dz - \frac{\dot{A}(t)}{A(t)} O_i(t). \tag{B.6}$$

Using $\omega_i(z,t) = O_i(t) o_i(z,t)$, this becomes

$$\frac{\dot{O}_i(t)}{O_i(t)} = \bar{s}_i(t) - \frac{\dot{A}(t)}{A(t)}, \tag{B.7}$$

where

$$\bar{s}_i(t) \equiv \int s_i(z,t)\, o_i(z,t)\, dz. \tag{B.8}$$

Step 4: Within-sector dynamics. Substituting (B.7) into (B.5) yields

$$\frac{\partial o_i(z,t)}{\partial t} = \left[s_i(z,t) - \bar{s}_i(t) \right] o_i(z,t) - \frac{\partial}{\partial z} \left[\mu_i(z) o_i(z,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial z^2} \left[\sigma_i^2(z) o_i(z,t) \right]. \tag{B.9}$$

Conclusion. The evolution of wealth shares can therefore be decomposed into: (i) within-sector dynamics governed by deviations from the sectoral average savings rate, and (ii) across-sector dynamics governed by differences in average savings rates across sectors.

C Quantitative Open Economy

This appendix extends the model to allow it to speak to the Peruvian data and to implement a quantitative application that tests the implications of the main model. This extension can be summarized in four changes.

First, the model introduces factors specific to each sector which are fixed in supply. This allows firm-level production functions to remain tractable with constant returns to scale in capital, labor, and the sector-specific factor. At the same time, this introduces a congestion mechanism that provides a foundation for decreasing returns to scale at the sector level and a structure which is consistent with decreasing returns to scale in empirical estimates of production functions based only on capital and labor.

Second, it explicitly includes nontradable sectors and excludes resource-intensive activities such as oil and mining. I incorporate a residual denoted $NX_{\text{res}}(t)$ to represent exogenous net exports generated by these omitted natural-resource-based sectors and to close the aggregate external-balance condition.

Third, for tradable sectors, I distinguish between domestic-origin and imported varieties which are not perfect substitutes.

Fourth, the prices at which these varieties are traded are subject to sector-specific tariffs. Given exogenous world prices $p_i^*(t)$, domestic demand can purchase imports at a tariff-inclusive price $p_i^M(t)$, and domestic producers can export receiving net-of-tariffs export prices $p_i^X(t)$. The domestic-origin prices are determined endogenously. When a sector's supply is enough for it to export, the domestic-origin price equals the net export price; when production is not enough to satisfy domestic demand at that price, exports are zero, and the domestic-origin price may exceed the net export price.

Both resource income and import tariff revenues are rebated as a lump sum to a separate hand-to-mouth household and contribute to aggregate domestic absorption without entering entrepreneurs' wealth accounts.

C.1 Static Problem of the Firm with Sector-Specific Factors and Solution

The entrepreneur's static problem is:

$$\Pi_i(a,z) = \max_{k,\ell,h} \left\{ p_i (zk)^{\alpha_i} \ell^{\beta_i} h^{\gamma_i} - w\ell - q_i h - (r+\delta)k \;\; \text{s.t.} \;\; k \leq \lambda_i a \right\}, \tag{C.1}$$

where $\alpha_i, \beta_i, \gamma_i > 0$ and

$$\alpha_i + \beta_i + \gamma_i = 1. \tag{C.2}$$

The variable h represents a sector-specific factor rented by firms at price q_i. The price p_i denotes the price received by domestic producers in sector i. For tradable sectors, this price is linked to international prices through the export channel. But when domestic goods are scare, the domestic-origin variety may be larger than the net export price in order to clear domestic demand. For nontradable sectors, p_i is always the price that clears domestic demand. Regardless of the specific sectoral context, individual firms take these prices as given.

Step 1: Optimal labor and specific-factor demand. For a given level of capital k, the first-order conditions with respect to ℓ and h are:

$$p_i\beta_i(zk)^{\alpha_i}\ell^{\beta_i-1}h^{\gamma_i} = w, \tag{C.3}$$

$$p_i\gamma_i(zk)^{\alpha_i}\ell^{\beta_i}h^{\gamma_i-1} = q_i. \tag{C.4}$$

Dividing the first-order condition for labor by the first-order condition for the specific factor gives

$$h = \frac{\gamma_i}{\beta_i}\frac{w}{q_i}\ell. \tag{C.5}$$

Substituting this expression into the labor first-order condition results in

$$p_i\beta_i z^{\alpha_i}k^{\alpha_i}\left(\frac{\gamma_i}{\beta_i}\frac{w}{q_i}\right)^{\gamma_i}\ell^{\beta_i+\gamma_i-1} = w. \tag{C.6}$$

Since $\alpha_i + \beta_i + \gamma_i = 1$, this becomes

$$p_i\beta_i z^{\alpha_i}k^{\alpha_i}\left(\frac{\gamma_i}{\beta_i}\frac{w}{q_i}\right)^{\gamma_i}\ell^{-\alpha_i} = w. \tag{C.7}$$

Therefore,

$$\ell_i(k,z) = \left[\frac{p_i\beta_i}{w}\left(\frac{\gamma_i}{\beta_i}\frac{w}{q_i}\right)^{\gamma_i}\right]^{1/\alpha_i} zk. \tag{C.8}$$

Replacing it back into Equation C.5,

$$h_i(k,z) = \frac{\gamma_i}{\beta_i}\frac{w}{q_i}\left[\frac{p_i\beta_i}{w}\left(\frac{\gamma_i}{\beta_i}\frac{w}{q_i}\right)^{\gamma_i}\right]^{1/\alpha_i} zk. \tag{C.9}$$

Equivalently, the optimal input demands can be written as

$$\ell_i(k,z) = zk\left[\frac{p_i\beta_i}{w}\left(\frac{\gamma_i w}{\beta_i q_i}\right)^{\gamma_i}\right]^{\frac{1}{\alpha_i}}, \tag{C.10}$$

$$h_i(k,z) = zk\left[\frac{p_i\gamma_i}{q_i}\left(\frac{\beta_i q_i}{\gamma_i w}\right)^{\beta_i}\right]^{\frac{1}{\alpha_i}}. \tag{C.11}$$

That is, both non-capital input demands are proportional to zk. Since optimal capital is at the same time proportional to wealth due to the collateral constraint, as it will be shown below, labor and specific-factor demand will also be linear in wealth.

Step 2: Reduced-form profit function. From Step 1, optimal non-capital input demands can be written as

$$\ell_i(k,z) = m_{\ell i}zk, \qquad h_i(k,z) = m_{hi}zk, \tag{C.12}$$

where

$$m_{\ell i} \equiv \left[\frac{p_i\beta_i}{w}\left(\frac{\gamma_i w}{\beta_i q_i}\right)^{\gamma_i}\right]^{\frac{1}{\alpha_i}}, \tag{C.13}$$

and

$$m_{hi} \equiv \left[\frac{p_i \gamma_i}{q_i} \left(\frac{\beta_i q_i}{\gamma_i w} \right)^{\beta_i} \right]^{\frac{1}{\alpha_i}}. \tag{C.14}$$

Substituting these expressions into output gives

$$y_i(k,z) = (zk)^{\alpha_i} (m_{\ell i} zk)^{\beta_i} (m_{hi} zk)^{\gamma_i} \tag{C.15}$$

$$= m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i} z^{\alpha_i+\beta_i+\gamma_i} k^{\alpha_i+\beta_i+\gamma_i}. \tag{C.16}$$

Since $\alpha_i + \beta_i + \gamma_i = 1$, this becomes

$$y_i(k,z) = m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i} zk. \tag{C.17}$$

Plugging these in the profits function results in:

$$\max_{\ell,h} \left\{ p_i (zk)^{\alpha_i} \ell^{\beta_i} h^{\gamma_i} - w\ell - q_i h \right\} = \left[p_i m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i} - w m_{\ell i} - q_i m_{hi} \right] zk. \tag{C.18}$$

Rearranging the first-order conditions, payments to labor and the specific factor must satisfy

$$w\ell_i = \beta_i p_i y_i, \qquad q_i h_i = \gamma_i p_i y_i. \tag{C.19}$$

Which implies that operating surplus equals the capital share of revenue:

$$p_i y_i - w\ell_i - q_i h_i = p_i y_i - \beta_i p_i y_i - \gamma_i p_i y_i = \alpha_i p_i y_i. \tag{C.20}$$

Therefore,

$$\max_{\ell,h} \left\{ p_i (zk)^{\alpha_i} \ell^{\beta_i} h^{\gamma_i} - w\ell - q_i h \right\} = \alpha_i p_i y_i = \pi_i zk, \tag{C.21}$$

where

$$\pi_i \equiv \alpha_i p_i m_{\ell i}^{\beta_i} m_{hi}^{\gamma_i} \equiv \alpha_i p_i \left(\frac{\beta_i p_i}{w} \right)^{\frac{\beta_i}{\alpha_i}} \left(\frac{\gamma_i p_i}{q_i} \right)^{\frac{\gamma_i}{\alpha_i}}. \tag{C.22}$$

Equivalently, since $\alpha_i + \beta_i + \gamma_i = 1$,

$$\pi_i = \alpha_i p_i^{1/\alpha_i} \left(\frac{\beta_i}{w} \right)^{\beta_i/\alpha_i} \left(\frac{\gamma_i}{q_i} \right)^{\gamma_i/\alpha_i}. \tag{C.23}$$

Thus, the entrepreneur's static problem becomes

$$\Pi_i(a,z) = \max_{k \le \lambda_i a} \left\{ \left[\pi_i z - (r+\delta) \right] k \right\}. \tag{C.24}$$

Step 3: Optimal capital choice. Since profits are linear in k, the optimal choice is at a corner:

$$k_i(a,z) = \begin{cases} \lambda_i a, & \text{if } z \ge \underline{z}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{C.25}$$

The productivity cutoff $\underline{z}_i$ is defined by

$$\pi_i \underline{z}_i = r + \delta. \tag{C.26}$$

Thus,

$$\underline{z}_i = \frac{r+\delta}{\pi_i}. \tag{C.27}$$

Input demands. Substituting the optimal capital choice into the input demands gives:

$$\ell_i(a,z) = \begin{cases} m_{\ell i} z \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise,} \end{cases} \tag{C.28}$$

and

$$h_i(a,z) = \begin{cases} m_{hi} z \lambda_i a, & \text{if } z \geq \underline{z}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{C.29}$$

Profit function. Substituting optimal capital demand into the reduced-form problem yields

$$\Pi_i(a,z) = \lambda_i \max\left\{\pi_i z - (r+\delta), 0\right\} a. \tag{C.30}$$

C.2 Specific-Factor Market Clearing and Sector-Level Decreasing Returns to Scale

The supply of the sector-specific factors for sector i is fixed. In equilibrium, its rental price q_i should be such that it guarantees clearing

$$H_i = \int h_i(a,z) g_i(a,z)\, da\, dz. \tag{C.31}$$

Using the linear policy rules derived above, this condition can be written as

$$H_i = \lambda_i A_i \int_{\underline{z}_i}^{\infty} m_{hi} z o_i(z)\, dz, \tag{C.32}$$

and more explicitly:

$$H_i = q_i^{-\left(\frac{\alpha_i+\gamma_i}{\alpha_i}\right)} \lambda_i \left[p_i \gamma_i \left(\frac{\beta_i}{\gamma_i w} \right)^{\beta_i} \right]^{\frac{1}{\alpha_i}} A_i \int_{\underline{z}_i}^{\infty} z o_i(z)\, dz, \tag{C.33}$$

When sector i expands, its demand for its specific factor increases. However, since H_i is fixed, the rental price q_i increases in equilibrium. This constitutes a congestion force that prevents corner solutions in which a sector expands indefinitely while preserving the linearity of firm policies in wealth.

Sector-level decreasing returns in scalable inputs. Since individual policies are linear in wealth, sectoral output can be written as

$$Y_i = Z_i K_i^{\alpha_i} L_i^{\beta_i} H_i^{\gamma_i}, \tag{C.34}$$

where

$$\alpha_i + \beta_i + \gamma_i = 1. \tag{C.35}$$

Because H_i is fixed at the sector level, equilibrium output can be interpreted as a decreasing-returns function of the scalable inputs K_i and L_i:

$$Y_i = \widetilde{Z}_i K_i^{\alpha_i} L_i^{\beta_i}, \qquad \widetilde{Z}_i \equiv Z_i H_i^{\gamma_i}, \tag{C.36}$$

with

$$\alpha_i + \beta_i = 1 - \gamma_i < 1. \tag{C.37}$$

C.3 Aggregation, Prices, and Domestic-Origin Prices

The aggregation follows the same steps as in Appendix A.3. I first aggregate firms' input demands and output, and then derive the sectoral wealth law of motion and factor-price conditions.

Throughout this subsection, $p_i^P(t)$ denotes the price received by domestic producers. It equals the domestic-origin price $p_i^H(t)$ in tradable sectors and the domestic market-clearing price $p_i^N(t)$ in nontradable sectors. While the details on the definitions and interactions between these prices will be introduced below, the results in this subsection depend only on the prices the producers face.

Useful definitions. To avoid conflict with the notation for gross exports and imports introduced below, I denote the active wealth share by $n_i(t)$ and the productivity-weighted active wealth share by $e_i(t)$ in this appendix.

Let

$$n_i(t) \equiv 1 - \Omega_i(\underline{z}_i(t)), \qquad e_i(t) \equiv \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz.$$

Here, $n_i(t)$ is the share of sectoral wealth held by active entrepreneurs, while $e_i(t)$ is the corresponding productivity-weighted active wealth share. Since active entrepreneurs choose $k_i(a,z;t) = \lambda_i a$, sectoral capital demand is

$$K_i(t) = \lambda_i A_i(t) n_i(t).$$

Define sectoral productivity as

$$Z_i(t) \equiv \left(\frac{e_i(t)}{n_i(t)}\right)^{\alpha_i} = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

C.3.1 Sectoral input demands

From the static problem with sector-specific factors, active entrepreneurs satisfy

$$k_i(a,z;t) = \lambda_i a,$$

$$\ell_i(a,z;t) = m_{\ell i}(t) z \lambda_i a, \qquad h_i(a,z;t) = m_{hi}(t) z \lambda_i a,$$

for $z \geq \underline{z}_i(t)$, where

$$m_{\ell i}(t) \equiv \left[\frac{p_i^P(t)\beta_i}{w(t)} \left(\frac{\gamma_i w(t)}{\beta_i q_i(t)} \right)^{\gamma_i} \right]^{1/\alpha_i},$$

and

$$m_{hi}(t) \equiv \left[\frac{p_i^P(t)\gamma_i}{q_i(t)} \left(\frac{\beta_i q_i(t)}{\gamma_i w(t)} \right)^{\beta_i} \right]^{1/\alpha_i}.$$

Aggregating capital demand gives

$$K_i(t) = \int_{\underline{z}_i(t)}^{\infty} \int_0^{\infty} \lambda_i a g_i(a, z; t)\, da\, dz = \lambda_i A_i(t) n_i(t).$$

Aggregating labor demand gives

$$L_i(t) = \int_{\underline{z}_i(t)}^{\infty} \int_0^{\infty} m_{\ell i}(t) z \lambda_i a g_i(a, z; t)\, da\, dz = m_{\ell i}(t) \lambda_i A_i(t) e_i(t).$$

Similarly, aggregate demand for the sector-specific factor is

$$H_i(t) = \int_{\underline{z}_i(t)}^{\infty} \int_0^{\infty} m_{hi}(t) z \lambda_i a g_i(a, z; t)\, da\, dz = m_{hi}(t) \lambda_i A_i(t) e_i(t).$$

In equilibrium, the sector-specific factor is in fixed supply, so

$$H_i(t) = \bar{H}_i.$$

This condition pins down the rental price $q_i(t)$. As sector i expands, demand for the fixed factor rises; because $\bar{H}_i$ is fixed, $q_i(t)$ increases, reducing sectoral profitability.

C.3.2 Sectoral output and measured productivity

For active entrepreneurs,

$$y_i(a, z; t) = (z k_i(a, z; t))^{\alpha_i} \ell_i(a, z; t)^{\beta_i} h_i(a, z; t)^{\gamma_i}.$$

Using the policy rules above and $\alpha_i + \beta_i + \gamma_i = 1$,

$$\begin{aligned} y_i(a, z; t) &= (z\lambda_i a)^{\alpha_i} \left[m_{\ell i}(t) z \lambda_i a \right]^{\beta_i} \left[m_{hi}(t) z \lambda_i a \right]^{\gamma_i} \\ &= m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} z \lambda_i a. \end{aligned}$$

Aggregating over active entrepreneurs,

$$Y_i(t) = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t).$$

Now use the aggregate input demand expressions

$$K_i(t) = \lambda_i A_i(t) n_i(t),$$

$$L_i(t) = m_{\ell i}(t) \lambda_i A_i(t) e_i(t),$$

and

$$H_i(t) = m_{hi}(t)\lambda_i A_i(t) e_i(t).$$

Then

$$\begin{aligned} K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} &= \left[\lambda_i A_i(t) n_i(t)\right]^{\alpha_i} \left[m_{\ell i}(t)\lambda_i A_i(t) e_i(t)\right]^{\beta_i} \left[m_{hi}(t)\lambda_i A_i(t) e_i(t)\right]^{\gamma_i} \\ &= m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) n_i(t)^{\alpha_i} e_i(t)^{\beta_i+\gamma_i}. \end{aligned}$$

Since $\beta_i + \gamma_i = 1 - \alpha_i$,

$$Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} = \left(\frac{e_i(t)}{n_i(t)}\right)^{\alpha_i} m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) n_i(t)^{\alpha_i} e_i(t)^{1-\alpha_i}.$$

Therefore,

$$Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t) = Y_i(t).$$

Thus, sectoral output can be written as

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i},$$

where

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i}.$$

Because $H_i(t) = \bar{H}_i$ is fixed in equilibrium, this can also be written as

$$Y_i(t) = \widetilde{Z}_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i}, \qquad \widetilde{Z}_i(t) \equiv Z_i(t) \bar{H}_i^{\gamma_i}.$$

Since

$$\alpha_i + \beta_i = 1 - \gamma_i < 1,$$

sectoral production exhibits decreasing returns in the scalable inputs $K_i(t)$ and $L_i(t)$.

C.3.3 Sectoral wealth law of motion

The reduced-form profit coefficient after optimizing over labor and the sector-specific factor is

$$\pi_i(t) = \alpha_i p_i^P(t) m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i}.$$

Equivalently,

$$\pi_i(t) = \alpha_i \left[p_i^P(t)\right]^{1/\alpha_i} \left(\frac{\beta_i}{w(t)}\right)^{\beta_i/\alpha_i} \left(\frac{\gamma_i}{q_i(t)}\right)^{\gamma_i/\alpha_i}.$$

The productivity cutoff satisfies

$$\pi_i(t)\underline{z}_i(t) = r(t) + \delta.$$

Using $s_i(z,t) = \Gamma_i(z,t) - \rho$, sectoral wealth growth is

$$\dot{A}_i(t) = A_i(t) \int \left[\lambda_i \max\{\pi_i(t) z - (r(t)+\delta), 0\} + r(t) - \rho\right] o_i(z,t)\, dz.$$

Splitting the integral at the cutoff,

$$\dot{A}_i(t) = \lambda_i \pi_i(t) A_i(t) e_i(t) - (r(t) + \delta) K_i(t) + (r(t) - \rho) A_i(t).$$

Since

$$Y_i(t) = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t),$$

and

$$\pi_i(t) = \alpha_i p_i^P(t) m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i},$$

we have

$$\lambda_i \pi_i(t) A_i(t) e_i(t) = \alpha_i p_i^P(t) Y_i(t).$$

Therefore,

$$\dot{A}_i(t) = \alpha_i p_i^P(t) Y_i(t) + r(t) \left[A_i(t) - K_i(t) \right] - \delta K_i(t) - \rho A_i(t).$$

The sectoral wealth law is thus unchanged in form relative to the baseline model, but $Y_i(t)$, $Z_i(t)$, and equilibrium profitability now depend on the fixed factor and its rental price.

C.3.4 Factor-price conditions

Labor price. From the Cobb-Douglas first-order condition for labor,

$$w(t) \ell_i(a, z; t) = \beta_i p_i^P(t) y_i(a, z; t).$$

Aggregating across entrepreneurs in sector i,

$$w(t) L_i(t) = \beta_i p_i^P(t) Y_i(t).$$

Using the aggregate production representation,

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i},$$

the real wage condition can be written as

$$\frac{w(t)}{p_i^P(t)} = \beta_i Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i - 1} H_i(t)^{\gamma_i}.$$

Specific-factor price. Similarly, the first-order condition for the sector-specific factor implies

$$q_i(t) h_i(a, z; t) = \gamma_i p_i^P(t) y_i(a, z; t).$$

Aggregating,

$$q_i(t) H_i(t) = \gamma_i p_i^P(t) Y_i(t).$$

Hence,

$$\frac{q_i(t)}{p_i^P(t)} = \gamma_i Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} H_i(t)^{\gamma_i - 1}.$$

Since $H_i(t) = \bar{H}_i$, this equation pins down the sector-specific rental rate $q_i(t)$ given aggregate sectoral allocations.

Interest-rate condition. The productivity cutoff satisfies

$$\pi_i(t)\underline{z}_i(t) = r(t) + \delta.$$

Using

$$\pi_i(t) = \alpha_i p_i^P(t) m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i},$$

and the aggregate production identity,

$$Y_i(t) = m_{\ell i}(t)^{\beta_i} m_{hi}(t)^{\gamma_i} \lambda_i A_i(t) e_i(t),$$

one can express the cutoff condition in terms of aggregate sectoral variables as

$$r(t) + \delta = p_i^P(t)\alpha_i \zeta_i(t) Z_i(t) K_i(t)^{\alpha_i - 1} L_i(t)^{\beta_i} H_i(t)^{\gamma_i},$$

where the selection wedge is

$$\zeta_i(t) \equiv \frac{\underline{z}_i(t)}{Z_i(t)^{1/\alpha_i}} = \frac{\underline{z}_i(t)}{\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]} \in [0,1].$$

Therefore,

$$r(t) = p_i^P(t)\alpha_i \zeta_i(t) Z_i(t) K_i(t)^{\alpha_i - 1} L_i(t)^{\beta_i} H_i(t)^{\gamma_i} - \delta.$$

Aggregate factor-price implications. Using labor-market clearing, $\sum_{i \in I} L_i(t) = \bar{L}$, and the labor payment condition, the aggregate wage satisfies

$$w(t) = \frac{1}{\bar{L}} \sum_{i \in I} \beta_i p_i^P(t) Y_i(t).$$

Using capital-market clearing, $K(t) = \sum_{i \in I} K_i(t)$, and the interest-rate condition, the aggregate interest rate satisfies

$$r(t) = \frac{1}{K(t)} \sum_{i \in I} \alpha_i p_i^P(t) \zeta_i(t) Y_i(t) - \delta.$$

Finally, the sector-specific factor price is pinned down sector by sector:

$$q_i(t) = \frac{\gamma_i p_i^P(t) Y_i(t)}{\bar{H}_i}.$$

Collecting the main aggregation results,

$$Y_i(t) = Z_i(t) K_i(t)^{\alpha_i} L_i(t)^{\beta_i} \bar{H}_i^{\gamma_i},$$

$$Z_i(t) = \left(\mathbb{E}_{o_i,t}[z \mid z \geq \underline{z}_i(t)]\right)^{\alpha_i},$$

and

$$\dot{A}_i(t) = \alpha_i p_i^P(t) Y_i(t) + r(t)\left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t).$$

C.3.5 Domestic-origin prices, tariffs, imports, and exports

In the open-economy environment, sectoral domestic absorption is interpreted as domestic use of a sectoral composite. For each tradable sector $i \in I^T$, let $D_i(t)$ denote domestic use: the quantity of sector i's composite used by resident households, firms, and the government, excluding exports. The sectoral composite combines a domestic-origin variety, $D_i^H(t)$, and an imported variety, $M_i(t)$, which are imperfect substitutes.

The domestic-use composite is given by the CES Armington aggregator

$$D_i(t) = \left[\varphi_i^{1/\eta_i^\star} \left(D_i^H(t)\right)^{(\eta_i^\star-1)/\eta_i^\star} + (1-\varphi_i)^{1/\eta_i^\star} M_i(t)^{(\eta_i^\star-1)/\eta_i^\star}\right]^{\eta_i^\star/(\eta_i^\star-1)}, \qquad i \in I^T,$$

where $\eta_i^\star > 0$ is the elasticity of substitution between domestic-origin and imported varieties, and $\varphi_i \in [0,1]$ is the Armington home-bias parameter. The Cobb-Douglas case $\eta_i^\star = 1$ is understood as the usual limiting case.

Let $p_i^*(t)$ denote the exogenous world price in sector i. Imports into the domestic economy are subject to the ad valorem tariff $\tau_i^M(t)$, while domestic exports face the ad valorem tariff $\tau_i^X(t)$ imposed abroad. The tariff-inclusive import price and the net-of-tariff price received by domestic exporters are, respectively,

$$p_i^M(t) = \left[1 + \tau_i^M(t)\right] p_i^*(t),$$

and

$$p_i^X(t) = \frac{p_i^*(t)}{1 + \tau_i^X(t)}.$$

The revenue generated by $\tau_i^M(t)$ accrues to the domestic economy, whereas the tariff on domestic exports is collected abroad.

Let $p_i^H(t)$ denote the domestic-origin price received by domestic producers. Given $p_i^H(t)$ and $p_i^M(t)$, the unit cost of the tradable domestic-use composite is

$$p_i^D(t) = \left[\varphi_i \left(p_i^H(t)\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^M(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, \qquad i \in I^T,$$

again with the Cobb-Douglas limiting expression when $\eta_i^\star = 1$. Cost minimization within the Armington nest implies

$$D_i^H(t) = \varphi_i \left(\frac{p_i^H(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T,$$

and

$$M_i(t) = (1-\varphi_i) \left(\frac{p_i^M(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T.$$

These demands satisfy the expenditure identity

$$p_i^D(t) D_i(t) = p_i^H(t) D_i^H(t) + p_i^M(t) M_i(t).$$

Domestic output is allocated between domestic-origin use and exports:

$$Y_i(t) = D_i^H(t) + X_i(t), \qquad i \in I^T.$$

Thus,

$$X_i(t) = Y_i(t) - D_i^H(t) \geq 0.$$

Because domestic producers can export at the net-of-tariff price $p_i^X(t)$, the domestic-origin price satisfies

$$p_i^H(t) \geq p_i^X(t),$$

together with the complementarity condition

$$\left[p_i^H(t) - p_i^X(t)\right] X_i(t) = 0.$$

If $X_i(t) > 0$, the sector exports and

$$p_i^H(t) = p_i^X(t).$$

If domestic-origin demand exhausts domestic output, exports are zero and the domestic-origin price may exceed the net export price:

$$p_i^H(t) > p_i^X(t) \quad \Longrightarrow \quad X_i(t) = 0.$$

This condition allows domestic scarcity to affect producer prices while preserving the discipline imposed by access to foreign markets.

For tradable sectors, the producer price used in the firm problem is

$$p_i^P(t) = p_i^H(t), \qquad i \in I^T.$$

When the sector exports, its producer price equals the net export price. When exports are zero, all domestic output is sold to domestic users at the domestic-origin price. Producer revenue can therefore be written as

$$p_i^P(t) Y_i(t) = p_i^H(t) Y_i(t)$$

in either case.

Sectoral net exports in quantities are

$$NX_i(t) = X_i(t) - M_i(t), \qquad i \in I^T.$$

Because $D_i(t)$ is a CES composite rather than a physical sum of domestic and imported units, $NX_i(t)$ is not generally equal to $Y_i(t) - D_i(t)$. Moreover, imports and exports need not have the same relevant transaction price. The value of sectoral trade accruing to the domestic economy is

$$p_i^X(t) X_i(t) - p_i^*(t) M_i(t).$$

Domestic import-tariff revenue in sector i is

$$T_i^M(t) = \left[p_i^M(t) - p_i^*(t)\right] M_i(t) = \tau_i^M(t) p_i^*(t) M_i(t).$$

The Armington expenditure identity, domestic-output feasibility, and export complementarity imply

$$p_i^H(t) Y_i(t) - p_i^D(t) D_i(t) = p_i^X(t) X_i(t) - p_i^M(t) M_i(t).$$

Adding domestic import-tariff revenue gives

$$p_i^H(t)Y_i(t) - p_i^D(t)D_i(t) + T_i^M(t) = p_i^X(t)X_i(t) - p_i^*(t)M_i(t).$$

Thus, tariff revenue reconciles domestic expenditure valued at the tariff-inclusive import price with external transactions valued at the amounts paid to foreign suppliers and received by domestic exporters.

For nontradable sectors $i \in I^N$, there is no world price or Armington nest. I set

$$M_i(t) = X_i(t) = NX_i(t) = 0, \qquad i \in I^N.$$

Their goods markets clear domestically:

$$Y_i(t) = D_i(t), \qquad i \in I^N,$$

and the producer and domestic-use prices coincide with the domestic market-clearing price:

$$p_i^P(t) = p_i^D(t) = p_i^N(t), \qquad i \in I^N.$$

In the quantitative implementation, φ_i is chosen to match the observed domestic-origin expenditure share conditional on baseline tariffs and prices. Let

$$s_i^{H,IO} \equiv 1 - \frac{M_i^{IO}}{D_i^{IO}},$$

where D_i^{IO} denotes total domestic-use expenditure in sector i, and M_i^{IO} denotes tariff-inclusive expenditure on imports used domestically. In the Peruvian input-output data described in Appendix E, D_i^{IO} corresponds to total demand net of exports and inventory changes.

At the baseline prices, the model-implied domestic-origin expenditure share is

$$s_i^H = \frac{\varphi_i \left(p_i^H\right)^{1-\eta_i^\star}}{\varphi_i \left(p_i^H\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^M\right)^{1-\eta_i^\star}}.$$

The parameter φ_i is chosen so that

$$s_i^H = s_i^{H,IO}.$$

Equivalently,

$$\varphi_i = \frac{s_i^{H,IO}\left(p_i^M\right)^{1-\eta_i^\star}}{(1-s_i^{H,IO})\left(p_i^H\right)^{1-\eta_i^\star} + s_i^{H,IO}\left(p_i^M\right)^{1-\eta_i^\star}}.$$

The resulting home-bias parameter is held fixed across tariff counterfactuals, so changes in trade flows arise from changes in the tariff wedges rather than changes in preferences.

C.3.6 Final composite demand with domestic-origin prices

The domestic final composite is produced from sectoral domestic-use composites:

$$D(t) = \left[\sum_{i\in I} \upsilon_i^{1/\eta} D_i(t)^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)}.$$

The sectoral domestic-use price is

$$p_i^D(t) = \begin{cases} \left[\varphi_i \left(p_i^H(t)\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^M(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, & i \in I^T, \\ p_i^N(t), & i \in I^N. \end{cases}$$

Thus, final users in tradable sectors face the unit cost of a composite of domestic-origin and imported varieties rather than either price separately.

Total domestic import-tariff revenue is

$$T^M(t) = \sum_{i\in I^T} T_i^M(t) = \sum_{i\in I^T} \tau_i^M(t) p_i^*(t) M_i(t).$$

This revenue is rebated lump-sum to a separate hand-to-mouth household and contributes to domestic absorption without entering entrepreneurs' wealth accounts. The tariff imposed abroad on domestic exports does not generate income for the domestic economy.

Total expenditure on sectoral domestic use is

$$E(t) = P^D(t)C(t) + \dot{A}(t) + \delta A(t).$$

Using aggregate factor payments, resource income, and the domestic tariff rebate, this expenditure can equivalently be written as

$$E(t) = w(t)\bar{L} + \sum_{i\in I} q_i(t)\bar{H}_i + \rho A(t) + NX_{\text{res}}(t) + T^M(t) + \dot{A}(t) + \delta A(t),$$

where

$$A(t) = \sum_{i\in I} A_i(t), \qquad \dot{A}(t) = \sum_{i\in I} \dot{A}_i(t).$$

Given the sectoral domestic-use prices, final demand across sectoral composites solves a standard CES expenditure problem. The corresponding price index is

$$P^D(t)^{1-\eta} = \sum_{i\in I} \upsilon_i \left[p_i^D(t)\right]^{1-\eta},$$

with the usual Cobb-Douglas limiting expression when $\eta = 1$. Sectoral domestic use is

$$D_i(t) = \upsilon_i \left(\frac{p_i^D(t)}{P^D(t)}\right)^{-\eta} \frac{E(t)}{P^D(t)}, \qquad i \in I.$$

For tradable sectors, domestic-origin use and imports are determined by the Armington de-

mand system derived in subsubsection C.3.5. The domestic-origin price adjusts subject to

$$p_i^H(t) \geq p_i^X(t), \qquad X_i(t) = Y_i(t) - D_i^H(t) \geq 0, \qquad \left[p_i^H(t) - p_i^X(t)\right] X_i(t) = 0.$$

For nontradable sectors, $p_i^N(t)$ adjusts so that

$$D_i(t) = Y_i(t).$$

Domestic scarcity therefore affects $p_i^H(t)$, which changes both the relative price of the sectoral domestic-use composite and the allocation between domestic-origin and imported varieties. Tariffs influence this adjustment through the import price $p_i^M(t)$, the net export price $p_i^X(t)$, and the domestic tariff rebate $T^M(t)$.

C.4 Equilibrium of the Quantitative Open Economy

Definition 9 (Small open economy equilibrium with nontradables, sector-specific factors, domestic-origin prices, and tariffs)**.** Let sectors be partitioned into tradable and nontradable sectors:

$$I = I^T \cup I^N, \qquad I^T \cap I^N = \emptyset.$$

Given initial wealth distributions $\{\omega_i(z,0)\}_{i\in I}$, initial wealth levels $\{A_i(0)\}_{i\in I}$, labor endowment $\bar{L}$, sector-specific factor endowments $\{\bar{H}_i\}_{i\in I}$, world-price sequences $\{p_i^*(t)\}_{i\in I^T, t\geq 0}$, domestic import-tariff sequences $\{\tau_i^M(t)\}_{i\in I^T, t\geq 0}$, foreign tariff sequences faced by domestic exports $\{\tau_i^X(t)\}_{i\in I^T, t\geq 0}$, Armington home-bias parameters $\{\varphi_i\}_{i\in I^T}$, Armington elasticities $\{\eta_i^\star\}_{i\in I^T}$, and a sequence of resource net exports $\{NX_{\text{res}}(t)\}_{t\geq 0}$, a competitive small open economy equilibrium with financial autarky is a path of factor prices $\{w(t), r(t), q_i(t)\}_{i\in I, t\geq 0}$, domestic-origin prices for tradable sectors $\{p_i^H(t)\}_{i\in I^T, t\geq 0}$, domestic goods prices for nontradable sectors $\{p_i^N(t)\}_{i\in I^N, t\geq 0}$, domestic-use prices $\{p_i^D(t)\}_{i\in I, t\geq 0}$, allocation rules $\{k_i(z,t), \ell_i(z,t), h_i(z,t), c_i(z,t)\}_{i\in I, t\geq 0}$, sectoral outputs, factor allocations, and wealth accumulation $\{Y_i(t), K_i(t), L_i(t), H_i(t), \dot{A}_i(t)\}_{i\in I, t\geq 0}$, sectoral domestic use $\{D_i(t)\}_{i\in I, t\geq 0}$, domestic-origin use, imports, exports, and net exports for tradable sectors $\{D_i^H(t), M_i(t), X_i(t), NX_i(t)\}_{i\in I^T, t\geq 0}$, domestic import-tariff revenue $\{T^M(t)\}_{t\geq 0}$, and wealth distributions $\{\omega_i(z,t)\}_{i\in I, t\geq 0}$, such that:

1. Entrepreneurs optimize given producer prices and factor prices. The producer price relevant for sector i is
$$p_i^P(t) = \begin{cases} p_i^H(t), & i \in I^T, \\ p_i^N(t), & i \in I^N. \end{cases}$$

2. Sectoral output is generated by the production technology with sector-specific factors:
$$Y_i(t) = A_i(t) \int y_i(z,t)\, \omega_i(z,t)\, dz,$$
where
$$y_i(z,t) = \left(z k_i(z,t)\right)^{\alpha_i} \ell_i(z,t)^{\beta_i} h_i(z,t)^{\gamma_i}, \qquad \alpha_i + \beta_i + \gamma_i = 1.$$

3. Factor markets clear:

$$\bar{L} = \sum_{i \in I} A_i(t) \int \ell_i(z,t)\, \omega_i(z,t)\, dz,$$
$$A(t) = \sum_{i \in I} A_i(t) \int k_i(z,t)\, \omega_i(z,t)\, dz,$$
$$\bar{H}_i = A_i(t) \int h_i(z,t)\, \omega_i(z,t)\, dz, \qquad \forall i \in I,$$

where

$$A(t) = \sum_{i \in I} A_i(t).$$

4. For each tradable sector, the tariff-inclusive import price and the net-of-tariff export price are

$$p_i^M(t) = \left[1 + \tau_i^M(t)\right] p_i^*(t), \qquad i \in I^T,$$

and

$$p_i^X(t) = \frac{p_i^*(t)}{1 + \tau_i^X(t)}, \qquad i \in I^T.$$

The tariff $\tau_i^M(t)$ is imposed domestically on imports, whereas $\tau_i^X(t)$ is imposed abroad on domestic exports.

5. Domestic-use prices are determined by the sectoral price structure. For tradable sectors,

$$p_i^D(t) = \left[\varphi_i \left(p_i^H(t)\right)^{1-\eta_i^\star} + (1-\varphi_i) \left(p_i^M(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, \qquad i \in I^T,$$

with the usual Cobb-Douglas limiting expression when $\eta_i^\star = 1$. For nontradable sectors,

$$p_i^D(t) = p_i^N(t), \qquad i \in I^N.$$

6. Domestic import-tariff revenue is

$$T^M(t) = \sum_{i \in I^T} \left[p_i^M(t) - p_i^*(t)\right] M_i(t) = \sum_{i \in I^T} \tau_i^M(t) p_i^*(t) M_i(t).$$

This revenue is rebated lump-sum to a separate hand-to-mouth household and contributes to domestic absorption without entering entrepreneurs' wealth accounts. The tariffs imposed abroad on domestic exports do not generate income for the domestic economy.

7. Aggregate expenditure on sectoral domestic use is

$$E(t) = w(t)\bar{L} + \sum_{i \in I} q_i(t) \bar{H}_i + \rho A(t) + NX_{\text{res}}(t) + T^M(t) + \dot{A}(t) + \delta A(t),$$

where

$$\dot{A}(t) = \sum_{i \in I} \dot{A}_i(t).$$

Given domestic-use prices, sectoral domestic use is determined by the CES final-composite demand system:

$$D(t) = \left[\sum_{i\in I} \upsilon_i^{1/\eta} D_i(t)^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)},$$

with domestic-use price index

$$P^D(t)^{1-\eta} = \sum_{i\in I} \upsilon_i \left[p_i^D(t)\right]^{1-\eta}.$$

Sectoral domestic use is therefore

$$D_i(t) = \upsilon_i \left(\frac{p_i^D(t)}{P^D(t)}\right)^{-\eta} \frac{E(t)}{P^D(t)}, \qquad i \in I.$$

8. For tradable sectors, domestic-use composites are allocated between domestic-origin and imported varieties according to the Armington demand system:

$$D_i^H(t) = \varphi_i \left(\frac{p_i^H(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T,$$

and

$$M_i(t) = (1-\varphi_i) \left(\frac{p_i^M(t)}{p_i^D(t)}\right)^{-\eta_i^\star} D_i(t), \qquad i \in I^T.$$

The corresponding expenditure identity is

$$p_i^D(t) D_i(t) = p_i^H(t) D_i^H(t) + p_i^M(t) M_i(t), \qquad i \in I^T.$$

9. Domestic output in tradable sectors is allocated between domestic-origin use and exports:

$$Y_i(t) = D_i^H(t) + X_i(t), \qquad i \in I^T.$$

Thus,

$$X_i(t) = Y_i(t) - D_i^H(t) \geq 0.$$

The domestic-origin price satisfies

$$p_i^H(t) \geq p_i^X(t), \qquad \left[p_i^H(t) - p_i^X(t)\right] X_i(t) = 0, \qquad i \in I^T.$$

Thus, if a tradable sector exports, its domestic-origin price equals the net export price. If its domestic-origin price exceeds the net export price, domestic output is entirely absorbed by domestic-origin use.

10. Sectoral net exports in quantities are

$$NX_i(t) = X_i(t) - M_i(t), \qquad i \in I^T.$$

Because $D_i(t)$ is a CES composite quantity, $NX_i(t)$ is not generally equal to $Y_i(t) - D_i(t)$. The value identity implied by the Armington demand system and the export complementarity

condition is

$$p_i^H(t)Y_i(t) - p_i^D(t)D_i(t) = p_i^X(t)X_i(t) - p_i^M(t)M_i(t), \qquad i \in I^T.$$

Equivalently, after adding domestic import-tariff revenue,

$$p_i^H(t)Y_i(t) - p_i^D(t)D_i(t) + T_i^M(t) = p_i^X(t)X_i(t) - p_i^*(t)M_i(t), \qquad i \in I^T,$$

where

$$T_i^M(t) = \tau_i^M(t)p_i^*(t)M_i(t).$$

11. Nontradable sectors satisfy

$$M_i(t) = X_i(t) = NX_i(t) = 0, \qquad i \in I^N.$$

Their goods markets clear domestically:

$$Y_i(t) = D_i(t), \qquad i \in I^N.$$

12. External balance holds for the complete economy:

$$\sum_{i \in I^T} \left[p_i^X(t)X_i(t) - p_i^*(t)M_i(t)\right] + NX_{\text{res}}(t) = 0.$$

Export receipts are valued at the net price received by domestic producers, whereas imports are valued at the world price paid to foreign suppliers. Domestic import-tariff revenue does not enter this condition because it is an internal transfer.

13. Sectoral wealth evolves according to individual optimization:

$$\dot{A}_i(t) = \alpha_i p_i^P(t)Y_i(t) + r(t)\left[A_i(t) - K_i(t)\right] - \delta K_i(t) - \rho A_i(t), \qquad i \in I.$$

14. Wealth distributions evolve consistently with individual wealth accumulation and the productivity process.

When tariffs are zero,

$$\tau_i^M(t) = \tau_i^X(t) = 0,$$

the tariff-adjusted prices satisfy

$$p_i^M(t) = p_i^X(t) = p_i^*(t),$$

and the equilibrium reduces to the quantitative open-economy environment without tariffs.

More generally, if a tradable sector exports positive amounts, then $X_i(t) > 0$ and the complementarity condition implies

$$p_i^H(t) = p_i^X(t).$$

The domestic-origin price then equals the net price available to domestic exporters. If domestic-origin demand exhausts domestic output, exports are zero and $p_i^H(t)$ may exceed $p_i^X(t)$. The resulting scarcity premium raises producer revenue and the domestic-use price $p_i^D(t)$, reducing demand for the sectoral composite and shifting demand towards the imported variety.

The closed economy and the fully tradable small open economy are special cases of this definition. The closed economy corresponds to $I^N = I$, $I^T = \emptyset$, and $NX_{\text{res}}(t) = 0$. In that case, all goods prices are determined domestically, tariff revenue is zero, and goods markets satisfy

$$Y_i(t) = D_i(t), \qquad i \in I.$$

The fully tradable small open economy corresponds to $I^T = I$ and $I^N = \emptyset$. In that case, world prices and tariffs determine the import and export parity prices in every sector, while domestic-origin prices equal net export prices whenever exports are positive and may exceed them when domestic-origin demand exhausts domestic output. The quantitative application uses the intermediate case with both tradable and nontradable sectors.

The resource-income term does not represent foreign borrowing or an exogenous current-account transfer. It is the net external income generated by a domestic sector omitted from the entrepreneurial production block. Financial autarky therefore continues to hold for the economy as a whole. Resource income and domestic import-tariff revenue are rebated to the hand-to-mouth household, affecting modeled sectors through aggregate demand, prices, wages, and factor rents without directly entering entrepreneurs' wealth accumulation equations. Tariffs imposed abroad on domestic exports generate no revenue for the domestic economy.

D Computation of Equilibrium and Dynamics

Equilibrium objects are computed in two steps at each point in time. First, conditional on the current distribution of wealth and productivity, I solve for the within-period equilibrium allocation. This fixed-point problem determines wages, interest rates, sector-specific factor rents, and the relevant goods prices, with the set of endogenous prices depending on the price regime. In the closed economy, all goods prices adjust to clear domestic goods markets. In the small open economy, world prices for tradable goods are taken as exogenous. In the quantitative open economy, exogenous world prices and sector-specific tariffs determine the prices relevant for imports and exports. Nontradable prices adjust to clear domestic markets, while tradable domestic-origin prices adjust subject to the net-export-price floor and export complementarity conditions. Sector-specific factor rents clear fixed-factor markets.

Domestic absorption is allocated across sectoral composites using a CES final-demand system, while tradable-sector domestic use is further allocated between domestic-origin and imported varieties through Armington nests. Domestic import-tariff revenue is rebated to the hand-to-mouth household and included in aggregate absorption. This step delivers factor prices, producer prices, domestic-use prices, sectoral output, domestic use, domestic-origin use, gross imports and exports, tariff revenue, and the implied sectoral wealth dynamics at time t.

Second, given these equilibrium objects, I update the distribution of wealth shares by solving the associated partial differential equation using a finite-difference approximation. The following subsections describe each step in detail.

D.1 Equilibrium at point t

D.1.1 Closed Economy

Given sectoral wealth $\{A_i(t)\}$ and within-sector wealth distributions $\{o_i(z,t)\}$, the task is to find prices $\{p_i(t)\}_{i\in I}$, wage $w(t)$, and interest rate $r(t)$ such that all markets clear.

Prices are normalized so that the CES price index satisfies $P(t) = 1$.
The equilibrium is computed as a fixed point using the following steps:

1. **Initial guesses.** Guess $(w^0, r^0, \{p_i^0\})$, and normalize prices so that $P^0 = 1$.

2. **Static firm block.** Given prices, compute:

   - Profits: $\pi_i^0 = p_i^0 \alpha_i \left(\frac{1-\alpha_i}{w^0/p_i^0}\right)^{\frac{1-\alpha_i}{\alpha_i}}$
   - Productivity cutoffs: $\underline{z}_i^0 = \frac{r^0+\delta}{\pi_i^0}$
   - Active-productivity integral: $X_i^0 = \int_{\underline{z}_i^0}^{\infty} z\, o_i(z,t)\, dz$
   - Active mass: $1 - \Omega_i(\underline{z}_i^0)$
   - Sectoral productivity: $Z_i^0 = \left(\frac{X_i^0}{1-\Omega_i(\underline{z}_i^0)}\right)^{\alpha_i}$
   - Selection wedge: $\zeta_i^0 = \frac{\underline{z}_i^0}{(Z_i^0)^{1/\alpha_i}}$

3. **Factor demands and output.**

   - Labor demand: $L_i^0 = \lambda_i A_i(t)\, X_i^0 \left(\frac{\pi_i^0}{p_i^0 \alpha_i}\right)^{\frac{1}{1-\alpha_i}}$
   - Capital demand: $K_i^0 = \lambda_i A_i(t)\, [1 - \Omega_i(\underline{z}_i^0)]$
   - Output: $Y_i^0 = Z_i^0 (K_i^0)^{\alpha_i} (L_i^0)^{1-\alpha_i}$

4. **Wealth accumulation and absorption.**

   - Sectoral wealth dynamics: $\dot{A}_i^0 = \alpha_i p_i^0 Y_i^0 + (r^0 - \rho) A_i(t) - (r^0 + \delta) K_i^0$
   - Total absorption: $P^0 D^0 = w^0 \bar{L} + \rho A(t) + \dot{A}(t) + \delta A(t)$
   - Sectoral demand: $D_i^0 = \upsilon_i \left(\frac{p_i^0}{P^0}\right)^{-\eta} D^0$

5. **Excess demands.**

   - Goods: $G_i^0 = Y_i^0 - D_i^0$
   - Labor: $L_{\text{diff}}^0 = \frac{\sum_i L_i^0 - \bar{L}}{\bar{L}}$
   - Capital: $K_{\text{diff}}^0 = \frac{\sum_i K_i^0 - A(t)}{A(t)}$

6. **Update prices.**

   - Wage (labor clearing): $w^1 = w^0 \left(1 + \gamma_w L_{\text{diff}}^0\right)$
   - Interest rate (capital clearing): $r^1 = r^0 + \gamma_r K_{\text{diff}}^0$
   - Goods prices (goods clearing): $p_i^1 = p_i^0 \left(1 - \gamma_p \frac{G_i^0}{Y_i^0}\right)$
   - Normalize prices so that $P^1 = 1$

7. **Iteration.** Repeat until convergence: $\max\{|L_{\text{diff}}|, |K_{\text{diff}}|, \max_i |G_i/Y_i|\} < \varepsilon$

D.1.2 Small Open Economy with Financial Autarky

Given sectoral wealth $\{A_i(t)\}$, within-sector wealth distributions $\{o_i(z,t)\}$, and exogenous goods prices $\{p_i^*\}_{i\in I}$, the task is to find the wage $w(t)$ and interest rate $r(t)$ such that factor markets clear.

Since goods prices are fixed, goods market clearing does not pin down prices, and the equilibrium reduces to clearing labor and capital markets given sectoral profitability.

The equilibrium is computed as follows:

1. **Initial guesses.** Guess (w^0, r^0).
2. **Static firm block.** Given (w^0, r^0) and world prices $\{p_i^*\}$, compute:
   - Profits: $\pi_i^0 = p_i^* \alpha_i \left(\frac{1-\alpha_i}{w^0/p_i^*}\right)^{\frac{1-\alpha_i}{\alpha_i}}$
   - Productivity cutoffs: $\underline{z}_i^0 = \frac{r^0+\delta}{\pi_i^0}$
   - Active-productivity integral: $X_i^0 = \int_{\underline{z}_i^0}^{\infty} z\, o_i(z,t)\, dz$
   - Active mass: $1 - \Omega_i(\underline{z}_i^0)$
   - Sectoral productivity: $Z_i^0 = \left(\frac{X_i^0}{1-\Omega_i(\underline{z}_i^0)}\right)^{\alpha_i}$
   - Selection wedge: $\zeta_i^0 = \frac{\underline{z}_i^0}{(Z_i^0)^{1/\alpha_i}}$
3. **Factor demands and output.**
   - Labor demand: $L_i^0 = \lambda_i A_i(t)\, X_i^0 \left(\frac{\pi_i^0}{p_i^* \alpha_i}\right)^{\frac{1}{1-\alpha_i}}$
   - Capital demand: $K_i^0 = \lambda_i A_i(t)\, [1 - \Omega_i(\underline{z}_i^0)]$
   - Output: $Y_i^0 = Z_i^0 (K_i^0)^{\alpha_i} (L_i^0)^{1-\alpha_i}$
4. **Factor market residuals.**
   - Labor: $L_{\text{diff}}^0 = \frac{\sum_i L_i^0 - \bar{L}}{\bar{L}}$
   - Capital: $K_{\text{diff}}^0 = \frac{\sum_i K_i^0 - A(t)}{A(t)}$
5. **Update factor prices.**
   - Wage: $w^1 = w^0 \left(1 + \gamma_w L_{\text{diff}}^0\right)$
   - Interest rate: $r^1 = r^0 + \gamma_r K_{\text{diff}}^0$
6. **Iteration.** Repeat until convergence: $\max\{|L_{\text{diff}}|, |K_{\text{diff}}|\} < \varepsilon$
7. **Post-solution objects.**

   Given equilibrium (w, r), compute:
   - Sectoral wealth dynamics: $\dot{A}_i(t) = \alpha_i p_i^* Y_i(t) + (r(t) - \rho) A_i(t) - (r(t) + \delta) K_i(t)$
   - Total absorption: $P^*(t) D(t) = w(t)\bar{L} + \rho A(t) + \dot{A}(t) + \delta A(t)$

- Sectoral absorption: $D_i(t) = \upsilon_i \left(\frac{p_i^*}{P^*(t)}\right)^{-\eta} D(t)$
- Net exports (residual): $NX_i(t) = Y_i(t) - D_i(t)$

The quantitative application uses a richer hybrid environment than the two benchmark cases above. The next subsection describes the fixed-point procedure used for this environment.

D.1.3 Quantitative Open Economy

The quantitative implementation combines small-open-economy price discipline for international transactions with endogenous domestic-origin prices for tradable goods used domestically. Sectors are partitioned into tradable and nontradable sectors:

$$I = I^T \cup I^N, \qquad I^T \cap I^N = \emptyset.$$

For each tradable sector, the world-price sequence $\{p_i^*(t)\}_{i\in I^T}$, the domestic import-tariff sequence $\{\tau_i^M(t)\}_{i\in I^T}$, and the foreign tariff sequence faced by domestic exports $\{\tau_i^X(t)\}_{i\in I^T}$ are exogenous. The tariff-inclusive import price and the net-of-tariff export price are

$$p_i^M(t) = \left[1 + \tau_i^M(t)\right] p_i^*(t), \qquad i \in I^T,$$

and

$$p_i^X(t) = \frac{p_i^*(t)}{1 + \tau_i^X(t)}, \qquad i \in I^T.$$

Domestic producers in tradable sectors receive a domestic-origin price $p_i^H(t)$, which is bounded below by $p_i^X(t)$ and may exceed it when domestic-origin goods are scarce. For nontradable sectors, prices $p_i^N(t)$ adjust endogenously to clear domestic markets. The economy also includes sector-specific factors in fixed supply $\{\bar{H}_i\}_{i\in I}$, Armington home-bias parameters $\{\varphi_i\}_{i\in I^T}$, Armington elasticities $\{\eta_i^\star\}_{i\in I^T}$, and omitted resource-sector net exports $NX_{\text{res}}(t)$.

Given sectoral wealth $\{A_i(t)\}$, within-sector wealth distributions $\{o_i(z,t)\}$, world prices, tariffs, and current resource net exports, the task is to find the wage $w(t)$, the interest rate $r(t)$, sector-specific factor rents $\{q_i(t)\}_{i\in I}$, domestic-origin prices $\{p_i^H(t)\}_{i\in I^T}$, and nontradable prices $\{p_i^N(t)\}_{i\in I^N}$. At the solution, factor markets and nontradable goods markets clear, domestic-origin price complementarity holds in tradable sectors, and external balance holds for the complete economy.

For notational clarity, let $n_i(t)$ denote the active wealth share and $e_i(t)$ the productivity-weighted active wealth share:

$$n_i(t) = 1 - \Omega_i(\underline{z}_i(t)), \qquad e_i(t) = \int_{\underline{z}_i(t)}^{\infty} z o_i(z,t)\, dz.$$

This avoids using $X_i(t)$ for both an active-productivity integral and gross exports.

The equilibrium at each point in time is computed as a fixed point using the following steps:

1. **Tariff-adjusted prices and initial guesses.**

   Construct

   $$p_i^M(t) = \left[1 + \tau_i^M(t)\right] p_i^*(t), \qquad i \in I^T,$$

and

$$p_i^X(t) = \frac{p_i^*(t)}{1+\tau_i^X(t)}, \qquad i \in I^T.$$

Guess

$$\left(w^0, r^0, \{q_i^0\}_{i\in I}, \{p_i^{H,0}\}_{i\in I^T}, \{p_i^{N,0}\}_{i\in I^N}\right),$$

with

$$p_i^{H,0} \geq p_i^X(t), \qquad i \in I^T.$$

The producer price in each sector is

$$p_i^{P,0} = \begin{cases} p_i^{H,0}, & i \in I^T, \\ p_i^{N,0}, & i \in I^N. \end{cases}$$

The domestic-use price is

$$p_i^{D,0} = \begin{cases} \left[\varphi_i \left(p_i^{H,0}\right)^{1-\eta_i^\star} + (1-\varphi_i)\left(p_i^M(t)\right)^{1-\eta_i^\star}\right]^{1/(1-\eta_i^\star)}, & i \in I^T, \\ p_i^{N,0}, & i \in I^N, \end{cases}$$

with the usual Cobb-Douglas limiting expression when $\eta_i^\star = 1$.

2. **Static firm block.**
   Given $p_i^{P,0}$, w^0, r^0, and q_i^0, compute the reduced-form profit coefficient

$$\pi_i^0 = \alpha_i \left(p_i^{P,0}\right)^{1/\alpha_i} \left(\frac{\beta_i}{w^0}\right)^{\beta_i/\alpha_i} \left(\frac{\gamma_i}{q_i^0}\right)^{\gamma_i/\alpha_i}.$$

   The productivity cutoff is

$$\underline{z}_i^0 = \frac{r^0+\delta}{\pi_i^0}.$$

   Then compute

$$n_i^0 = 1 - \Omega_i(\underline{z}_i^0), \qquad e_i^0 = \int_{\underline{z}_i^0}^{\infty} z o_i(z,t)\, dz,$$

   and sectoral productivity

$$Z_i^0 = \left(\frac{e_i^0}{n_i^0}\right)^{\alpha_i}.$$

3. **Input demands and output.**
   The input demand coefficients are

$$m_{\ell i}^0 = \left[\frac{p_i^{P,0}\beta_i}{w^0}\left(\frac{\gamma_i w^0}{\beta_i q_i^0}\right)^{\gamma_i}\right]^{1/\alpha_i},$$

and

$$m_{hi}^0 = \left[\frac{p_i^{P,0}\gamma_i}{q_i^0} \left(\frac{\beta_i q_i^0}{\gamma_i w^0} \right)^{\beta_i} \right]^{1/\alpha_i}.$$

Sectoral factor demands are

$$K_i^0 = \lambda_i A_i(t) n_i^0,$$
$$L_i^0 = m_{\ell i}^0 \lambda_i A_i(t) e_i^0,$$

and

$$H_i^{d,0} = m_{hi}^0 \lambda_i A_i(t) e_i^0.$$

Sectoral output is

$$Y_i^0 = Z_i^0 (K_i^0)^{\alpha_i} (L_i^0)^{\beta_i} (H_i^{d,0})^{\gamma_i}.$$

4. **Wealth accumulation and expenditure before the tariff rebate.**

   Sectoral wealth accumulation is

$$\dot{A}_i^0 = \alpha_i p_i^{P,0} Y_i^0 + (r^0 - \rho) A_i(t) - (r^0 + \delta) K_i^0.$$

   Let

$$A(t) = \sum_{i \in I} A_i(t), \qquad \dot{A}^0 = \sum_{i \in I} \dot{A}_i^0.$$

   Omitted resource net exports are treated as exogenous income rebated to a hand-to-mouth resource owner. In calibrated exercises, I write

$$NX_{\text{res}}^0(t) = \chi_{\text{res}}(t) \sum_{i \in I} p_i^{P,0} Y_i^0,$$

   where $\chi_{\text{res}}(t)$ is the resource net-export share.

   Define expenditure before the domestic import-tariff rebate as

$$B^0 = w^0 \bar{L} + \sum_{i \in I} q_i^0 \bar{H}_i + \rho A(t) + NX_{\text{res}}^0(t) + \dot{A}^0 + \delta A(t).$$

5. **Domestic import-tariff revenue and final demand.**

   Given domestic-use prices, first compute the price index

$$(P^{D,0})^{1-\eta} = \sum_{i \in I} \upsilon_i \left(p_i^{D,0} \right)^{1-\eta},$$

   with the usual Cobb-Douglas limiting expression when $\eta = 1$.

   The expenditure share of sectoral composite i is

$$s_i^{D,0} \equiv \frac{p_i^{D,0} D_i^0}{E^0} = \upsilon_i \left(\frac{p_i^{D,0}}{P^{D,0}} \right)^{1-\eta}.$$

For each tradable sector, the import expenditure share within the Armington composite is

$$s_i^{M,0} \equiv \frac{p_i^M(t) M_i^0}{p_i^{D,0} D_i^0} = (1-\varphi_i) \left(\frac{p_i^M(t)}{p_i^{D,0}} \right)^{1-\eta_i^\star}.$$

Domestic import-tariff revenue as a share of total expenditure is therefore

$$\theta_T^0 = \sum_{i \in I^T} \frac{\tau_i^M(t)}{1+\tau_i^M(t)} s_i^{M,0} s_i^{D,0}.$$

Because tariff revenue is rebated to the hand-to-mouth household,

$$E^0 = B^0 + T^{M,0}, \qquad T^{M,0} = \theta_T^0 E^0.$$

Hence total expenditure and tariff revenue can be computed directly as

$$E^0 = \frac{B^0}{1-\theta_T^0},$$

and

$$T^{M,0} = \frac{\theta_T^0}{1-\theta_T^0} B^0 = \theta_T^0 E^0.$$

Aggregate domestic use is

$$D^0 = \frac{E^0}{P^{D,0}},$$

and sectoral domestic use is

$$D_i^0 = \upsilon_i \left(\frac{p_i^{D,0}}{P^{D,0}} \right)^{-\eta} D^0, \qquad i \in I.$$

6. **Armington allocation and gross trade flows.**

   For tradable sectors, allocate sectoral domestic use between domestic-origin and imported varieties:

$$D_i^{H,0} = \varphi_i \left(\frac{p_i^{H,0}}{p_i^{D,0}} \right)^{-\eta_i^\star} D_i^0, \qquad i \in I^T,$$

   and

$$M_i^0 = (1-\varphi_i) \left(\frac{p_i^M(t)}{p_i^{D,0}} \right)^{-\eta_i^\star} D_i^0, \qquad i \in I^T.$$

   The corresponding expenditure identity is

$$p_i^{D,0} D_i^0 = p_i^{H,0} D_i^{H,0} + p_i^M(t) M_i^0.$$

Domestic import-tariff revenue by sector is

$$T_i^{M,0} = \left[p_i^M(t) - p_i^*(t)\right] M_i^0 = \tau_i^M(t) p_i^*(t) M_i^0.$$

The resulting revenue satisfies

$$T^{M,0} = \sum_{i \in I^T} T_i^{M,0}.$$

Domestic output is allocated between domestic-origin use and exports:

$$X_i^0 = Y_i^0 - D_i^{H,0}, \qquad i \in I^T.$$

Sectoral net exports in quantities are

$$NX_i^0 = X_i^0 - M_i^0, \qquad i \in I^T.$$

For nontradable sectors,

$$M_i^0 = X_i^0 = NX_i^0 = T_i^{M,0} = 0.$$

7. **Equilibrium residuals.**

The factor-market residuals are

$$L_{\text{diff}}^0 = \frac{\sum_{i \in I} L_i^0 - \bar{L}}{\bar{L}},$$

$$K_{\text{diff}}^0 = \frac{\sum_{i \in I} K_i^0 - A(t)}{A(t)},$$

and

$$H_{\text{diff},i}^0 = \frac{H_i^{d,0} - \bar{H}_i}{\bar{H}_i}, \qquad i \in I.$$

For nontradable sectors, goods-market residuals are

$$G_i^{N,0} = Y_i^0 - D_i^0, \qquad i \in I^N.$$

For tradable sectors, the domestic-origin residual is

$$G_i^{H,0} = D_i^{H,0} - Y_i^0, \qquad i \in I^T.$$

A positive value of $G_i^{H,0}$ indicates excess demand for the domestic-origin variety. A negative value indicates excess domestic output, which can be exported if the domestic-origin price is at the net-export-price floor.

The external-balance residual is

$$TB^0 = \sum_{i \in I^T} \left[p_i^X(t) X_i^0 - p_i^*(t) M_i^0\right] + NX_{\text{res}}^0(t).$$

In a feasible equilibrium, this residual is zero up to numerical error. At an equilibrium satis-

fying export complementarity, the value identity also implies

$$TB^0 = \sum_{i \in I^T} \left[p_i^{H,0} Y_i^0 - p_i^{D,0} D_i^0 + T_i^{M,0} \right] + N X_{\text{res}}^0(t).$$

8. **Update factor prices, nontradable prices, and domestic-origin prices.**

   Factor prices are updated using damped fixed-point steps:

$$w^1 = w^0 \exp\left(\gamma_w L_{\text{diff}}^0\right),$$

$$r^1 = r^0 + \gamma_r K_{\text{diff}}^0,$$

   and

$$q_i^1 = q_i^0 \exp\left(\gamma_q H_{\text{diff},i}^0\right).$$

   Nontradable prices are updated using the nontradable goods-market residuals:

$$p_i^{N,1} = p_i^{N,0} \exp\left(-\gamma_N \frac{G_i^{N,0}}{\max\{Y_i^0, D_i^0\}} \right), \qquad i \in I^N.$$

   If $Y_i^0 > D_i^0$, there is excess supply of the nontradable good and its price falls. If $Y_i^0 < D_i^0$, there is excess demand and its price rises.

   For tradable sectors, domestic-origin prices are updated using the domestic-origin residual:

$$p_i^{H,1} = \max\left\{ p_i^X(t), p_i^{H,0} \exp\left(\gamma_H \frac{D_i^{H,0} - Y_i^0}{\max\{Y_i^0, D_i^{H,0}\}} \right) \right\}, \qquad i \in I^T.$$

   If domestic-origin demand exceeds domestic output, $D_i^{H,0} > Y_i^0$, the domestic-origin price rises. If domestic output exceeds domestic-origin demand, $D_i^{H,0} < Y_i^0$, the domestic-origin price falls, but never below $p_i^X(t)$. When $p_i^{H,0} = p_i^X(t)$ and domestic output exceeds domestic-origin demand, the surplus is exported.

9. **Iteration.**

   Repeat the previous steps until factor-market residuals, nontradable goods-market residuals, and tradable domestic-origin complementarity residuals are sufficiently small. The tradable complementarity conditions are

$$p_i^H(t) \geq p_i^X(t), \qquad X_i(t) = Y_i(t) - D_i^H(t) \geq 0, \qquad \left[p_i^H(t) - p_i^X(t)\right] X_i(t) = 0.$$

   A normalized residual for the tradable domestic-origin block is

$$R_i^H = \frac{\max\{-X_i(t), 0\}}{\max\{Y_i(t), D_i^H(t)\}} + \frac{\max\{p_i^X(t) - p_i^H(t), 0\}}{p_i^X(t)} + \frac{\left[p_i^H(t) - p_i^X(t)\right] \max\{X_i(t), 0\}}{p_i^X(t) \max\{Y_i(t), D_i^H(t)\}}, \qquad i \in I^T.$$

The fixed point is declared converged when

$$\max\left\{|L_{\text{diff}}|, |K_{\text{diff}}|, \max_i |H_{\text{diff},i}|, \max_{i\in I^N}\left|\frac{Y_i - D_i}{\max\{Y_i, D_i\}}\right|, \max_{i\in I^T} R_i^H\right\} < \varepsilon.$$

The external-balance residual and the tariff-revenue identity are reported as diagnostics. When factor markets clear, nontradable goods markets clear, and the domestic-origin complementarity conditions hold, aggregate external balance follows from the absorption budget, the resource net-export term, the domestic tariff rebate, and the value identity linking producer revenue, domestic absorption expenditure, and international transactions.

10. **Post-solution objects.**

    At the converged prices, recompute all equilibrium quantities and verify

$$T^M(t) = \sum_{i\in I^T} \tau_i^M(t) p_i^*(t) M_i(t),$$

and

$$\sum_{i\in I^T} \left[p_i^X(t) X_i(t) - p_i^*(t) M_i(t)\right] + NX_{\text{res}}(t) = 0$$

up to numerical tolerance.

Record sectoral wealth growth $\{\dot{A}_i(t)\}$, domestic use $\{D_i(t)\}$, domestic-origin use $\{D_i^H(t)\}$, gross imports $\{M_i(t)\}$, gross exports $\{X_i(t)\}$, net exports $\{NX_i(t)\}$, domestic tariff revenue $T^M(t)$, and the saving policy

$$s_i(z,t) = \lambda_i \max\{\pi_i(t) z - (r(t) + \delta), 0\} + r(t) - \rho.$$

These objects are then used to update the distribution of sectoral wealth shares through the finite-difference approximation described below.

D.2 Solving the PDE via Approximation

1. Write equation B.1 as a generic second order PDE:

$$\frac{\partial \omega_i(z,t)}{\partial t} = u_i(z,t)\omega_i(z,t) + v_i(z)\frac{\partial \omega_i(z,t)}{\partial z} + x_i(z)\frac{\partial^2 \omega_i(z,t)}{\partial z^2} \tag{D.1}$$

2. Apply forward-difference approximation in time dimension and central-difference approximations in the productivity domain dimension:

$$\begin{aligned}
\frac{\partial \omega_i(z_k,t^n)}{\partial t} &\approx \frac{\omega_i(z_k,t^{n+1}) - \omega_i(z_k,t^n)}{\Delta t} \\
\frac{\partial \omega_i(z_k,t^n)}{\partial z} &\approx \frac{\omega_i(z_{k+1},t^n) - \omega_i(z_{k-1},t^n)}{2\Delta z} \\
\frac{\partial^2 \omega_i(z_k,t^n)}{\partial z^2} &\approx \frac{\omega_i(z_{k+1},t^n) - 2\omega_i(z_k,t^n) + \omega_i(z_{k-1},t^n)}{(\Delta z)^2}
\end{aligned}$$

where $\{z_k\}_{k=1}^K$ are the elements of the discrete grid for z and (t^n) is the discretized time

sequence such that $t^{n+1} = t^n + \Delta t$.

3. Using these to approximate (D.1) in an implicit scheme (right-hand side evaluated in $n+1$):

$$\begin{aligned}\frac{\omega_i(z_k, t^{n+1}) - \omega_i(z_k, t^n)}{\Delta t} &= u_i(z_k, t)\omega_i(z, t^n) \\ &+ v_i(z_k)\frac{\omega_i(z_{k+1}, t^{n+1}) - \omega_i(z_{k-1}, t^{n+1})}{2\Delta z} \\ &+ x_i(z_k)\frac{\omega_i(z_{k+1}, t^{n+1}) - 2\omega_i(z_k, t^{n+1}) + \omega_i(z_{k-1}, t^{n+1})}{(\Delta z)^2}\end{aligned} \tag{D.2}$$

Rearranging:

$$\omega_i(z_{k-1}, t^{n+1})f_i(z_k) + \omega_i(z_k, t^{n+1})g_i(z_k) + \omega_i(z_{k+1}, t^{n+1})h_i(z_k) = \omega_i(z_k, t^n) \tag{D.3}$$

Where $f_i(z_k)$, $g_i(z_k)$, and $h_i(z_k)$ are the following functions of $a_i(z_k, t^n)$, $b_i(z_k)$, $c_i(z_k)$, Δt, and Δz:

$$\begin{aligned}f_i(z_k) &= v_i(z_k)\frac{\Delta t}{2\Delta z} - x_i(z_k)\frac{\Delta t}{(\Delta z)^2} \\ g_i(z_k) &= 1 - u_i(z_k)\Delta t + 2x_i(z_k)\frac{\Delta t}{(\Delta z)^2} \\ h_i(z_k) &= -v_i(z_k)\frac{\Delta t}{2\Delta z} - x_i(z_k)\frac{\Delta t}{(\Delta z)^2}\end{aligned}$$

4. With boundary conditions $\omega_i(z_1, t^{n+1}) = 0$ and $\Delta z \sum_{k=1}^{K} \omega_i(z_k, t^{n+1}) = O_i(t^{n+1})$. The problem now becomes:

$$\begin{bmatrix} g_i(z_2) & h_i(z_2) & 0 & 0 & \dots & 0 \\ f_i(z_3) & g_i(z_3) & h_i(z_3) & 0 & \dots & 0 \\ 0 & f_i(z_4) & g_i(z_4) & h_i(z_4) & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ \dots & \dots & f_i(z_{K-2}) & g_i(z_{K-2}) & h_i(z_{K-2}) & 0 \\ \dots & \dots & \dots & f_i(z_{K-1}) & g_i(z_{K-1}) & h_i(z_{K-1}) \\ \Delta z & \Delta z & \dots & \dots & \Delta z & \Delta z \end{bmatrix} \begin{bmatrix} \omega_i(z_2, t^{n+1}) \\ \omega_i(z_3, t^{n+1}) \\ \omega_i(z_4, t^{n+1}) \\ \vdots \\ \omega_i(z_{K-2}, t^{n+1}) \\ \omega_i(z_{K-1}, t^{n+1}) \\ \omega_i(z_K, t^{n+1}) \end{bmatrix} = \begin{bmatrix} \omega_i(z_2, t^n) \\ \omega_i(z_3, t^n) \\ \omega_i(z_4, t^n) \\ \vdots \\ \omega_i(z_{K-2}, t^n) \\ \omega_i(z_{K-1}, t^n) \\ O_i(t^{n+1}) \end{bmatrix}$$

Which can be solved for $\omega_i(t^{n+1})$ by inverting the matrix in the left.

D.2.1 Specific Solution

From the OU process for log productivity described in Appendix E, subsection E.1,

$$dz_i = \frac{1}{\theta_i}\left(\bar{\mu}_i - \log z_i + \frac{\sigma_i^2}{2}\right) z_i\, dt + \sigma_i \sqrt{\frac{1}{\theta_i}} z_i\, dW.$$

Thus,

$$\mu_i(z) = \frac{1}{\theta_i}\left(\bar{\mu}_i - \log z + \frac{\sigma_i^2}{2}\right) z, \qquad \sigma_i(z) = \sigma_i \sqrt{\frac{1}{\theta_i}} z.$$

Plugging these expressions into the wealth-share dynamics equation gives

$$\begin{aligned}\frac{\partial \omega_i(z,t)}{\partial t} = & \left[s_i(z,t) - \frac{\dot{A}(t)}{A(t)} + \frac{1}{\theta_i}\left(1 - \bar{\mu}_i + \log z + \frac{\sigma_i^2}{2}\right)\right]\omega_i(z,t) \\ & + \frac{1}{\theta_i}\left(\log z - \bar{\mu}_i + \frac{3}{2}\sigma_i^2\right) z\frac{\partial \omega_i(z,t)}{\partial z} + \frac{\sigma_i^2}{2\theta_i} z^2 \frac{\partial^2 \omega_i(z,t)}{\partial z^2}.\end{aligned}$$

Therefore,

$$u_i(z,t) = s_i(z,t) - \frac{\dot{A}(t)}{A(t)} + \frac{1}{\theta_i}\left(1 - \bar{\mu}_i + \log z + \frac{\sigma_i^2}{2}\right),$$

$$v_i(z,t) = \frac{1}{\theta_i}\left(\log z - \bar{\mu}_i + \frac{3}{2}\sigma_i^2\right) z, \text{and}$$

$$x_i(z,t) = \frac{\sigma_i^2}{2\theta_i} z^2.$$

E Numerical Implementation, Parametrization, and Measurement

E.1 Productivity Process

The theoretical model allows productivity to follow a general sector-specific ergodic diffusion,

$$dz_i(t) = \mu_i(z_i(t))\, dt + \sigma_i(z_i(t))\, dW_t.$$

In the quantitative exercises, I specialize this process to an Ornstein–Uhlenbeck process in log productivity. This specification captures two features that are useful for the simulations: productivity is persistent, but shocks are mean-reverting, so the cross-sectional distribution of productivity remains stationary over time.

Let

$$x_i(t) \equiv \log z_i(t).$$

Specifically, I assume that log productivity follows:

$$dx_i(t) = -\frac{1}{\theta_i}\left[x_i(t) - \bar{\mu}_i\right] dt + \sigma_i \sqrt{\frac{1}{\theta_i}}\, dW_t,$$

where $\theta_i > 0$ governs persistence and $\sigma_i > 0$ governs the dispersion of productivity shocks. A larger θ_i implies slower mean reversion and therefore more persistent productivity. The autocorrelation of log productivity over horizon s is

$$\operatorname{corr}\left(x_i(t), x_i(t+s)\right) = e^{-s/\theta_i}.$$

The stationary distribution of log productivity is

$$x_i \sim \mathcal{N}\left(\bar{\mu}_i, \frac{\sigma_i^2}{2}\right).$$

Equivalently, productivity $z_i = \exp(x_i)$ has stationary density

$$\psi_i(z) = \frac{1}{z\sigma_i\sqrt{\pi}} \exp\left(-\frac{(\log z - \bar{\mu}_i)^2}{\sigma_i^2}\right),$$

Thus, $\bar{\mu}_i$ determines the long-run mean of log productivity in sector i, the parameter σ_i controls the stationary dispersion of log productivity, while θ_i controls the speed at which firms revert towards the sectoral mean.

To express the process in terms of $z_i(t)$, apply Itô's lemma to

$$z_i(t) = \exp(x_i(t)).$$

Since $f(x) = \exp(x)$ satisfies $f'(x) = f''(x) = \exp(x) = z_i(t)$, Itô's lemma[21] gives

$$dz_i(t) = z_i(t)\, dx_i(t) + \frac{1}{2} z_i(t)\, (dx_i(t))^2.$$

Using Itô multiplication rules[22] in the term $(dx_i(t))^2$, I obtain

$$dz_i(t) = \frac{1}{\theta_i}\left[\bar{\mu}_i - \log z_i(t) + \frac{\sigma_i^2}{2}\right] z_i(t)\, dt + \sigma_i \sqrt{\frac{1}{\theta_i}} z_i(t)\, dW_t.$$

Therefore, in the notation of the main text,

$$\mu_i(z) = \frac{1}{\theta_i}\left[\bar{\mu}_i - \log z + \frac{\sigma_i^2}{2}\right] z,$$

and

$$\sigma_i(z) = \sigma_i \sqrt{\frac{1}{\theta_i}} z.$$

[21] For a scalar Itô process $dx = \mu(x,t)\, dt + \sigma(x,t)\, dW_t$, Itô's lemma gives

$$df(x) = f'(x)\, dx + \frac{1}{2} f''(x)(dx)^2,$$

with $(dx)^2 = \sigma(x,t)^2 dt$ under the Itô rules $(dt)^2 = 0$, $dt\, dW_t = 0$, and $(dW_t)^2 = dt$.

[22] Starting from

$$dx_i(t) = -\frac{1}{\theta_i}\left[x_i(t) - \bar{\mu}_i\right] dt + \sigma_i \sqrt{\frac{1}{\theta_i}}\, dW_t,$$

we have

$$(dx_i(t))^2 = \left[-\frac{1}{\theta_i}\left(x_i(t) - \bar{\mu}_i\right) dt + \sigma_i \sqrt{\frac{1}{\theta_i}}\, dW_t\right]^2.$$

Using the rules

$$(dt)^2 = 0, \qquad dt\, dW_t = 0, \qquad (dW_t)^2 = dt,$$

only the diffusion term remains:

$$(dx_i(t))^2 = \sigma_i^2 \frac{1}{\theta_i} (dW_t)^2 = \frac{\sigma_i^2}{\theta_i}\, dt.$$

E.2 Illustrative Model Parametrization

Table E.1: Parameters for Illustrative Simulations

Parameter / Object	Description	Baseline value
Panel A: Environment		
I	Number of sectors	2
$\bar{L}$	Aggregate labor endowment	1.0
Panel B: Preferences, Technology, and Constraints		
η	Elasticity of substitution across sectoral goods	2.0
υ_i	CES weights in final-good aggregator	$1/I$
ρ	Discount rate	0.05
δ	Depreciation rate	0.05
α_i	Capital share in sector i	0.50 (each sector)
λ_i	Collateral parameter in sector i	1.20 (each sector)
Panel C: Productivity Process and Initial Conditions		
θ_i	Mean-reversion parameter of log productivity	6.25 (each sector)
σ_i	Volatility of log productivity	0.65 (each sector)
$\bar{\mu}_i$	Long-run mean of log productivity	0.0 (baseline)
$\mathrm{Corr}_i = e^{-1/\theta_i}$	One-period persistence	Implied
$\mathrm{std}_i = \sigma_i/\sqrt{2}$	Stationary std. dev. of log productivity	Implied
$\rho_{a,z,i}(0)$	Initial correlation between wealth and productivity	0.50 (each sector)
$\mu_{\log z,i,0}$	Initial mean of log productivity	0.0
$\sigma_{\log z,i,0}$	Initial std. dev. of log productivity	std_i
$\mu_{\log a,i,0}$	Initial mean of log wealth	0.0
$\sigma_{\log a,i,0}$	Initial std. dev. of log wealth	std_i
$A_{i,0}$	Initial wealth in sector i	Implied by initialization
Panel D: Computational Parameters		
$z_{\min}$	Lower bound of productivity grid	10^{-4}
$z_{\max}$	Upper bound of productivity grid	$\max_{i\in\mathcal{I}} Q_{0.99}\left[\mathrm{LogNormal}\left(\bar{\mu}_i, \frac{\sigma_i}{\sqrt{2}}\right)\right]$
N_z	Number of productivity grid points	1000
T	Simulation horizon	125
N_t	Number of time steps	1250
Δt	Time step size	$T/(N_t-1)$
$p_i(0)$	Initial guess for sectoral prices	1.0
$w(0)$	Initial wage guess	1.0
$r(0)$	Initial interest rate guess	0.03

Notes: Parameters follow standard values, and the productivity process generates persistent and dispersed firm-level heterogeneity. In the baseline exercise, sectors share the same long-run productivity process and marginal distributions, and asymmetry arises only from differences in the initial correlation between wealth and productivity. In the comparative-advantage exercise, sectors instead begin from identical initial distributions but differ in the long-run mean of log productivity, $\bar{\mu}_i$.

The illustrative parameterization is chosen to provide a transparent and symmetric benchmark for the model exercises. Preference and technology parameters follow standard values in the macroeconomic literature. The parameters governing the productivity process are selected to

generate persistent and dispersed productivity across entrepreneurs.

Initial conditions are constructed using sector-specific joint distributions over wealth and productivity. For each sector i, I assume a lognormal specification for (a, z) characterized by

$$(\mu_{\log a,i,0}, \sigma_{\log a,i,0}, \mu_{\log z,i,0}, \sigma_{\log z,i,0})$$

and a correlation parameter $\rho_{a,z,i}(0)$. This determines both the marginal distributions of wealth and productivity and the initial allocation of wealth across productivity levels.

To ensure comparability across exercises, total initial wealth $A(0)$ and initial sectoral wealth shares are held fixed. The first exercise isolates the role of initial allocation by giving sectors identical marginal distributions and identical long-run productivity processes, while allowing their initial wealth-productivity correlations $\rho_{a,z,i}(0)$ to differ. Thus, the sectors have the same long-run productivity potential but differ in how effectively wealth is initially allocated across entrepreneurs.

The second exercise instead isolates differences in sectoral fundamentals. Sectors begin from identical wealth and productivity distributions and the same initial wealth-productivity alignment, but differ in the long-run mean of log productivity, $\bar{\mu}_i$. This generates a fundamental long-run productivity advantage for one sector while holding initial conditions fixed, allowing the analysis to study how financial constraints affect the speed of adjustment towards the specialization pattern implied by sectoral fundamentals.

E.3 Data Sources, Measurement, and Quantitative Model Calibration

E.3.1 Sector Definitions and Concordance

The quantitative economy contains seven sectors divided into two groups:

- Tradable: Agroindustry, Fishing, Machinery and Transport Equipment, and Non-Machinery Manufacturing.
- Non-tradable: Commerce, Construction, and Market Services.

Sector definitions are harmonized across the EEA, ENAHO, and the 2007 Input–Output and Supply and Use Tables using the EEA industry classification as the main reference. When national-accounts categories combine activities assigned to different sectors, I use the more detailed product classification whenever possible and otherwise assign the aggregate according to its predominant activity.

Primary agriculture, livestock, and forestry are excluded from the modeled Agroindustry sector since they are not covered by the EEA. Agroindustry instead includes the food-processing activities classified by the EEA as such. Likewise, Fishing includes the fish-processing activities classified within the EEA fishing sector. Activities outside the seven-sector economy therefore include primary agriculture, extractive and hydrocarbon industries, financial services, and public or non-market services.

E.3.2 Firm-Side Parameters and Initial Distributions

I use the EEA to discipline the sector-specific production parameters, the stochastic process for entrepreneurial productivity, and the initial joint distribution of entrepreneurial wealth and productivity. Unless otherwise indicated, these moments are computed using observations from the initial years of the sample (2000-2003), as they are meant to represent initial conditions.

Production parameters. Firm production in sector i is

$$y_{fi} = (z_{fi}k_{fi})^{\alpha_i}\ell_{fi}^{\beta_i}h_{fi}^{\gamma_i}, \qquad \alpha_i + \beta_i + \gamma_i = 1.$$

The labor share, β_i, is measured as the sectoral median of firm labor remuneration relative to value added,

$$\beta_i = \text{median}_{f,t}\left(\frac{w_{fit}\ell_{fit}}{va_{fit}}\right).$$

To measure the capital share, I assume the user-cost condition $(r+\delta)k_{fit}$ and construct

$$\alpha_i = \text{median}_{f,t}\left[(r+\delta)\frac{k_{fit}}{va_{fit}}\right],$$

where the required net return on capital is set to $r = 0.07$, a value consistent with estimates of the cost of capital for Peruvian industries in Vásquez Cordano and Aguirre Zurita (2017), and the annual depreciation rate is set to $\delta = 0.10$, a standard value also used in Peruvian firm-level applications, including the construction of firm capital stocks by Castellares (2015).

Finally, because production is constant returns to scale at the firm level, the share of the sector-specific factor is obtained residually as

$$\gamma_i = 1 - \alpha_i - \beta_i.$$

The resulting sector-specific moments are reported in Table E.2.

Productivity. I construct a within-sector-year productivity proxy using the optimal input choices implied by the model. That is, conditional on sector-year output and factor prices, firms' optimal labor and sector-specific-factor demands are assumed to satisfy

$$\ell_{fit} = m_{it}^L z_{fit}k_{fit}, \qquad h_{fit} = m_{it}^H z_{fit}k_{fit},$$

where m_{it}^L and m_{it}^H are common across firms within a sector and year. Combining these conditions with constant returns to scale gives

$$\log va_{fit}^{nom} - \log k_{fit} = \log z_{fit} + \kappa_{it},$$

where

$$\kappa_{it} = \log p_{it} + \beta_i \log m_{it}^L + \gamma_i \log m_{it}^H$$

is common to firms in sector i and year t. Then, I approximate measured relative firm log-productivity using

$$\widehat{x}_{fit} = \log va_{fit}^{nom} - \log k_{fit} - \overline{\left(\log va_{fit}^{nom} - \log k_{fit}\right)}_{it}.$$

This transformation removes sector-year variation in output and factor prices and identifies the cross-sectional component of productivity relevant for firms' relative returns.

In the quantitative model, log productivity follows

$$d\log z_{fi} = -\frac{1}{\theta_i}\log z_{fi}\,dt + \frac{\sigma_i}{\sqrt{\theta_i}}\,dW_{fi}.$$

For this process,

$$\mathrm{sd}(\log z_{fi}) = \frac{\sigma_i}{\sqrt{2}}, \qquad \mathrm{corr}\left(\log z_{fi,t}, \log z_{fi,t-1}\right) = \exp\left(-\frac{1}{\theta_i}\right).$$

I therefore use the cross-sectional dispersion and annual persistence of the EEA productivity proxy to recover the sector-specific parameters σ_i and θ_i.

Table E.2: Sector-specific moments used in the calibration

Sector	α_i	β_i	γ_i	$\mathrm{sd}(\log z_{fi})$	σ_i	$\mathrm{corr}(\log z_{fi,t}, \log z_{fi,t-1})$	θ_i	$\widehat{\frac{A_i}{w_i L_i}}$	ρ_i^{az}
Agroindustry	0.399	0.410	0.191	1.234	1.745	0.720	3.044	10.898	-0.360
Commerce	0.226	0.340	0.434	1.431	2.024	0.869	7.122	3.077	-0.562
Construction	0.283	0.420	0.297	1.632	2.308	0.781	4.046	3.464	-0.572
Fishing	0.411	0.340	0.249	1.565	2.213	0.653	2.346	20.726	-0.421
Machinery & Transport	0.275	0.440	0.285	1.276	1.805	0.845	5.938	16.759	-0.664
Non-Machinery Manufacturing	0.282	0.430	0.288	1.294	1.830	0.856	6.431	9.995	-0.625
Market Services	0.381	0.470	0.149	1.626	2.300	0.873	7.363	12.268	-0.419

Notes: α_i and β_i are measured using EEA firm data and $\gamma_i = 1 - \alpha_i - \beta_i$. Productivity follows $d\log z_{fi} = -(1/\theta_i)\log z_{fi}\,dt + (\sigma_i/\sqrt{\theta_i})\,dW_{fi}$, so that $\mathrm{sd}(\log z_{fi}) = \sigma_i/\sqrt{2}$ and $\mathrm{corr}(\log z_{fi,t}, \log z_{fi,t-1}) = \exp(-1/\theta_i)$. $\widehat{A}_i/(w_i L_i)$ denotes the sectoral assets-to-labor-income ratio, computed as the median across 2000–2003 of the ratio of aggregate firm assets to aggregate remuneration within each sector-year. ρ_i^{az} denotes the initial within-sector wealth–productivity correlation, proxied using firm capital and productivity residuals.

Initial wealth–productivity dependence. The model allows the initial distribution of entrepreneurial wealth to be correlated with productivity within sectors. I discipline this dependence using the initial within-sector correlation

$$\rho_i^{az} = \mathrm{corr}\left(\log a_{fi,0}, \log z_{fi,0}\right),$$

proxied in the EEA using firm capital and the productivity residual described above. Table E.2 summarizes these and the other moments measured from the EEA.

Initial entrepreneurial wealth. For each sector-year in 2000–2003, I first calculate the ratio of aggregate firm assets to aggregate remuneration,

$$(A/wL)_{it} = \frac{\sum_f A_{fit}}{\sum_f \mathrm{Remuneration}_{fit}},$$

and define the sectoral assets-to-labor-income ratio

$$\left(\frac{A}{wL}\right)_i^{EEA} = \mathrm{median}_{t\in\{2000,\ldots,2003\}}\,(A/wL)_{it}\,.$$

I then combine these ratios with the sectoral labor-income bill measured in the 2004 ENAHO. Let $(w_i L_i)^{ENAHO}$ denote weighted labor income among dependent workers in modeled sector i. Implied sectoral entrepreneurial assets are

$$\widehat{A}_i = \left(\frac{A}{wL}\right)_i^{EEA} \times (w_i L_i)^{ENAHO}.$$

The initial sectoral wealth shares used in the model are therefore

$$O_i(0) = \frac{\widehat{A}_i}{\sum_j \widehat{A}_j}.$$

The results of these procedures are in Table E.3. The nominal level of implied assets in soles is not used directly in the quantitative exercise. The construction disciplines both the relative allocation of initial wealth across sectors, through $O_i(0)$, and the aggregate wealth scale. The aggregate scale of initial model wealth is calibrated separately, as described below, to satisfy $A(0)/[w(0)L(0)] = 9.639$.

Table E.3: Initial entrepreneurial wealth calibration

Sector	$A_i/(w_iL_i)$	w_iL_i (bn. soles)	L_i (thousands)	$\widehat{A}_i$ (bn. soles)	$O_i(0)$ (%)
Agroindustry	10.898	1.360	66.5	14.826	5.96
Commerce	3.077	4.203	574.7	12.935	5.20
Construction	3.464	2.764	305.9	9.574	3.85
Fishing	20.726	1.108	73.9	22.965	9.23
Machinery & Transport Equipment	16.759	0.203	21.3	3.403	1.37
Non-Machinery Manufacturing	9.995	5.860	590.3	58.571	23.54
Market Services	12.268	10.314	1093.7	126.528	50.85

Notes: $A_i/(w_iL_i)$ is the sectoral assets-to-labor-income ratio measured from the EEA as the median across 2000–2003 of the ratio of aggregate firm assets to aggregate remuneration within each sector-year. w_iL_i and L_i are the sectoral labor-income bill and weighted number of dependent workers measured using the 2004 ENAHO. Implied sectoral entrepreneurial assets are constructed as $\widehat{A}_i = [A_i/(w_iL_i)]^{EEA}(w_iL_i)^{ENAHO}$, and initial wealth shares are $O_i(0) = \widehat{A}_i/\sum_j \widehat{A}_j$. Across the seven modeled sectors, the procedure implies aggregate entrepreneurial assets of 248.802 billion soles, 2.726 million dependent workers, and aggregate labor income of 25.813 billion soles. The corresponding ratios are $A(0)/L(0) = 91261.54$ soles per worker and $A(0)/[w(0)L(0)] = 9.639$. The latter dimensionless ratio is used to discipline the aggregate scale of initial entrepreneurial wealth in the quantitative model.

Initial joint distribution. Initial log productivity in each sector is drawn from the stationary distribution implied by the sector-specific productivity process. Within-sector log wealth is initialized with a common dispersion, normalized to $\mathrm{sd}(\log a) = 1$, while its mean is scaled so that sectoral aggregate wealth is consistent with $O_i(0)$. Conditional on this common wealth dispersion, differences in the initial within-sector alignment between wealth and productivity are characterized by the sector-specific correlation ρ_i^{az} reported in Table E.2.

E.3.3 National Accounts and Validation

I use Peru's 2007 Input–Output Table at current prices to discipline the cross-sectional sectoral structure of the quantitative economy. The original tables are aggregated to the seven modeled sectors using the concordance described above. I assume that the cross-sectional structure observed in 2007 provides an appropriate approximation to the initial economy in 2004, while firm-side initial conditions and exogenous transition paths are anchored to information corresponding more closely to the initial year.

For each modeled sector i, let Y_i, M_i, and X_i denote gross output, imports, and exports, respec-

tively. I define domestic demand as

$$D_i = TD_i - X_i - \Delta Inv_i,$$

where TD_i denotes total demand and ΔInv_i denotes inventory changes. I then construct the sectoral gross-output share and the expenditure share in the outer CES aggregator as

$$s_i^Y = \frac{Y_i}{\sum_j Y_j}, \qquad s_i^D = \frac{D_i}{\sum_j D_j}.$$

For tradable sectors, I additionally construct the trade-balance target

$$tb_i = \frac{X_i - M_i}{\sum_j Y_j},$$

and the domestic-origin share of domestic expenditure,

$$s_i^H = 1 - \frac{M_i}{D_i}.$$

The trade-balance target is normalized by total production in the seven modeled sectors. For sectors treated as nontradable in the quantitative model, I set $tb_i = 0$ and $s_i^H = 1$.

Table E.4 reports the resulting moments. The domestic-demand shares, tradable-sector trade balances, and domestic-origin shares discipline the initial-equilibrium calibration described in Section E.3.4. As an additional independent measure of sectoral size, I report the share of dependent-worker labor income in each modeled sector from the 2004 ENAHO. Gross-output shares and labor-income shares are not directly targeted in the calibration and are used instead as validation.

Table E.4: Sectoral national-accounts moments and validation data

Sector	Output share (%)	D share (%)	tb_i (%)	$1 - M_i/D_i$ (%)	Labor-income share (%)
Agroindustry	7.19	7.24	-0.27	88.51	5.27
Commerce	11.98	11.63	–	100.00	16.28
Construction	9.45	9.43	–	100.00	10.71
Fishing	2.90	1.33	1.50	97.52	4.29
Machinery & Transport Equipment	1.55	6.67	-5.37	18.43	0.79
Non-Machinery Manufacturing	27.08	24.47	1.68	73.26	22.70
Market Services	39.85	39.23	–	100.00	39.96

Notes: Output shares and domestic-demand shares are calculated relative to the seven modeled sectors using the 2007 Input–Output Table. Domestic demand is total demand net of exports and inventory changes. For tradable sectors, $tb_i = (X_i - M_i)/\sum_j Y_j$ and $1 - M_i/D_i$ measures the domestic-origin share of domestic demand. Commerce, Construction, and Market Services are treated as nontradable in the quantitative model, so their trade-balance targets are set to zero and their domestic-origin shares to one. Labor-income shares are calculated using weighted labor income of dependent workers in the 2004 ENAHO.

E.3.4 Demand Parameters and Initial-Equilibrium

Outer CES elasticity. I estimate the elasticity of substitution across sectoral domestic-use composites using annual Supply and Use Tables from INEI for 2007–2024. Let E_{it}^N and E_{it}^R denote nominal and real domestic absorption in sector i and year t. I construct the sectoral expenditure share and implicit price as

$$s_{it} = \frac{E_{it}^N}{\sum_j E_{jt}^N}, \qquad p_{it} = \frac{E_{it}^N}{E_{it}^R}.$$

Under the outer CES aggregator,

$$s_{it} = \frac{v_i p_{it}^{1-\eta}}{\sum_j v_j p_{jt}^{1-\eta}}.$$

Taking logs therefore motivates the estimating equation

$$\log s_{it} = \mu_i + \delta_t + \beta \log p_{it} + \varepsilon_{it}, \qquad \eta = 1 - \beta,$$

where sector fixed effects absorb the time-invariant CES weights v_i, while year fixed effects absorb variation in aggregate expenditure and the CES price index.

The baseline specification is estimated in levels with sector and year fixed effects. The sample contains six sectors observed over 18 years, for a total of 108 observations. Commerce is excluded because measured final use for this sector is zero in the SUT after the adjustment of commerce margins. As robustness checks, I estimate the equation in first differences, consider alternative treatments of the standard errors and price normalization, and incorporate inventory changes in the measure of domestic absorption.

The baseline estimate is $\widehat{\beta} = -0.014$, implying an outer elasticity of substitution of

$$\widehat{\eta} = 1 - \widehat{\beta} = 1.01.$$

The estimate is imprecise and the hypothesis of unit elasticity cannot be rejected. I therefore use $\eta = 1.01$ in the quantitative baseline. Table E.5 reports the baseline estimate and the alternative specifications.

Armington elasticity and home bias. Armington differentiation at the sector level implies the following domestic-use price index for tradable sectors:

$$p_i^D = \left[\phi_i (p_i^H)^{1-\eta_i^\star} + (1-\phi_i)(p_i^M)^{1-\eta_i^\star}\right]^{\frac{1}{1-\eta_i^\star}},$$

where p_i^H is the price of the domestic-origin variety and $p_i^M = (1+\tau_i^M)p_i^*$ is the tariff-inclusive import price.

I set the elasticity of substitution to a common value $\eta_i^\star = 1.76$ for all tradable sectors, based on the aggregate short-run trade elasticity estimated in Boehm et al. (2023). The home-bias parameters ϕ_i are calibrated using the domestic-origin expenditure shares reported in Table E.4.

In particular, the Armington system implies that the domestic-origin expenditure share is

$$s_i^H = \frac{\phi_i (p_i^H)^{1-\eta_i^\star}}{\phi_i (p_i^H)^{1-\eta_i^\star} + (1-\phi_i)(p_i^M)^{1-\eta_i^\star}}.$$

Table E.5: Estimates of the outer CES elasticity of substitution.

Specification	Inference	N	$\hat{\beta}$ (SE)	$\hat{\eta}$ [95% CI]
Levels (main)	HC1	108	-0.014 (0.315)	1.01 [0.39, 1.64]
Levels (main)	Cluster: sector	108	-0.014 (0.353)	1.01 [0.11, 1.92]
Levels (main)	Cluster: year	108	-0.014 (0.398)	1.01 [0.17, 1.85]
Levels (main)	Two-way	108	-0.014 (0.426)	1.01 [-0.08, 2.11]
Levels, price normalized to 2009	HC1	108	-0.014 (0.315)	1.01 [0.39, 1.64]
First differences	HC1	102	-0.154 (0.448)	1.15 [0.26, 2.04]
First differences	Cluster: sector	102	-0.154 (0.324)	1.15 [0.32, 1.99]
First differences	Cluster: year	102	-0.154 (0.516)	1.15 [0.06, 2.25]
First differences	Two-way	102	-0.154 (0.435)	1.15 [0.04, 2.27]
Levels, absorption includes inventories	HC1	108	-0.777 (0.672)	1.78 [0.44, 3.11]
First differences, absorption includes inventories	HC1	102	-1.220 (1.301)	2.22 [-0.37, 4.81]

Notes: The table reports estimates of the outer CES elasticity of substitution. The estimating equation relates the log sectoral expenditure share to the log sectoral price, with coefficient $\beta = 1-\eta$; the implied elasticity is therefore $\widehat{\eta} = 1-\widehat{\beta}$. The baseline levels specification includes sector and year fixed effects, while first-difference specifications include year fixed effects. Alternative rows vary the treatment of standard errors and whether total absorption is included in the construction of sectoral expenditure. Standard errors for $\widehat{\beta}$ are reported in parentheses, and 95% confidence intervals for $\widehat{\eta}$ are reported in brackets. The baseline calibration uses the first-row estimate.

I invert this expression to recover ϕ_i from the observed s_i^H, conditional on initial trade-policy wedges and relative prices. For this calibration, the domestic-origin price is evaluated at the initial net export price,

$$p_i^X(0) = \frac{p_i^*(0)}{1+\tau_i^X(0)}.$$

The resulting ϕ_i are held fixed throughout the quantitative transition.

Initial-equilibrium calibration. I normalize the initial world price of Agroindustry to one and choose the remaining tradable-sector world prices $\{p_i^*(0)\}$ together with the outer-CES expenditure weights $\{v_i\}$ to reproduce the domestic-demand shares and tradable-sector trade-balance shares reported in Table E.4.

Conditional on a vector of initial world prices, I solve the within-period equilibrium and obtain the domestic-use prices $\{p_i^D(0)\}$. Given these prices and the target expenditure shares s_i^D, the outer-CES weights satisfy

$$v_i \propto s_i^D \left[p_i^D(0)\right]^{\eta-1}, \qquad \sum_i v_i = 1.$$

I then resolve the equilibrium and adjust tradable-sector relative world prices in the direction implied by the gaps between model-implied and targeted trade-balance shares. After each price update, world prices are renormalized relative to Agroindustry and the CES weights are recalibrated at the resulting domestic-use prices. I iterate between these steps until the domestic-demand and trade-balance moments match their empirical targets under a numerical tolerance.

Throughout this procedure, nontradable prices, tradable domestic-origin prices, wages, the interest rate, and sector-specific-factor rents are endogenous equilibrium objects. Tradable domestic-origin prices satisfy the export-complementarity condition $p_i^H \geq p_i^X$, with equality whenever the

sector exports.

E.3.5 Financial Development Measurement and Aggregate Calibration

Collateral multiplier. Financial frictions are summarized by the collateral multiplier λ in the constraint $k \leq \lambda a$. This implies an entrepreneur operating at the borrowing limit can finance a fraction

$$\frac{k-a}{k} = 1 - \frac{1}{\lambda}$$

of operated capital externally. To measure λ across countries, I follow Moll (2014) to approximate the externally financed share of the aggregate capital stock using private credit relative to the capital stock,

$$1 - \frac{1}{\lambda} \approx \frac{D}{K} = \frac{D/Y}{K/Y},$$

where D denotes domestic credit to the private sector and K the aggregate capital stock. This gives the mapping

$$\lambda = \left[1 - \frac{D/Y}{K/Y}\right]^{-1}.$$

I measure D/Y using the World Development Indicators series *Domestic credit to private sector (% of GDP)* (`FS.AST.PRVT.GD.ZS`). The capital-output ratio is constructed from Penn World Table data as

$$\frac{K}{Y} = \frac{\texttt{cn}}{\texttt{cgdpo}},$$

where `cn` is the capital stock at current purchasing-power parities and `cgdpo` is output-side real GDP at current purchasing-power parities. For each country, I combine these series to obtain an annual implied value of λ and summarize its level over 2007-2023, when data is available for all countries, and is a period relevant for the quantitative application. The resulting values are reported in Table E.6.

Table E.6: Financial development and implied collateral multipliers.

Country	D/Y (%)	K/Y (%)	D/K	λ
Peru	39.3	356.7	0.111	1.13
Colombia	41.2	326.9	0.127	1.15
Chile	107.6	382.7	0.282	1.39
Korea	136.4	468.4	0.292	1.41
Malaysia	116.2	375.9	0.310	1.45
Thailand	138.5	394.7	0.351	1.55
United States	190.4	340.9	0.559	2.29

Notes: The table reports averages over the common 2007–2023 sample. D/Y is domestic credit to the private sector as a share of GDP from the World Development Indicators, and K/Y is the capital-output ratio constructed from Penn World Table variables `cn` and `cgdpo`. For each country-year, I compute $D/K = (D/Y)/(K/Y)$ and $\lambda = [1 - D/K]^{-1}$. The reported λ is the average of the annual implied collateral multipliers over 2007–2023.

Baseline parameters and normalizations. I set the baseline collateral multiplier to a common $\lambda = 1.13$ as implied by Table E.6. The discount rate is set to $\rho = 0.05$ and the depreciation rate to $\delta = 0.10$. Aggregate labor supply is normalized to $\bar{L} = 1$. I also normalize the effective supply of the sector-specific factor so that $\bar{H}_i^{\gamma_i} = 1$ in every sector.

Aggregate initial wealth. The construction in Section E.3.2 determines the initial allocation of entrepreneurial wealth across sectors, $O_i(0)$, separately from its aggregate scale. Given aggregate initial wealth $A(0)$, sectoral initial wealth is therefore

$$A_i(0) = O_i(0)A(0).$$

I choose $A(0)$ numerically so that the solved initial equilibrium reproduces the empirical aggregate assets-to-labor-income ratio measured above:

$$\frac{A(0)}{w(0)\bar{L}} = \left(\frac{A}{wL}\right)_{\text{data}} = 9.639.$$

The equilibrium wage $w(0)$ depends on the aggregate wealth endowment, so $A(0)$ cannot be obtained be imposed directly. Instead, I adjust the aggregate wealth scale until the initial equilibrium satisfies this condition. With $\bar{L} = 1$, this is equivalent to targeting $A(0)/w(0) = 9.639$. This procedure yields $A(0) = 24.14$. At this value, the initial equilibrium implies $w(0) = 2.505$ and $A(0)/w(0) = 9.637$, effectively matching the empirical target.

E.3.6 Exogenous Transition Paths and External Balance

The quantitative transition treats 2004 as the start year. After solving for the initial equilibrium, I treat international prices, trade-policy wedges, and the net-export contribution of activities outside the modeled economy as exogenous time-varying objects.

International price paths. For each tradable sector i, the initial world price $p_i^*(0)$ is determined as part of the initial-equilibrium calibration described in Section E.3.4. I then use import-price series from PWT to discipline the subsequent evolution of relative world prices. Let $g_i^*(t)$ denote the cumulative change in the corresponding PWT price index relative to its 2004 value. The world-price path used in the model is constructed as

$$p_i^*(t) = p_i^*(0)g_i^*(t).$$

Relative-price changes are used instead of the levels because the initial model prices are calibrated internally. World prices are normalized relative to Agroindustry, which serves as the numeraire.

Tariff paths. Trade-policy wedges enter separately on imports and exports. Imports of sector i are purchased at $p_i^M(t) = \left[1 + \tau_i^M(t)\right] p_i^*(t)$, while domestic exporters receive $p_i^X(t) = \frac{p_i^*(t)}{1+\tau_i^X(t)}$. The paths for $\tau_i^M(t)$ are informed by average Most-Favored-Nation tariff rates. In the baseline exercise, the corresponding tariff path is also used to parameterize the foreign trade-policy wedge faced by Peruvian exports.

External balance. I parameterize the residual $NX_{\text{res}}(t)$ as a share of nominal production in the modeled sectors. For the initial year, the residual is calculated directly from the 2004 sectoral trade-balance targets. Let tb_i^{2004} denote the trade balance of tradable sector i as a share of modeled nominal production. The initial residual share is

$$nx_{\text{res},2004} = -\sum_{i \in \mathcal{I}_T} tb_i^{2004}.$$

Using the sectoral targets in the initial calibration gives

$$nx_{\text{res},2004} = -\left(-0.0027 + 0.0150 - 0.0537 + 0.0168\right) = 0.0246.$$

That is, omitted activities contribute net exports equal to approximately 2.5 percent of modeled-sector production in the initial year.

For 2007–2024, I measure the residual directly from the annual INEI Supply and Use Tables. In each year t, I first calculate the aggregate trade balance of the seven modeled sectors and define

$$nx_{\text{res},t} = -\frac{TB_t^{\text{modeled}}}{Y_t^{N,\text{modeled}}},$$

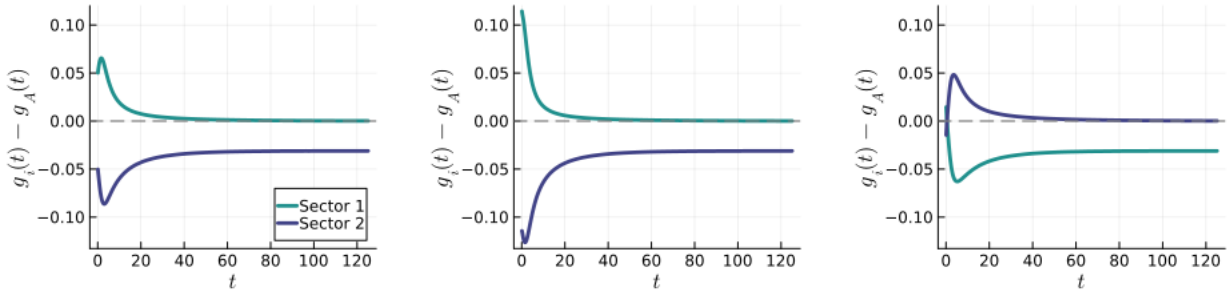
where $Y_t^{N,\text{modeled}}$ denotes nominal production in the modeled sectors. The resulting series varies from approximately 2.5 percent in 2007 to 6.8 percent in 2024, with larger values in years in which the modeled sectors jointly run larger trade deficits in the data.

Because annual SUT information is first available in 2007, I linearly interpolate the residual share between the calibrated 2004 value and the first observed SUT value in 2007, and between annual observations thereafter. After 2024, the residual share is held fixed at its final observed value.

F Additional Results and Robustness

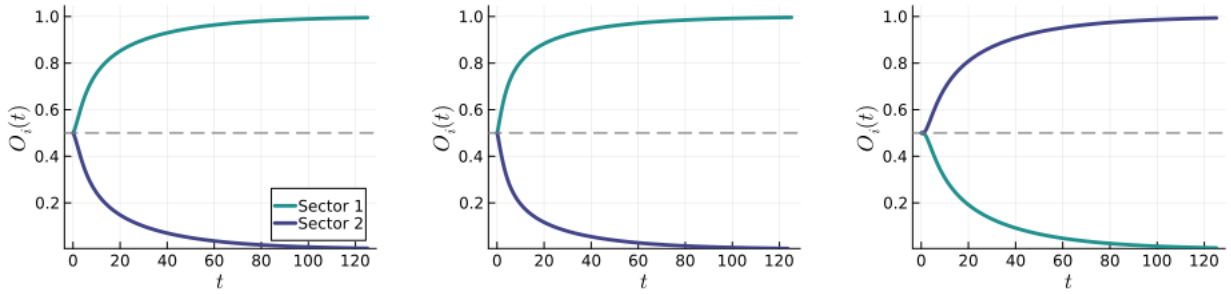
F.1 Price shocks and endogenous specialization

Panel A. Saving-rate gap



(a) Symmetric with $p_1^* \uparrow$ (b) Asymmetric with $p_1^* \uparrow$ (c) Asymmetric with $p_2^* \uparrow$

Panel B. Wealth shares

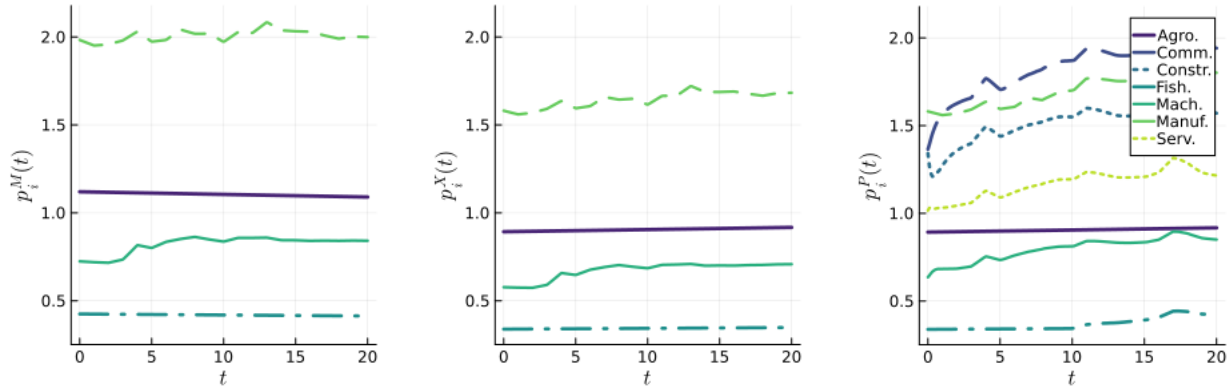


(d) Symmetric with $p_1^* \uparrow$ (e) Asymmetric with $p_1^* \uparrow$ (f) Asymmetric with $p_2^* \uparrow$

Figure F.1: Dynamic response to world price shocks.

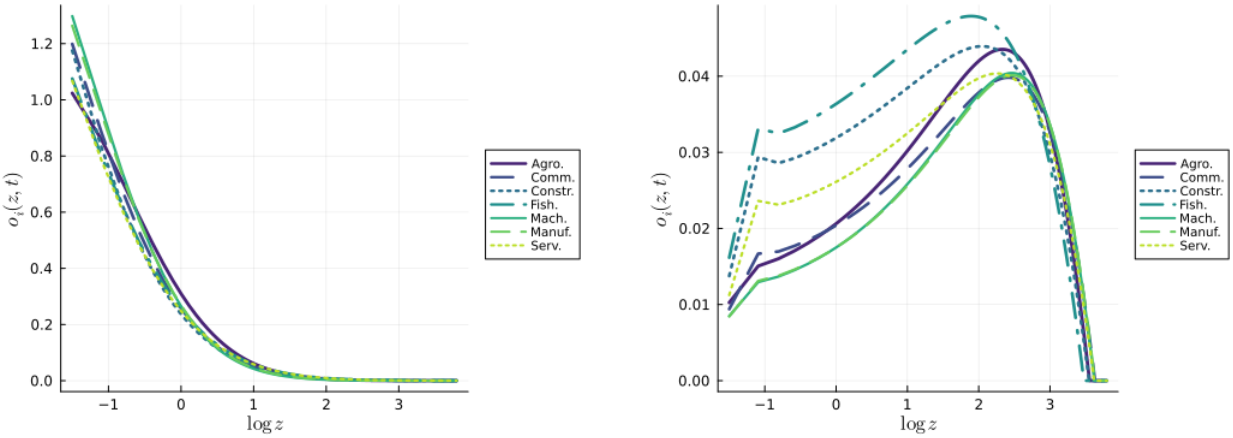
The figure reports the response of a small open economy to permanent changes in world prices. Panel A plots sectoral saving-rate gaps $g_i(t) - g_A(t)$, and Panel B reports sectoral wealth shares $O_i(t)$. Colors identify sectors. Columns correspond to symmetric and asymmetric environments, with price increases in sector 1 (columns (a)–(b)) and sector 2 (column (c)). In each case, the sector facing the higher price exhibits a positive saving-rate gap, while the other exhibits a negative gap, with corresponding changes in wealth shares.

F.2 Additional Figures for Baseline Quantitative Exercise



(a) Import Prices (b) Net Export Prices (c) Producer Prices

Figure F.2: Prices Shocks and Domestic Adjustment



(a) Initial (b) Final

Figure F.3: Within-sector Wealth Distributions: Baseline initial vs. final

Notes: The figure truncates $\log z < -1.5$. The numerical grid is uniform in z and therefore sparse in $\log z$ near zero.

F.3 Additional Financial Frictions Counterfactuals Summary

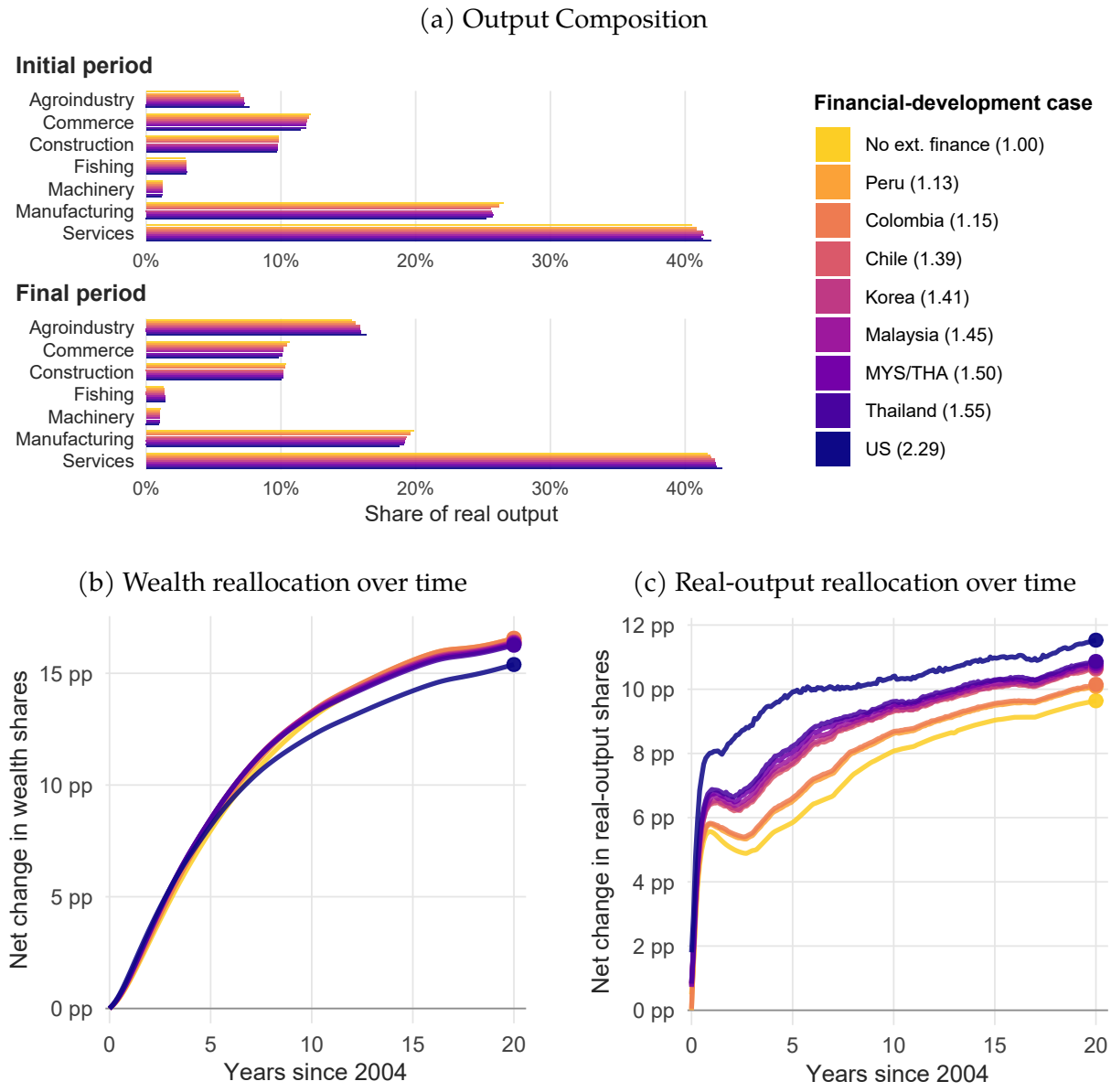
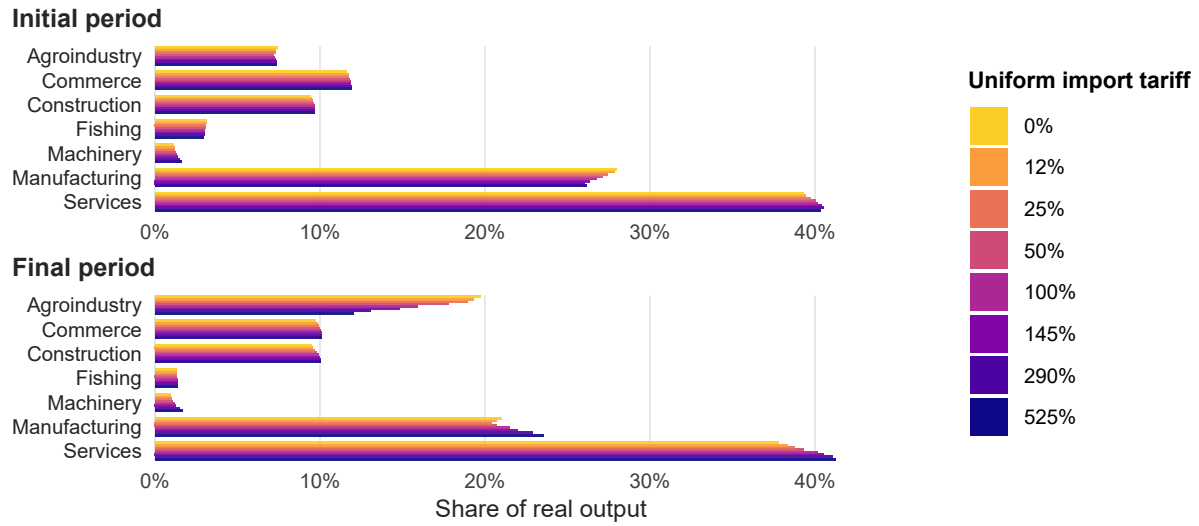


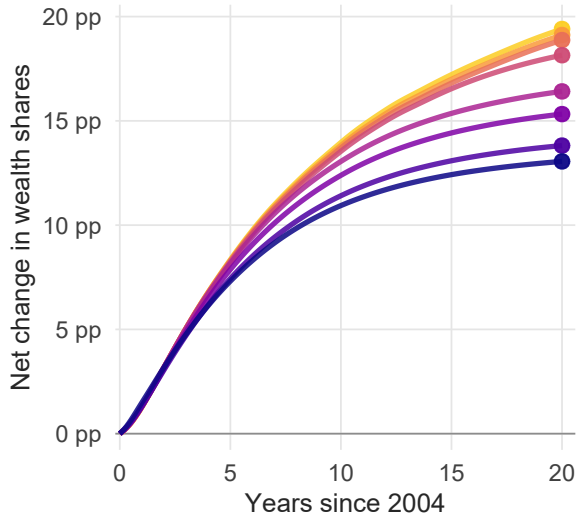
Figure F.4: Sectoral specialization across financial-development regimes. Common initial state, world prices, tariffs, and NX_{res} path; output valued at the Peru baseline's initial producer prices.

F.4 Additional Import Tariffs Counterfactuals Summary

(a) Output Composition



(b) Wealth reallocation over time



(c) Real-output reallocation over time

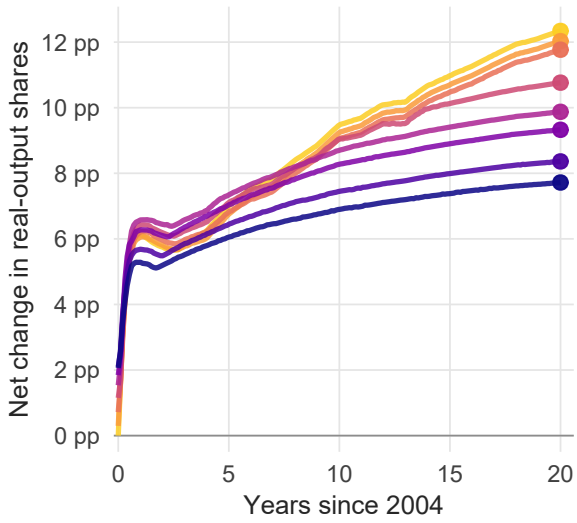


Figure F.5: Sectoral specialization across import tariff regimes. Common initial state, world prices, export tariffs, and $NX_{\text{res}} = 0$ path; output valued at the free trade case's initial producer prices.

G Persistent Specialization in a Small Open Economy

The purpose of this appendix is to provide a stripped-down example of the mechanism studied in the paper. The model has two sectors, mobile labor, fixed world prices, and sector-specific capital accumulation. The goal is to derive a closed-form characterization of how the interaction between capital accumulation and fixed world prices can generate persistent specialization patterns.

The appendix considers two exercises. First, if productivity differences across sectors are permanent, the economy converges to full specialization in the more productive sector. Second, if productivity differences are only temporary, aggregate capital eventually converges to the standard Solow steady state once technologies become identical again. However, the sectoral composition of capital continues to reflect the history of those temporary productivity differences. As a result, temporary productivity advantages can generate persistent differences in sectoral capital and output even after productivity differences disappear.

Importantly, the appendix shows that persistent specialization patterns can arise even without heterogeneous entrepreneurs, financial frictions, or a CES demand system. The key "trick" is that labor is mobile across sectors, while capital is accumulated within sectors. Mobile labor links the two sectors in general equilibrium. However, because the economy is small and open, goods prices are fixed at world levels. As a result, sectoral expansion does not lower the relative price of the expanding sector. This removes the closed-economy force that would otherwise reduce initial asymmetries.

The appendix therefore should not be interpreted as a literal reduced-form version of the main model. In particular, capital in this environment is accumulated within sectors, rather than being allocated through an integrated economy-wide financial market as in the heterogeneous-agent framework studied in the paper. As a result, persistence here operates through sectoral accumulation dynamics rather than through the endogenous wealth reallocation mechanism emphasized in the main model.

The purpose of the exercise is instead to isolate, in a transparent environment, the role of fixed world prices in shaping long-run sectoral dynamics. By shutting down the closed-economy relative price adjustment mechanism, temporary asymmetries in productivity translate into persistent differences in sectoral capital stocks and production patterns. The appendix therefore provides an illustrative benchmark for the broader mechanism emphasized in the main text, where similar forces operate through endogenous wealth accumulation and financial frictions in an integrated multisector economy.

G.1 Setup

Environment. Time is continuous. There are two sectors indexed by $i \in \{1, 2\}$. Sector i produces using capital $K_i(t)$ and labor $L_i(t)$ according to

$$Y_i(t) = A_i(t) K_i(t)^{\alpha} L_i(t)^{1-\alpha}, \qquad 0 < \alpha < 1. \tag{G.1}$$

Total labor supply is fixed and normalized to one:

$$\sum_{i \in \{1,2\}} L_i(t) = 1. \tag{G.2}$$

Labor is perfectly mobile across sectors. Goods prices are fixed at world levels. Without loss of

generality, prices can be normalized to one so that all quantities are expressed in real terms:

$$p_i^* = 1, \qquad i \in \{1, 2\}.$$

Capital is accumulated by sector-specific entrepreneurs. To keep the model as close as possible to the Solow model, assume that a constant fraction $s_i \in (0, 1)$ of sectoral output is invested in the same sector. Capital depreciates at rate $\delta > 0$. Thus,

$$\dot{K}_i(t) = s_i Y_i(t) - \delta K_i(t). \tag{G.3}$$

Labor Allocation. At each point in time, labor is allocated competitively across sectors. Hence, the wage equals the value marginal product of labor in every active sector:

$$w(t) = (1 - \alpha) A_i(t) K_i(t)^{\alpha} L_i(t)^{-\alpha}. \tag{G.4}$$

For any two active sectors, this implies

$$(1 - \alpha) A_1(t) K_1(t)^{\alpha} L_1(t)^{-\alpha} = (1 - \alpha) A_2(t) K_2(t)^{\alpha} L_2(t)^{-\alpha}. \tag{G.5}$$

Canceling common terms and rearranging gives

$$\frac{L_1(t)}{L_2(t)} = \left(\frac{A_1(t)}{A_2(t)} \right)^{1/\alpha} \frac{K_1(t)}{K_2(t)}. \tag{G.6}$$

Using $\sum_{i \in \{1,2\}} L_i(t) = 1$, sectoral labor allocations are

$$L_i(t) = \frac{A_i(t)^{1/\alpha} K_i(t)}{\sum_{j \in \{1,2\}} A_j(t)^{1/\alpha} K_j(t)}. \tag{G.7}$$

Labor flows to sectors with higher productivity and larger capital stocks.

Reduced-Form Sectoral Output. Substituting the labor allocation into production gives

$$Y_i(t) = A_i(t) K_i(t)^{\alpha} \left[\frac{A_i(t)^{1/\alpha} K_i(t)}{\sum_{j \in \{1,2\}} A_j(t)^{1/\alpha} K_j(t)} \right]^{1-\alpha}. \tag{G.8}$$

This simplifies to

$$Y_i(t) = \frac{A_i(t)^{1/\alpha} K_i(t)}{\left[\sum_{j \in \{1,2\}} A_j(t)^{1/\alpha} K_j(t) \right]^{1-\alpha}}. \tag{G.9}$$

Define aggregate effective capital as

$$X(t) \equiv \sum_{i \in \{1,2\}} A_i(t)^{1/\alpha} K_i(t). \tag{G.10}$$

Then sectoral output can be written compactly as

$$Y_i(t) = A_i(t)^{1/\alpha} X(t)^{-(1-\alpha)} K_i(t). \tag{G.11}$$

Capital Accumulation. The law of motion for sectoral capital becomes

$$\dot{K}_i(t) = s_i A_i(t)^{1/\alpha} X(t)^{-(1-\alpha)} K_i(t) - \delta K_i(t). \tag{G.12}$$

Therefore, the growth rate of sectoral capital is

$$g_{K,i} = \frac{\dot{K}_i(t)}{K_i(t)} = s_i A_i(t)^{1/\alpha} X(t)^{-(1-\alpha)} - \delta. \tag{G.13}$$

Taking the difference across sectors gives the law of motion for relative capital:

$$\frac{d}{dt} \log \left(\frac{K_1(t)}{K_2(t)} \right) = X(t)^{-(1-\alpha)} \left[s_1 A_1(t)^{1/\alpha} - s_2 A_2(t)^{1/\alpha} \right]. \tag{G.14}$$

This equation summarizes the mechanism. Relative sectoral capital grows towards the sector with higher productivity or a higher reinvestment rate.

G.2 A Persistent Productivity Gap

First, this toy model shows that if a productivity gap persists across sectors, the economy eventually specializes in the more productive sector.

Suppose both sectors begin with positive capital stocks,

$$K_1(0) > 0, \qquad K_2(0) > 0,$$

and have the same investment rate,

$$s_1 = s_2 = s.$$

Let sector 1 be permanently more productive:

$$A_1(t) = A, \qquad A_2(t) = (1-\tau)A, \qquad 0 < \tau < 1.$$

Using the relative capital growth derived above (Equation G.14),

$$\frac{d}{dt} \log \left(\frac{K_1(t)}{K_2(t)} \right) = s X(t)^{-(1-\alpha)} A^{1/\alpha} \left[1 - (1-\tau)^{1/\alpha} \right] > 0. \tag{G.15}$$

Hence, sector 1 accumulates capital faster than sector 2, and its relative size increases over time.

In this case, the economy converges to a corner allocation in which only the more productive sector remains active:

$$K_1^* > 0, \qquad K_2^* = 0. \tag{G.16}$$

At the corner, aggregate capital is equal to capital in sector 1:

$$K^* = K_1^*. \tag{G.17}$$

Since sector 1 has productivity A, aggregate capital satisfies the standard Solow steady-state condition:

$$0 = sA(K^*)^\alpha - \delta K^*. \tag{G.18}$$

Thus,

$$K^* = K_1^* = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}, \qquad K_2^* = 0. \tag{G.19}$$

This benchmark shows that with a permanent productivity gap, fixed world prices imply full specialization in the more productive sector. Aggregate capital converges to the standard Solow steady state, but all capital is eventually concentrated in the more productive sector.

G.3 A Temporary Productivity Gap

The main model shows that under identical fundamentals, the self-financing mechanism tends to reduce initial productivity differences across sectors over time. To capture this idea in the toy model, now suppose that the productivity gap between sectors disappears after a finite period of time.

Suppose the two sectors begin with the same capital stock:

$$K_1(0) = K_2(0) = K_0. \tag{G.20}$$

Assume equal investment rates:

$$s_1 = s_2 = s. \tag{G.21}$$

As before, let sector 1 have productivity

$$A_1(t) = A,$$

while sector 2 has a temporary productivity disadvantage:

$$A_2(t) = \begin{cases} (1-\tau)A, & t < T, \\ A, & t \geq T, \end{cases}, \qquad \text{with } 0 < \tau < 1 \text{ and finite } T. \tag{G.22}$$

Then, for $t < T$, the same logic from the previous subsection implies

$$\frac{d}{dt}\log\left(\frac{K_1(t)}{K_2(t)}\right) > 0. \tag{G.23}$$

Hence, while the productivity gap persists, sector 1 accumulates capital faster than sector 2.

Now suppose that at time T, the productivity gap disappears:

$$A_1(t) = A_2(t) = A, \qquad t \geq T. \tag{G.24}$$

From that moment onward, both sectors grow at the same rate as the factor $\left[1-(1-\tau)^{1/\alpha}\right]$ goes to zero:

$$\frac{d}{dt}\log\left(\frac{K_1(t)}{K_2(t)}\right) = 0 \qquad \Leftrightarrow \qquad g_{K,1} = g_{K,2}. \tag{G.25}$$

Therefore, the relative capital allocation reached at time T persists:

$$R_T \equiv \frac{K_1(t)}{K_2(t)} = \frac{K_1(T)}{K_2(T)} > 1 \qquad \text{for all } t \geq T. \tag{G.26}$$

Even though productivity differences disappear in finite time, the asymmetry generated during the transition remains permanently embedded in the sectoral composition of capital.

Steady State. After time T, both sectors have the same productivity A and the same investment rate s. Therefore, aggregate capital evolves according to:

$$\dot{K}(t) = sAK(t)^{\alpha} - \delta K(t), \tag{G.27}$$

which is exactly the same law of motion as in the standard Solow model.[23]

Hence, aggregate capital converges again to the standard Solow steady state:

$$K^* = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}} \qquad \Leftrightarrow \qquad 0 = sA(K^*)^{\alpha} - \delta K^*. \tag{G.28}$$

However, the sectoral composition of capital depends on the history of temporary productivity differences. Since relative capital remains constant after time T,

$$\frac{K_1(t)}{K_2(t)} = R_T \qquad \text{for all } t \geq T, \tag{G.29}$$

[23] After time T, both sectors share the same productivity level A, so

$$X(t) = A^{1/\alpha} \sum_{i \in \{1,2\}} K_i(t).$$

The sectoral capital accumulation equation becomes

$$g_{K,i} = sA^{1/\alpha} \left[A^{1/\alpha} \sum_{j \in \{1,2\}} K_j(t) \right]^{-(1-\alpha)} - \delta.$$

Using

$$\left[A^{1/\alpha}\right]^{-(1-\alpha)} = A^{-(1-\alpha)/\alpha},$$

and defining aggregate capital as

$$K(t) \equiv \sum_{i \in \{1,2\}} K_i(t),$$

this implies

$$g_{K,i} = sAK(t)^{-(1-\alpha)} - \delta.$$

Since both sectors grow at the same rate after time T, aggregate capital evolves according to

$$\dot{K}(t) = \sum_{i \in \{1,2\}} g_{K,i} K_i(t) = \left[sAK(t)^{-(1-\alpha)} - \delta\right] K(t),$$

which simplifies to

$$\dot{K}(t) = sAK(t)^{\alpha} - \delta K(t).$$

the steady-state sectoral capital stocks must satisfy

$$K_1^* = \frac{R_T}{1+R_T}K^*, \qquad K_2^* = \frac{1}{1+R_T}K^*, \tag{G.30}$$

with

$$\frac{K_1^*}{K_2^*} = R_T > 1. \tag{G.31}$$

This shows that even though both sectors eventually share identical technologies, the long-run allocation of capital depends on the history of temporary productivity differences. Aggregate capital converges to the standard Solow steady state, but the sectoral composition of that steady state inherits the asymmetries generated during the transition.

The Persistence of Specialization. This toy model delivers a simplified version of the history dependence mechanism in the paper. With a permanent productivity gap, the economy converges to full specialization in the more productive sector. However, even when productivity differences are only temporary, the sectoral composition of capital continues to reflect the history of those differences in the long run. Once technologies become identical again, aggregate capital converges to the standard Solow steady state, but sectoral capital shares remain pinned down by the relative accumulation that occurred during the transition.

The key mechanism is that world prices are fixed. As a result, temporary productivity differences are not offset by relative-price adjustments, allowing them to translate into persistent differences in capital accumulation. In the full model, the relevant state variable is not only sectoral capital, but the wealth-weighted distribution of entrepreneurs within each sector. The full model therefore endogenizes the accumulation mechanism through wealth heterogeneity and financial frictions, allowing the magnitude and persistence of sectoral differences to depend on the evolving distribution of wealth and productivity within sectors.